

Thinking about the Economy, Deep or Shallow?*

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November 25, 2025

Abstract

We propose a theory of *shallow thinking* to capture people's limited understanding of general equilibrium. We cast general equilibrium as a causal chain among variables. We assume that people only think through finite steps, which our estimation using a customized survey finds at about 2.6. This dampens longer causal chains and generates differential belief misreaction across variables and shocks. Applying shallow thinking to a New Keynesian model with active monetary policy suggests that long-term interest rates underreact to cost-push shocks but overreact to monetary policy shocks; and news about future cost-push shocks triggers inflation, rather than deflation under rational expectations.

Keywords: causal reasoning, mental models, general equilibrium, inflation expectations, monetary policy, return predictability, business cycle fluctuations.

JEL: D50, D83, D84, E12, E30, E52, E62, E70

*We thank Peter Andre, Francesca Bastianello, Adrien Bilal, John Campbell, Gabriel Chodorow-Reich, Ben Enke, Maryam Farboodi (discussant), Joel Flynn, Xavier Gabaix, Thomas Graeber, Sam Hanson, Oleg Itskhoki, Baiyun Jing, Spencer Kwon, David Laibson, Chen Lian, Avi Lipton, Matthew Rabin, Chris Roth, Karthik Sastry, Josh Schwartzstein, Dmitriy Sergeyev, Andrei Shleifer, Stefanie Stantcheva, Jeremy Stein, Ludwig Straub, Adi Sunderam, Alireza Tahbaz-Salehi, Chris Wolf, and audiences at BC Carroll, Brown, Chicago Booth Asset Pricing Conference, Columbia Business School, Harvard, HBS, IZA Beliefs Workshop, Kellogg, LSE, New York Fed Junior Macro Workshop, NYU Stern, UCLA Anderson, USC Marshall, UT Austin McCombs, UW Foster, Wharton, and Yale for their helpful comments. We thank Sam Cohen for research assistance and Roberto Colarieti for help with the survey design. The study received IRB approval from Harvard (IRB22-1403, IRB24-0959). We acknowledge financial support from the Michael S. Chae Macroeconomic Policy Fund and the Molly and Domenic Ferrante Fund, both awarded through Harvard. Wu is grateful for support from the Alfred P. Sloan Foundation Pre-Doctoral Fellowship in Behavioral Macroeconomics, awarded through the NBER. First version: November 2024.

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1 Introduction

It is well documented that ordinary people underappreciate “equilibrium effects” across micro- and macroeconomic settings (Nagel, 1995; Dal Bó, Dal Bó and Eyster, 2018; Andre et al., 2022). For example, many people fail to appreciate the directional effects of macro shocks on unemployment and inflation, and think of propagation channels that differ from those emphasized by economists. As all macro outcomes adjust in response to any shock, how do we formalize equilibrium effects in macro models and underappreciation thereof? Among all these equilibrium effects, are some harder to appreciate than others?

We propose a theory of *shallow thinking* to model people’s limited understanding of shock propagation in general equilibrium, measure it, and uncover its implications. We present a systematic way to unpack a general equilibrium model as a causal chain among multiple variables with feedback loops. We assume that, in forming beliefs, people think through only finite steps of shock propagation along the causal chain, and hence, longer chains are less understood.¹ Our customized survey finds that, on average, people understand about 2.6 steps, far below infinity that corresponds to rational expectations.

By truncating people’s reasoning along the causal chain, shallow thinking predicts belief under- and overreaction across variables and shocks, with implications that align with a range of evidence. In a New Keynesian model with an active Taylor rule, shallow thinking predicts that long-term nominal interest rates overreact to monetary policy shocks and underreact to supply shocks, consistent with empirical findings. It generates inflation in response to news about future supply shocks, in contrast to deflation under rational expectations. In an RBC model, shallow thinking amplifies and prolongs the economy’s responses to productivity shocks and leads to a stock market boom and crash.

In Section 2, we introduce shallow thinking by unpacking the textbook New Keynesian model as a system of causal relations. We construct a 2-period setting by considering transitory news shocks (i.e., period-1 shocks that become known in period 0), before generalizing to persistent shocks in Section 5. Under the standard assumption that agents

¹By distilling all “equilibrium effects” and suggesting differential difficulty in understanding them, we enrich previous work that focuses only on strategic interactions in general equilibrium (García-Schmidt and Woodford, 2019, Farhi and Werning, 2019, Iovino and Sergeyev, 2023, Bianchi-Vimercati, Eichenbaum and Guerreiro, 2024, Greenwood and Hanson, 2015, Bastianello and Fontanier, 2025a,b).

observe contemporaneous variables, the period-1 equilibrium is a static general equilibrium, with no role for beliefs. However, the period-0 equilibrium depends on agents' beliefs about period-1 outcomes, as households and firms make forward-looking decisions. We introduce shallow thinking of period-1 general equilibrium and explore its implications for macro outcomes and asset prices in period 0.

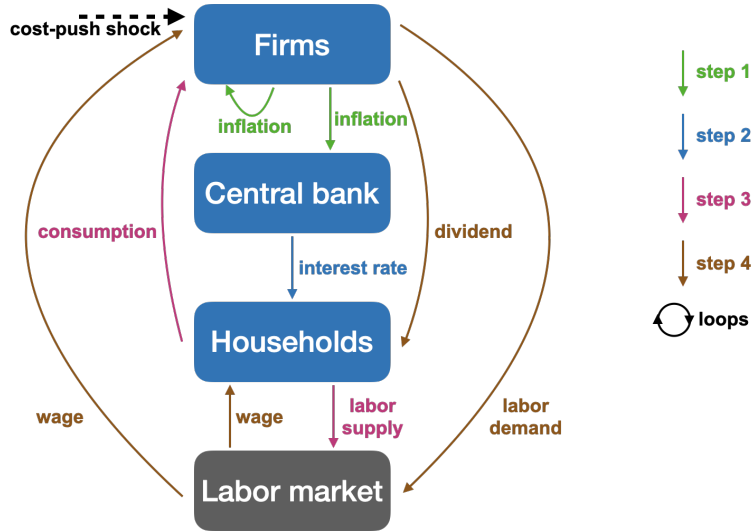


Figure 1: Period-1 New Keynesian economy as directed cyclic graph of causal relations

Notes: This figure depicts the causal relations among macroeconomic variables in the period-1 New Keynesian economy as a directed cyclic graph. Each node represents an agent type (firms, the central bank, households) or a competitive market (the labor market). For a competitive market, the price is interpreted as being set by a fictitious Walrasian auctioneer in response to supply and demand shifts, as explained in Section 2.1. We define a *causal relation* as the dependence of one variable on another (i.e., a partial derivative), arising from agents' best responses. Dashed arrows indicate shocks affecting some agents (e.g., cost-push shocks that trigger firms to raise prices in the absence of any changes in marginal cost), while solid arrows represent variables—decisions by either actual agents (e.g., inflation, interest rate, and consumption) or fictitious auctioneers (e.g., real wage). Each arrow points to the agents (including fictitious ones) responding to it. Variables are color-coded to illustrate the step-by-step propagation of a cost-push shock.

Figure 1 depicts the causal relations among macro variables in the period-1 economy using a directed cyclic graph, which is key to introducing shallow thinking. To draw this graph, we start from the basics of general equilibrium: agents' best responses and price determination in competitive markets. We cast each competitive market as a fictitious auctioneer who sets the price in response to changes in demand and supply, capturing Walras's idea of price adjustment via tâtonnement. With such auctioneers as fictitious agents, we represent general equilibrium as a system of best response functions of all the real and fictitious agents. We define a *causal relation* as the dependence of one variable on another (i.e., a partial derivative) driven by these best responses. As reflected in

Figure 1, each solid arrow represents a variable chosen by some agents, pointing to the agents it directly influences, and a dashed arrow indicates a shock that directly changes some agents' behavior. The graph is cyclic, representing the equilibrium as a fixed point, appreciated by rational agents who take infinite steps starting from the shock.

Deviating from rational expectations, we hypothesize that individuals foresee only a finite number of steps in shock propagation on the graph, which we refer to as their *depth of thinking* d . For example, we color-code Figure 1 to illustrate the propagation of a cost-push shock, i.e., a supply shock that triggers firms to raise prices without a change in marginal cost. A depth-1 agent acknowledges only the most obvious implication—firms will raise their prices, causing inflation—while overlooking changes in all other variables. A depth-2 agent further appreciates that the central bank will raise the interest rate in response to higher inflation according to the Taylor rule, but fails to foresee additional implications. A depth-3 agent understands that a higher interest rate will discourage household consumption and incentivize labor supply. A depth-4 agent recognizes that changes in household behavior will affect the firms and the labor market, and so on. This iterative process captures the idea that more distant causal effects are harder to grasp. It nests rational expectations as infinite depth of thinking, under which agents correctly assess the strength of all causal relations and loops embedded in the graph.

We allow for belief heterogeneity parsimoniously—assuming that, in the population, the depth of thinking d follows a geometric distribution with continuation rate $\lambda \in [0, 1]$. A higher λ means that people think more deeply, with $\frac{1}{1-\lambda}$ being the average depth of thinking and $\lambda = 1$ nesting rational expectations (i.e., infinite depth). The average expectations, which drive the equilibrium outcomes in a large class of models (beyond those analyzed in this paper), are tractably parametrized by λ —they are *as if* generated by a representative agent who knows all causal relations but dampens them by λ . Hence, λ is the only parameter we need to measure in order to apply our theory to macro models.

In Section 3, we develop a survey to test shallow thinking and measure λ based on a novel prediction. Our theory predicts that impulse responses of variables that are more distant from shocks are understood by fewer people, and this declining relationship informs λ . Crucially, this distance between variables and shocks is in the causal sense that captures people's iterative thinking process, even if all variables respond to any shock in equilibrium. This prediction arises because each shallow agent perceives changes only in a subset of variables that are close to the shocks in the directed graph. It allows us to estimate λ from a systematic pattern of belief heterogeneity at the population level.

To test this prediction, we survey US households and present them with hypothetical scenarios involving classic macro shocks, such as oil supply shocks and monetary policy shocks, building on [Andre et al. \(2022\)](#). For each shock, respondents provide directional forecasts for key macro variables, such as inflation, unemployment, and interest rates, in 12 months. We use directional responses from the empirical literature as benchmarks to assess the correctness of their forecasts. We confirm that variable-shock combinations that are further apart on the causal chain are forecasted correctly by fewer people. This finding is robust to including a rich set of fixed effects, such as individual-by-variable and individual-by-shock fixed effects, suggesting that it is not driven by certain individuals understanding certain variables or shocks better than others.

Our estimation strongly rejects the null hypothesis of $\lambda = 1$ and suggests $\lambda \approx 0.61$, implying an average depth of thinking ($\frac{1}{1-\lambda}$) of only about 2.6—far below infinity assumed under rational expectations. Our findings keenly support our parametrization of belief formation in deviation from rational expectations. While we nest rational expectations as a limit case of infinite depth of thinking, an alternative view is that rational agents may have learned the responses of variables to shocks through experience, without necessarily understanding the workings of the economy. Our survey does not require respondents to explain their predictions and finds that many fail to provide correct directional responses, casting doubt on this view. Further, our estimation of λ builds upon a systematic belief error: for the same variable, it is less understood when it is more distant from shocks, rather than resting on an arbitrary belief error that may result from noisy measurement.

In Sections 4 and 5, we show that shallow thinking generates belief under- and overreaction across variables and shocks, and leads to several implications in macro and finance that align with empirical evidence. Specifically, we show that, under shallow thinking, long-term nominal interest rates overreact to monetary policy shocks—as documented by [Cochrane and Piazzesi \(2002\)](#), [Gürkaynak, Sack and Swanson \(2005\)](#), [Hanson and Stein \(2015\)](#), and others—but underreact to supply shocks, consistent with findings by [Bauer, Pflueger and Sunderam \(2024\)](#) and [Cieslak \(2018\)](#). Moreover, macro variables have additional predictive power for bond excess returns beyond current yields, as found by [Ludvigson and Ng \(2009\)](#), [Cooper and Priestley \(2009\)](#), [Joslin, Priebsch and Singleton \(2014\)](#), and [Cieslak and Povala \(2015\)](#). Our model further predicts that news about future cost-push shocks leads to inflation—consistent with [Känzig \(2021\)](#)—rather than deflation as predicted by rational expectations. Finally, more persistent cost-push shocks lead to higher inflation, which contrasts with lower inflation predicted by rational expectations.

Shallow thinking reconciles all these findings through dampening perceived causal relations and generating variable-by-shock belief misreaction, which has two layers of meaning. First, beliefs of the same variable misreact differently to different shocks. In the 2-period setting, the period-0 long-term interest rate *underreacts* to cost-push news shocks, because agents underperceive the Taylor rule—that the central bank will raise the short-term interest rate in response to inflation in period 1. In contrast, the period-0 long-term interest rate *overreacts* to monetary policy news shocks, as agents underappreciate an offsetting loop: a positive monetary policy shock will lower inflation, prompting the central bank to slightly lower the short-term rate per the Taylor rule. When the economy is subject to multiple shocks, current yields (driven by interest rate expectations) mainly reflect monetary policy shocks, while macro variables such as inflation expectations load more on other shocks, providing additional predictive information on bond excess returns.

Second, in response to the same shock, beliefs of different variables misreact differently, which collectively modulate the equilibrium outcomes. In particular, cost-push news shocks lead to an overreaction of inflation expectations and an underreaction of expectations of other variables to various degrees, resulting in current inflation. The overreaction of inflation expectations arises from misperceiving the relative strength of two counteracting feedback loops. Inflation amplifies itself via a self-loop of *pricing complementarity*, as firms raise prices in response to higher aggregate inflation. It is offset by a longer loop via *monetary policy response*, whereby the interest rate hike in response to inflation lowers the real wage, exerting downward pressure on inflation. Shallow agents better understand the short amplification loop and thus perceive net amplification, though the long offset loop objectively dominates.

Hence, in response to a cost-push news shock, because of the overestimation of future inflation and the underappreciation of the future interest rate hike and associated economic downturn, firms set higher prices and households consume more in period 0, leading to inflation. In contrast, rational agents fully anticipate the future interest rate hike and the accompanying economic downturn, and their responses lead to current deflation. We further show that, in an infinite-horizon setting, more persistent cost-push shocks cause higher inflation, in contrast to lower inflation predicted by rational expectations. This difference in comparative statics arises because a more persistent shock strengthens general equilibrium feedbacks, particularly boosting the long offset loop of inflation via *monetary policy response* discussed previously. A rational agent recognizes that a more persistent cost-push shock is offset more, resulting in lower inflation. In contrast, shallow

agents anticipate higher future inflation through the short amplification loop of *pricing complementarity*, and their responses bring that about. Taking these results together, shallow thinking suggests that supply shock is harder to combat with monetary policy, and more so with more persistent shocks.

In extending shallow thinking to infinite-horizon models with persistent shocks, we define cross-variable causal relations as the *sequence-space Jacobians* à la [Auclert et al. \(2021\)](#), generalizing the partial derivatives in the case of static general equilibrium. This technique is applicable to other models that can be represented in the sequence space. As an example, we consider an RBC economy in Appendix C and illustrate that shallow thinking amplifies and prolongs the economy’s responses to productivity shocks and results in a stock market boom and crash. Hence, shallow thinking brings to the RBC model amplification and persistence which are realistic features of the data ([King and Rebelo, 1999](#)), in a way consistent with the classic [Kindleberger \(1978\)](#) narrative of crises, where innovations are followed by asset market booms and crashes.

Literature review. By studying limited *causal reasoning* of the full complexity of general equilibrium (GE), our theory broadens existing research on limited *strategic sophistication* in GE ([García-Schmidt and Woodford, 2019](#), [Farhi and Werning, 2019](#), [Iovino and Sergeyev, 2023](#), [Bianchi-Vimercati, Eichenbaum and Guerreiro, 2024](#), [Greenwood and Hanson, 2015](#), [Bastianello and Fontanier, 2025a,b](#)). They introduce level- k thinking ([Nagel, 1995](#); [Stahl and Wilson, 1994, 1995](#); [Camerer, Ho and Chong, 2004](#)) and competition neglect ([Camerer and Lovo, 1999](#)) from experimental and game-theoretical settings into GE.

By focusing on strategic interactions, previous work collapses workhorse macro models into simpler representations. For example, [García-Schmidt and Woodford \(2019\)](#) consider a 3-equation representation of the New Keynesian model. [Farhi and Werning \(2019\)](#) study the Keynesian cross whereby consumption amplifies itself. With only one feedback loop, level- k thinking predicts a uniform under- or overreaction of beliefs (depending on the sign of the feedback loop), and shallow thinking would yield similar predictions.²

We argue that workhorse macro models are better represented as a long causal chain with multiple variables and feedback loops. Conceptually, limitations in human reasoning about complex causal systems are well established in cognitive psychology, even absent strategic interaction.³ Pragmatically, preserving all the causal relations has realistic

²Indeed, [Angeletos and Lian \(2023a\)](#) show that this property is shared across several prominent models of bounded rationality and incomplete information.

³For example, people’s knowledge of causally complex systems—whether mechanical devices or natural

implications. For example, interest rate expectations underreact to a supply shock even if inflation expectations overreact, as agents underappreciate the Taylor rule (Bianchi, Lettau and Ludvigson, 2022; Bauer, Pflueger and Sunderam, 2024; Wu, 2023). Moreover, with multiple feedback loops, shallow thinking dampens them differentially by length, potentially reversing the sign of the perceived net effect. In summary, we suggest if agents only partially understand the economy, we as modelers should not simplify for them.

Our theory concerns bounded rationality in the absence of information frictions, and hence complements a large theoretical literature on information frictions (Lucas, 1972; Gabaix and Laibson, 2001; Mankiw and Reis, 2002; Woodford, 2003a; Nimark, 2008; Angeletos and La’O, 2009; Angeletos and Lian, 2018; Angeletos and Huo, 2021; Angeletos and Sastry, 2021) and inattention (Sims, 2003; Maćkowiak and Wiederholt, 2009; Gabaix, 2020). We find that respondents make directional forecast errors even with full information about the shocks, in support of our theory.⁴ We further complement papers on learning under misspecified models (Evans and Honkapohja, 2001; Eusepi and Preston, 2018; Farmer, Nakamura and Steinsson, 2024; Molavi, Tahbaz-Salehi and Vedolin, 2024; Molavi, 2025) or model uncertainty (Hansen and Sargent, 2007). We characterize misspecified mental models based on the true model analytically, measure it, and uncover implications that align with evidence. We also offer a tractable implementation in workhorse macro models using sequence-space Jacobian popularized by Auclert et al. (2021).

Our theory contributes to the belief overreaction literature, by generating under- and overreaction in a way that reconciles several bond market puzzles as previously discussed. Barberis (2018) reviews earlier papers in this literature and Bordalo et al. (2020), Afrouzi et al. (2023), and Azeredo da Silveira, Sung and Woodford (2024) present recent developments. In contrast to papers that generate under- or overreaction hinging on the sequence of shocks (Barberis, Shleifer and Vishny, 1998; Daniel, Hirshleifer and Subrahmanyam, 1998; Rabin, 2002; Wachter and Kahana, 2024; Kwon and Tang, 2024; Bianchi, Ludvigson and Ma, 2025), our prediction only depends on the propagation mechanisms of shocks.

phenomena—is sparse and shallow (Rozenblit and Keil, 2002; Wilson and Keil, 1998). They pay special attention to earlier nodes in causal chains (Ahn et al., 2000) and their predictions based on causal networks are insufficiently sensitive to the strengths of causal links (see Rottman and Hastie, 2014 for a review). While our primary contribution is in macro and finance, our study also advances the causal reasoning literature (Waldmann, 2017), which typically feature simple examples in short experiments, by examining a complex real-world domain involving many variables and long-term data accumulation.

⁴Moreover, prior work generates *horizon-dependent dampening* due to bounded rationality (Gabaix, 2020; Farhi and Werning, 2019) or information frictions (Angeletos and Lian, 2018; Angeletos and Huo, 2021). We focus on *cross-variable dampening* as a complementary aspect, and discuss the connection in Section 5.1.

We provide a theory of heterogeneous mental models, contributing to the literature that measures mental models in specific scenarios (Stantcheva 2021, 2023b; Andre et al. 2022, 2025; Andre, Schirmer and Wohlfart 2024; Bastianello, Décaire and Guenzel, 2025). Andre et al. (2022) show that people’s macro forecasts in response to hypothetical shocks are highly heterogeneous. Our theory predicts belief heterogeneity in a way confirmed in our survey which builds on theirs. Prior to this paper, Wu (2023) measures people’s imperfect mental models using existing forecasts of professional forecasters and policymakers. This paper tailors a survey for households and derives implications for macro and finance.

Our use of directed *cyclic* graphs connects to work using directed *acyclic* graphs (DAGs) for different purposes. Building on Auclert et al. (2021), who use DAGs to solve macro models and break cycles for technical convenience, we draw a cyclic graph to represent mental models and emphasize loops to capture equilibrium feedback. We complement Spiegel (2016, 2020), who examines agents fully utilizing a perceived DAG with causal links that differ from the true DAG (e.g., reverse causality, ignoring confounders). We focus on the consequences of not thinking through all causal links, predicting and confirming that agents consider different links depending on the shock.⁵ Further afield, Pearl (2009) and others in causal inference use DAGs for identification.⁶ Decades of macro research have been devoted to identifying the true model, which we take as given.

2 Shallow Thinking in a New Keynesian Economy

We set up the textbook New Keynesian model in Section 2.1 with transitory news shocks, i.e. shocks that realize in period 1 and become known in period 0. In Section 2.2, we represent the period-1 general equilibrium (GE) as a system of causal relations in a directed cyclic graph. This causal representation of GE is our broader theoretical contribution, encompassing the New Keynesian model and other applications. In Section 2.3, we introduce and discuss shallow thinking based on our representation of GE.

⁵Andre et al. (2025) suggest that people give narratives about the macroeconomy that can be represented by DAGs. However, our general equilibrium models feature cyclic dependencies. Our theory suggests that people may think with DAGs *because* they do not take enough steps to appreciate feedback loops.

⁶For reviews, see Heckman and Pinto (2015) and Imbens (2020). Our treatment of equilibrium prices as determined by demand and supply shifts aligns with Imbens (2020)’s suggested representation in his Figure 12B, drawing on seminal work on simultaneous equations (Tinbergen, 1930; Haavelmo, 1943).

2.1 The New Keynesian Economy

We consider the New Keynesian model à la [Woodford \(2003b\)](#) and [Galí \(2015\)](#). The economy consists of three types of agents (firms, households, and a central bank) and a competitive labor market. We take a log-linear approximation around the steady state for simplicity and use lower-case letters for log-linear deviations.

We study period-1 shocks that become known in period 0. Since the New Keynesian model is purely forward-looking, the economy returns to its steady state from period 2 onwards. Appendix [D.1](#) provides the full details of the infinite-horizon model. Here, we focus on period 1, obtained by setting all log deviations from period 2 onwards to zero.

We represent the period-1 GE as a system of causal relations as follows: (i) we interpret price determination in a competitive market as a fictitious Walrasian auctioneer setting a price in response to supply and demand shifts; (ii) we maintain the structural form of *best responses* of all agents (including Walrasian auctioneers), expressing their decisions as functions of decision-relevant variables. The guiding principle is that in each equation, the variable on the left-hand side is what an agent chooses and the variables on the right-hand side are external to their control but influence their decisions.

Firms. There is a continuum of firms indexed by $j \in [0, 1]$ that produce using labor to satisfy demand and set prices subject to Calvo rigidity. In period 1, firms choose labor demand, pay dividends, and reset prices if possible, taking as given the aggregate inflation rate π_1 , real wage w_1 , and aggregate demand c_1 . Each firm produces a differentiated good, which collectively forms a constant-elasticity bundle that households consume, and charges a markup μ in the steady state.

All firms produce to satisfy demand using the same linear technology in labor, giving rise to the aggregate dividend and labor demand

$$\begin{aligned} div_1 &= c_1 - \frac{1}{\mu - 1} w_1 \\ n_1^d &= c_1 \end{aligned} \tag{1}$$

To anticipate our analysis of the labor market, we interpret labor demand n_1^d as a demand curve $n_1^d = \hat{n}_1^d + e^d w_1$, which shifts by

$$\hat{n}_1^d = c_1 \tag{2}$$

and has an elasticity $e^d = 0$ in this model, as firms only use labor in production.

A $(1 - \theta)$ share of firms can reset their prices in period 1 to maximize dividends, and each chooses

$$p_{j1}^* = p_0 + (1 - \beta\theta) \left[w_1 + \sum_{k=0}^{\infty} (\beta\theta)^k \pi_1 \right]$$

where β is the household time discount rate, the inverse of which equals the steady-state interest rate. Period-0 aggregate inflation results from the pricing behavior of the $(1 - \theta)$ share of resetting firms as $\pi_1 = (1 - \theta)(p_{j1}^* - p_0)$. Following the tradition at least since [Clarida, Galí and Gertler \(1999\)](#), we consider a cost-push shock ϵ_1^π , and thus inflation is determined by the Phillips curve

$$\pi_1 = \theta\kappa w_1 + (1 - \theta)\pi_1 + \epsilon_1^\pi \quad (3)$$

with $\kappa \equiv \frac{(1-\theta)(1-\beta\theta)}{\theta}$ capturing the slope with respect to the real wage w_1 . Importantly, we do not move π_1 on the right-hand side to the left. We intentionally preserve the dependence of π_1 on itself, which encapsulates the within-period complementarity in individual price-setting, as each firm takes aggregate inflation as given.

Households. There is a continuum of households who live infinitely and maximize their lifetime utility, discounted by β , which is separable in consumption and labor supply. In period 1, households choose consumption and labor supply, taking as given the nominal interest rate i_1 , the real wage w_1 , and dividend div_1 . Their optimal consumption and labor supply decisions are given by

$$\begin{aligned} c_1 &= -\sigma^{-1}\beta i_1 + \frac{(1-\beta)(\mu-1)\nu}{\sigma + \mu\nu} div_1 + \frac{(1-\beta)(1+\nu)}{\sigma + \mu\nu} w_1 \\ n_1^s &= \nu^{-1}\beta i_1 - \frac{(1-\beta)(\mu-1)\sigma}{\sigma + \mu\nu} div_1 + \nu^{-1} \left[1 - \sigma \frac{(1-\beta)(1+\nu)}{\sigma + \mu\nu} \right] w_1 \end{aligned} \quad (4)$$

where σ^{-1} is the elasticity of intertemporal substitution, and ν^{-1} is the the Frisch elasticity of labor supply.

Similar to labor demand, we interpret labor supply n_1^s as a supply curve $n_1^s = \hat{n}_1^s + e^s w_1$, which shifts by

$$\hat{n}_1^s = \nu^{-1}\beta i_1 - \frac{(1-\beta)(\mu-1)\sigma}{\sigma + \mu\nu} div_1 \quad (5)$$

and has an elasticity $e^s = \nu^{-1} \left[1 - \sigma \frac{(1-\beta)(1+\nu)}{\sigma + \mu\nu} \right]$.

Central bank. The central bank follows a Taylor rule with a monetary policy shock ϵ_1^i ,

$$i_1 = \phi\pi_1 + \epsilon_1^i \quad (6)$$

Labor market. Finally, to close the model, the wage is determined by equilibrating labor supply and demand $n_1^s = n_1^d$. We introduce a fictitious labor market auctioneer who sets the wage as the intersection of supply and demand curves,

$$\begin{aligned} \hat{n}_1^s + e^s w_1 &= \hat{n}_1^d + e^d w_1 \\ w_1 &= (e^s - e^d)^{-1} (\hat{n}_1^d - \hat{n}_1^s) \end{aligned} \quad (7)$$

This captures Walras's idea of tâtonnement, where the auctioneer raises the wage if there is excess demand for labor and reduces it otherwise, as illustrated in Figure 2.

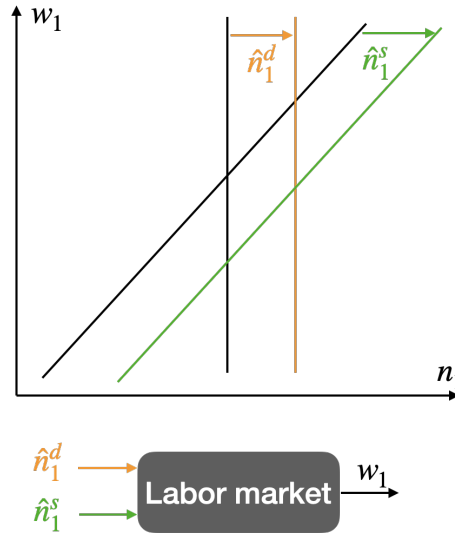


Figure 2: Wage determination in the competitive labor market

Notes: This figure depicts the labor market which gives rise to the real wage w_1 that balances supply shift $\hat{n}_1^s$ and demand shift $\hat{n}_1^d$. In the New Keynesian model, the labor demand curve is inelastic since firms only use labor in production.

Period-1 equilibrium. In period 1, agents observe all variables, best respond, and the labor market clears. The equilibrium is characterized by (1-7). This period-1 equilibrium is a static general equilibrium, where beliefs play no role.

2.2 Static General Equilibrium as a System of Causal Relations

We represent the period-1 general equilibrium in the New Keynesian economy as a system of causal relations, and draw it as a directed graph.

The period-1 equilibrium in response to the cost-push shock ϵ_1^π and the monetary policy shock ϵ_1^i is fully characterized by (1-7). We collect all macro variables in a vector $V_1 \equiv (i_1, \pi_1, div_1, \hat{n}_1^d, c_1, \hat{n}_1^s, w_1)'$ and the two shocks correspondingly in $S_1 \equiv (\epsilon_1^i, \epsilon_1^\pi, 0, 0, 0, 0, 0)'$.⁷

To capture the causal relations in the economy, we define M as a matrix of all partial derivatives among the macro variables in (1-7), detailed in (D17). Each element of M is a *partial derivative* that describes how one variable responds to another variable in period 1, which we define as a *causal relation*. A causal relation arises from either agents' best responses to decision-relevant variables (e.g., consumption c_1 responding to the interest rate i_1) or the determination of prices from supply and demand (i.e., the Walrasian auctioneer's response). We characterize the period-1 equilibrium as follows.

Proposition 1. (GE as a system of causal relations) *The period-1 New Keynesian economy is characterized by the fixed point to the system of causal relations among all agents' actions and competitive prices, $V_1 \equiv (i_1, \pi_1, div_1, \hat{n}_1^d, c_1, \hat{n}_1^s, w_1)'$, as*

$$\underbrace{V_1}_{\text{variables}} = \underbrace{M}_{\text{causal relations}} V_1 + \underbrace{S_1}_{\text{shocks}} \quad (8)$$

The equilibrium can be solved to yield

$$V_1 = (I - M)^{-1} S_1 = \sum_{n=1}^{\infty} M^{n-1} S_1 \quad (9)$$

which is a sum of all equilibrium effects of varying distance, where each $M^{n-1} S_1$ term is an n -step effect of a shock on a variable via $n - 1$ intermediate variables.

Each equation in system (8) describes how an outcome on the left depends on a set of causes on the right, where S_1 represents the direct (or partial equilibrium) effects of shocks and MV_1 captures the indirect (or general equilibrium) effects.⁸ Formula (9) expresses the solution to (8) as the sum of all equilibrium effects of varying distance. The 1-step effect S_1

⁷While we focus on these two shocks in this paper, it is straightforward to incorporate additional shocks.

⁸To incorporate shocks that directly affect multiple variables (e.g., a household preference shock impacting both consumption and labor supply), we can extend (8) as $V_1 = MV_1 + J^{direct} S_1$ in which J^{direct} captures the direct effects of each shock on a subset of variables and S_1 may differ in dimensionality from V_1 .

is the direct (or partial equilibrium) effect of shocks, while all subsequent terms represent indirect (or general equilibrium) effects of varying distance.

Mapping the system of causal relations to graph. We visualize this causal system (8) as a directed graph in Figure 1, formally supporting the intuition outlined in the introduction. Each node represents a type of agent (including the fictitious labor market auctioneer), and each arrow indicates a macro variable—either an agent’s decision or the real wage determined in the competitive labor market.

2.3 Shallow Thinking of Static General Equilibrium

Motivated by evidence from behavioral economics and psychology on limitations of causal reasoning, we assume that agents foresee only a finite number of steps in shock propagation in the directed graph. We outline two key assumptions that shape agents’ heterogeneous beliefs and lead to a parsimonious characterization of the average beliefs.

In period 0, we assume that agents form expectations about period-1 outcomes only based on the news shocks S_1 as follows.

Assumption 1. Individuals each have a finite *depth of thinking* $d \in \mathbb{N}^+$, with expectations

$$\mathbb{E}_0^d[V_1] \equiv \sum_{n=1}^d M^{n-1} S_1 \quad (10)$$

which implies an iterative formula for $d > 1$, starting from $\mathbb{E}_0^0[V_1] = 0$,

$$\mathbb{E}_0^d[V_1] = M\mathbb{E}_0^{d-1}[V_1] + S_1 \quad (11)$$

Definition (10) formalizes the idea that a depth- d agent only understands the effects of shocks that take no more than d steps. The iterative formula (11) suggests that a depth- d agent can think one step further compared to a depth- $(d - 1)$ agent.

For example, in response to a cost-push shock ϵ_1^π (Figure 1), a depth-1 agent acknowledges only the most obvious implication: firms will raise their prices (causing inflation π_1). A depth-2 agent can further appreciate that the central bank will raise the interest rate i_1 in response to higher inflation. A depth-3 agent understands that a higher interest rate i_1 will discourage household consumption and incentivize labor supply. A depth-4 agent

recognizes that changes in household behavior will affect the firms and the labor market. This iteration continues infinitely, and only a depth- ∞ (i.e., rational) agent correctly assesses the strength of all loops and accurately forecasts the period-1 equilibrium, i.e., $\mathbb{E}_0^{rational}[V_1] = V_1$.⁹ In this sense, the rational expectations hypothesis is nested by letting the agents take infinite steps in the causal graph to converge to the fixed point.

Shallow thinking and representation of GE. By definition, shallow agents do not appreciate the fixed point, and their beliefs depend on the causal representation of GE. We emphasize a specific representation of GE underlying shallow thinking—once we cast competitive markets as fictitious Walrasian auctioneers that determine prices, there is one systematic representation of GE that consists of the responses of all agents to decision-relevant variables, as captured by (8) in Proposition 1. There are alternative representations that lead to the same fixed point as (8). However, all these alternatives, including the classic 3-variable representation in terms of (i_1, π_1, c_1) which we discuss in Section D.2, implicitly mix in relations that *only hold in equilibrium* and consequently fail to reflect best responses. Hence, our later test of shallow thinking is a joint test of our assumption of iterative belief formation (11) and the specific representation M for iteration.

Two types of belief distortions. A finite-depth agent suffers from two types of belief distortions. First, if the agent only think through a small number of steps, they zero out changes in variables that are farther away from the shock. For those variables, the agent would not even understand their correct directional responses, which will be the basis of our later empirical measurement of shallow thinking. Second, they generally misforecast the impulse responses in level, due to underappreciation of feedback loops in the economy, even if such an agent takes enough steps to perceive changes in all variables.

Belief heterogeneity and average beliefs. We allow for rich belief heterogeneity with a parsimonious parametrization, which leads to an analytical characterization of the average beliefs that are comparable to the true equilibrium outcomes (8, 9).

⁹Strictly speaking, under the assumption behind static GE that agents observe all contemporaneous variables in period 1, they do not have to understand others' behavior. As a result, level- k thinking agents would correctly predict the period-1 equilibrium too, and thus the period-0 equilibrium under level- k thinking coincides with the rational expectations equilibrium.

Assumption 2. In the population, individual depth of thinking d follows a geometric distribution over $\mathbb{N}^+$ with continuation rate $\lambda \in [0, 1]$, i.e.,

$$\mathbb{P}(d \geq n) = \lambda^{n-1}, \forall n \in \mathbb{N}^+ \quad (12)$$

We assume that everyone can take at least one step, meaning that they understand the shock directly affects some variables. A λ share of them take at least two, a λ^2 share take at least three, and so on. A higher shallow thinking parameter λ means that individuals are deeper on average, with $\frac{1}{1-\lambda}$ being the average depth of thinking,¹⁰ and $\lambda = 1$ nesting the rational expectations hypothesis (i.e., people having infinite depth of thinking). With this parametric assumption, we could aggregate heterogenous beliefs into average beliefs, which will drive the period-0 equilibrium as we analyze in Section 4.

Proposition 2. (Average beliefs) *The average beliefs $\bar{\mathbb{E}}_0[V_1] \equiv \sum_{n=1}^{\infty} \mathbb{P}(d = n) \cdot \mathbb{E}_0^n[V_1]$ are sums of all effects of shocks of varying distance*

$$\bar{\mathbb{E}}_0[V_1] = \sum_{n=1}^{\infty} \lambda^{n-1} M^{n-1} S_1 \quad (13)$$

Further, the average beliefs satisfy a fixed point

$$\bar{\mathbb{E}}_0[V_1] = \underbrace{\lambda M}_{\text{average perceived causal relations}} \bar{\mathbb{E}}_0[V_1] + S_1 \quad (14)$$

Equation (13) is comparable to (9) that expresses the equilibrium as a sum of all effects of varying distance, but with more distant effects dampened more, since fewer people appreciate them. Equation (14) further suggests that, the average beliefs are formed *as if* as a fixed point by a representative agent who knows all causal relations in M but underappreciates them by a factor λ , parallel to (8) that characterizes the equilibrium as a fixed point. The proof is simple, by summing all n -step effects with decaying weights.¹¹

In summary, the backbone of shallow thinking is that people understand only a limited number of steps in shock propagation, as captured by Assumption 1. Assumption 2 serves

¹⁰The expectation of a geometric distribution is $\mathbb{E}[d] \equiv \sum_{n=1}^{\infty} \mathbb{P}(d = n) \cdot n = \sum_{n=1}^{\infty} \mathbb{P}(d \geq n) = \frac{1}{1-\lambda}$.

¹¹Moreover, formula (14) is exactly the formula of imperfect mental models studied in Wu (2023). That paper extracts an empirical moment using comovements in multivariate forecasts data and rejects the null of $\lambda = 1$ (rational expectations). That exercise is *quantitative*, comparing forecasts to the true conditional expectations. This paper provides *qualitative* evidence based on beliefs being directionally correct, elicited in a customized survey. The estimates from two papers are remarkably close.

as a convenient aggregator to generate average beliefs. Its nature is parametric rather than conceptual, akin to how Calvo pricing is a useful parametrization of nominal rigidity but not essential. With these two assumptions, one can generate heterogeneous and average beliefs in a macro model with a single additional parameter, λ . If the model also satisfies an additional assumption introduced next, we can estimate λ with a panel regression using a survey that elicits directional forecasts of shock propagation.

3 Measuring Shallow Thinking

In this section, we test shallow thinking and estimate λ using a survey. Readers more interested in the consequences of shallow thinking may choose to skip this section.

First we provide a brief overview of our survey design, which examines the general public’s understanding of shock propagation, with details in Appendix A.1. We formalize the theoretical prediction that changes in variables more distant from shocks are perceived by fewer people in Section 3.1, which enables us to measure λ in Section 3.2.

Overview of survey design. We assess the general public’s understanding of shock propagation by asking them to forecast the directional responses of key macro variables to hypothetical shocks. We conducted an online survey of 1,000 respondents with demographics quotas following the census distribution.¹²

We study six classic macro shocks in three groups: oil supply shocks (oil) and monetary policy shocks (MP) as group 1, government spending shocks (G) and personal income tax shocks (PIT) as group 2, and corporate income tax shocks (CIT) and transfer payment shocks (TP) as group 3. Half of the respondents are randomly assigned to group 1, which includes the two shocks analyzed in the model, while a quarter are assigned to each of groups 2 and 3 for additional evidence. For each shock, respondents provide directional forecasts of a set of macro variables, such as inflation and the interest rate. For each variable, respondents select “up,” “down,” “unchanged,” or “I don’t know” to indicate the expected change in response to the shock over the next 12 months.

¹²Due to resource constraints, we survey US households and apply the calibrated λ to both households and firms in our model. Notably, firm managers are largely uninformed about recent aggregate inflation or monetary policy and have inflation expectations that are far from anchored, similar to households (Candia, Coibion and Gorodnichenko, 2024; Coibion, Gorodnichenko and Kumar, 2018). Furthermore, professional forecasters also underappreciate the causal relations in the economy (Wu, 2023), including the response of monetary policy to economic conditions (Cieslak, 2018; Bauer, Pflueger and Sunderam, 2024).

Table 1 presents the baseline specification with eleven macro variables and their true directional responses to shocks according to the empirical literature reviewed in Table B2. In addition, it lists the *causal distance* D_{vs} , which is a model-implied measure of the distance between a variable v and a shock s on the directed graph that we formally define shortly. Based on the true directional responses, we determine the correctness of each respondent's forecasts accordingly. This baseline specification comprises variable-shock combinations whose responses are empirically and intuitively uncontroversial. The baseline causal distance D_{vs} is derived from our New Keynesian model, enriched with decreasing-returns production and a Taylor rule dependent on both inflation and unemployment. We consider various robustness checks for the selection of variable-shock combinations (v, s) and causal distance D_{vs} in Tables B3 and B4.

Table 1: Baseline version of causal distance and correct directions

	Group 1 (50%)		Group 2 (25%)		Group 3 (25%)	
	Oil ↑	MP ↑	G ↑	PIT ↑	CIT ↑	TP ↑
Output	3↓	2↓	1↑	2↓	3↓	2↑
Interest rate	2↑	1↑				
Price	1↑	3↓	2↑	3↓		
Unemployment	3↑	2↑	2↓	2↑	3↑	3↓
Labor hours	3↓	2↓	2↑	2↓	3↓	3↑
Durable consumption	3↓	2↓	2↓	2↓	3↓	2↑
Non-durable consumption	3↓	2↓	2↓	2↓	3↓	2↑
Dividend					2↓	2↑
Personal income tax			1↑	1↑		
Corporate income tax			1↑		1↑	
Government borrowing			1↑	1↓	1↓	

Notes: Six shocks are oil supply shock (oil), monetary policy shock (MP), government spending shock (G), personal income tax shock (PIT), corporate income tax shock (CIT), and transfer payment shock (TP). The latter four all concern the federal government. Each cell indicates the causal distance D_{vs} , which is formally defined shortly, and the directional response (up or down) in the baseline specification. The directional responses are from the empirical literature reviewed in Table B2, and robustness versions of selection of variable-shock combinations and causal distance are in Tables B3 and B4.

3.1 Empirical Implications of Shallow Thinking

We formally establish the theoretical prediction that changes in more distant variables are understood by fewer people, with an additional assumption.

To set the stage, we index respondents by n , variables by v , and shocks by s in our survey. We capture the correctness of respondents' directional forecasts, based on the true directional changes from the empirical literature, as an indicator 1_{nvs} . Among the four possible options ("up," "down," "unchanged," or "I don't know"), only the correct direction ("up" or "down") counts as correct directional belief, since we restrict to variable-shock combinations with statistically significant impulse responses.

Definition 1. We define *correct directional belief* 1_{nvs} as 1 if respondent n correctly forecasts the directional response of variable v to shock s , and 0 otherwise.

As discussed earlier, a shallow agent perceives changes only in variables that are close to shocks. When aggregated across the population, since people are of heterogeneous depths of thinking, changes in variables further removed from shocks are understood by fewer people. This is a prediction about the correct directional belief 1_{nvs} at the population level, without the need to determine the depth of thinking d for each individual, which facilitates our empirical test.

In order to formally define the distance of a variable relative to a shock for our test, we introduce some additional notation. Notice that beliefs $\mathbb{E}^d[V_1]$ are linear in the shocks S_1 , as determined in (10). With only a slight abuse of notation, we let $v \in V_1$ be a variable in our model and $s \in S_1$ be a shock. Thus, $\frac{\partial v}{\partial s}$ is the true sensitivity of variable v to shock s , whereas $\frac{\partial \mathbb{E}^d[v]}{\partial s}$ is the perceived sensitivity by a depth- d individual. And their signs indicate the true and perceived directional responses of variables to shocks, respectively.

Definition 2. We define the *causal distance* D_{vs} as the minimum d such that $\frac{\partial \mathbb{E}^d[v]}{\partial s}$ has the same sign as $\frac{\partial v}{\partial s}$, for each variable $v \in V_1$ and shock $s \in S_1$.

That is, causal distance D_{vs} corresponds to the depth of the shallowest individual who can correctly perceive the directional response of v to s . In our example with transitory cost-push and monetary policy shocks, D_{vs} equals the depth of the shallowest agent who perceives any change of v in response to s . That is, $\frac{\partial \mathbb{E}^d[v]}{\partial s}$ is 0 for all $d < D_{vs}$. Nonetheless, Definition 2 is more general when applied to persistent shocks and other models.

Assumption 3. Model parameters M are such that $\frac{\partial \mathbb{E}^d[v]}{\partial s}$ has the same sign as $\frac{\partial v}{\partial s}$ for all $d \geq D_{vs}$.

This assumption holds true when more distant causal relations either amplify or offset the impact of the shock, once the correct direction is established, but do not overturn it. This

assumption is true in the New Keynesian model we study under a standard calibration and is only useful to establish a reduced-form estimation of λ next.¹³ For instance, in response to the cost-push shock ϵ_1^π , the central bank will raise the interest rate i_1 to offset the shock, but does not lead to deflation. That is, the perceived inflation response, $\frac{\partial \mathbb{E}^d[\pi_1]}{\partial \epsilon_1^\pi}$, is positive for all $d \geq 1$. And since $\mathbb{E}^d[i_1] = \phi \mathbb{E}^{d-1}[\pi_1]$ from (11), the perceived interest rate response, $\frac{\partial \mathbb{E}^d[i_1]}{\partial \epsilon_1^\pi}$, is positive for all $d \geq 2$.

Under Assumption 3, any agent with a depth of thinking d greater than or equal to the causal distance D_{vs} will correctly forecast the directional change of variable v in response to shock s , while any agent with $d < D_{vs}$ will not. Thus, the expectation of 1_{nvs} conditional on causal distance D_{vs} is the share of respondents with depth $d \geq D_{vs}$, which is $\lambda^{D_{vs}-1}$ given Assumption 2, suggesting a reduced-form estimation of λ as follows.

Proposition 3. (Heterogeneity in correct directional beliefs) *The expectation of correct directional belief 1_{nvs} , conditional on causal distance D_{vs} , in the population is*

$$\mathbb{E}^{population}[1_{nvs}|D_{vs} = D] = \lambda^{D-1}, \forall D \in \mathbb{N}^+ \quad (15)$$

where $\mathbb{E}^{population}$ denotes the expectation in the population of survey respondents. Consequently,

1. an ordinary least squares estimation of $1_{nvs} = \gamma D_{vs} + \alpha + \epsilon_{nvs}$ yields a negative slope γ ;
2. a nonlinear least squares estimation of $1_{nvs} = b_1 \cdot b_2^{D_{vs}-1} + b_0 + \epsilon_{nvs}$ identifies λ with b_2 .

We consider both the linear and the nonlinear specifications. The null of $\gamma = 0$ and $b_2 = 1$ includes rational expectations and any other theories of beliefs that do not correlate with causal distance D_{vs} . Our estimation results in Section 3.2 will show that γ is negative and b_2 is lower than 1, both with high levels of statistical significance.

The nonlinear specification offers a direct estimation of λ from a panel regression. That occurs because, under our three assumptions, the conditional expectation of 1_{nvs} (i.e., the share of correct directional forecasts in the survey population) decays exponentially in causal distance D_{vs} with rate λ , as indicated by (15).¹⁴

A linear specification is valuable for two reasons. First, it allows us to empirically control for fixed effects to purge confounding sources of belief heterogeneity, and we will

¹³While we impose Assumption 3 on the model, it is much less restrictive than it may appear. In models where Assumption 3 does not hold for all variable-shock combinations, Proposition 3 can still be applied to the subset of variable-shock combinations where it holds.

¹⁴As noted earlier, if Assumption 3 does not hold for all variable-shock combinations, as long as one can find a subset of such combinations with varying D_{vs} , one can still estimate λ with the nonlinear regression on this subset. If even that is not possible, one can estimate λ by minimizing distance between the distribution of measured 1_{nvs} and the corresponding theory-implied distribution that is parametrized by λ .

show that our slope estimate γ is indeed robust to such controls. Second, it does not hinge on the parametric Assumption 2, since a negative γ by itself indicates that some agents only perceive changes in variables closer to shocks. When Assumption 2 holds, the estimated slope γ is a weighted average of the local slopes of exponential decay λ^{D-1} .

3.2 Correctness of Belief Declines in Causal Distance

We examine the predictability of correct directional belief 1_{nvs} by causal distance D_{vs} as prescribed by Proposition 3.

Figure 3 shows the expectation of correct directional belief 1_{nvs} , conditional on causal distance D_{vs} , together with the 99.9% confidence interval. The blue dot indicates the conditional expectation. The red diamond represents the conditional expectation, after controlling for individual-by-variable and individual-by-shock fixed effects δ_{nv}, δ_{ns} , the purpose of which we discuss soon.¹⁵ The conditional expectation of 1_{nvs} declines drastically in causal distance D_{vs} in both cases. Most respondents correctly forecasts step-1 variables (those that are directly affect by shocks). Only half of them understand variables that are 2 steps away. For variables that are 3 steps away, the correct share is around one quarter, which is as good as randomly selecting an answer from the four options.

Table 2 presents various specifications of the ordinary least squares (OLS) regression

$$1_{nvs} = \gamma D_{vs} + \alpha + \delta_{nv} + \delta_{ns} + \epsilon_{nvs} \quad (16)$$

and the nonlinear least squares (NLS) regression

$$1_{nvs} = b_1 \cdot b_2^{D_{vs}-1} + b_0 + \epsilon_{nvs} \quad (17)$$

of correct directional belief 1_{nvs} on causal distance D_{vs} as in Proposition 3. Standard errors are clustered at the individual level, since they may correlate across all answers submitted by an individual.

The coefficient γ from the linear regression tests for the theory-implied pattern that changes in more distant variables are understood by fewer people. The null of $\gamma = 0$ includes rational expectations and any other theory of beliefs that does not correlate with distance D_{vs} . It also means, if respondents give random answers in our survey, that will

¹⁵The conditional expectations and confidence intervals, with and without fixed effects, are produced following Cattaneo et al. (2024).

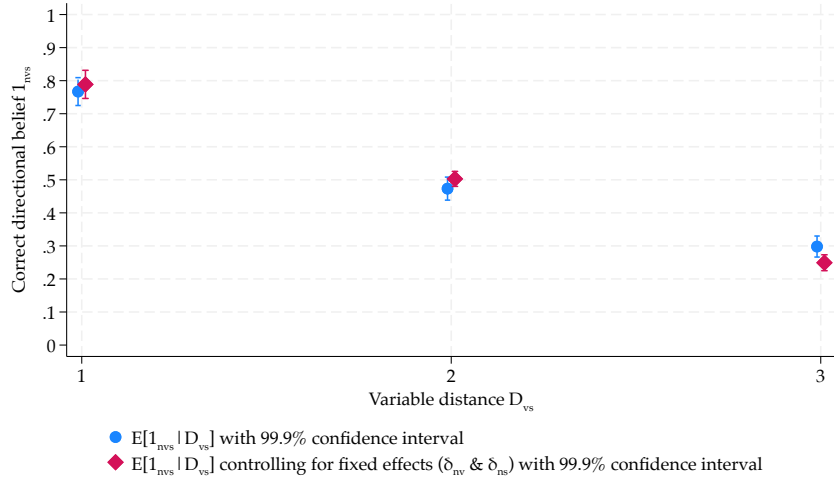


Figure 3: Expectation of correct directional belief 1_{nvs} conditional on causal distance D_{vs}

Notes: This figure shows the expectation of correct directional belief 1_{nvs} , conditional on causal distance D_{vs} , together with the 99.9% confidence interval. The blue dot indicates the conditional expectation. The red diamond represents the conditional expectation, controlling for individual-by-variable and individual-by-shock fixed effects δ_{nv} , δ_{ns} . Correct directional belief 1_{nvs} equals 1 if respondent n correctly forecasts the directional response of variable v to shock s , and 0 otherwise. Causal distance D_{vs} is derived from the New Keynesian model. Table 1 lists the correct directional responses and causal distance.

not be reflected in γ . That is, a negative γ not only implies that people make mistakes, but they do so in a distance-dependent way.

Column (1) uses causal distance D_{vs} as the only predictor and finds a statistically significant coefficient with a R^2 of 10%. Column (2) shows that individual fixed effects matter too, increasing the R^2 to 23%. That means some people are more likely to be correct than others, consistent with our assumption.

Column (3) shows that the slope estimate and its statistical significance are robust to the inclusion of variable and shock fixed effects. Controlling for these additional fixed effects addresses two important concerns. First, people may understand some variables or some shocks better in ways that happen to correlate with distance. For example, theories of optimal inattention (Sims, 2003; Gabaix, 2014) would imply that agents optimize their perceptions of variables or shocks that are more relevant for their utility.¹⁶ Our findings show that, for the same variable, its response is understood more poorly when it is further

¹⁶In Table B5, we run the same regression on subsamples of decision-relevant and decision-irrelevant variables separately. We find that, while the intercept on the decision-relevant subsample is indeed higher, indicating a higher unconditional probability of correct directional forecast, the slope is also higher, which suggests a faster decline in causal distance. Further, once we include individual-variable and individual-shock fixed effects, the intercepts and slopes on the two subsamples become statistically indistinguishable.

Table 2: Regression of correct directional belief 1_{nvs} against causal distance D_{vs}

Correct directional belief 1_{nvs}	OLS				NLS	
	(1)	(2)	(3)	(4)	(5)	(6)
Variable distance D_{vs}	-0.22*** (0.01)	-0.24*** (0.01)	-0.24*** (0.01)	-0.27*** (0.01)		
$1_{D_{vs}=2}$					-0.29*** (0.02)	
$1_{D_{vs} \geq 3}$					-0.54*** (0.02)	
$b_2 - 1$						-0.39*** (0.05)
Observations	10763	10763	10763	10763	10763	10763
R^2	0.10	0.23	0.30	0.63	0.63	0.11
Individual FE		Yes	Yes	Absorbed	Absorbed	
Variable FE			Yes	Absorbed	Absorbed	
Shock FE			Yes	Absorbed	Absorbed	
Individual-variable FE				Yes	Yes	
Individual-shock FE				Yes	Yes	

Standard errors in parentheses

Standard errors clustered at individual level

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Notes: This table shows the regression results of correct directional belief 1_{nvs} against causal distance D_{vs} , using the ordinary least squares (OLS) specification (16) and the nonlinear least squares (NLS) specification (17). Correct directional belief 1_{nvs} equals 1 if respondent n correctly forecasts the directional response of variable v to shock s and 0 otherwise. Causal distance D_{vs} is derived from a New Keynesian model. The OLS specification tests the null hypothesis that the slope is 0, controlling for individual-by-variable and individual-by-shock fixed effects δ_{nv}, δ_{ns} . The NLS specification identifies λ with b_2 , with a null hypothesis of $b_2 = 1$. Hence, we show the estimate $b_2 - 1$ and the associated p -value.

away from a shock. A second concern is that some shocks (like monetary policy shocks) may take longer to transmit into the economy or some variables may respond more slowly, which may lead people to predict no changes over a fixed horizon. Again, these possibilities are absorbed by shock and variable fixed effects.¹⁷

Column (4) further controls for individual-by-variable or individual-by-shock fixed effects. They absorb potentially confounding sources of individual heterogeneity. Such individual heterogeneity may stem from individual-specific cognitive capacity under optimal inattention, or individual-specific experience (Malmendier and Nagel, 2016). These fixed effects also absorb heterogeneity in mental models that do not hinge on distance. For

¹⁷We also note that we purposefully select variables that have statistically significant impulse responses to shocks at the 12-month horizon for our specification.

example, if a person believes in a post-pandemic quantity-constrained model of the economy, they will predict that prices respond to all shocks but quantities are fixed. Another person can instead believe in a price-constrained model.

Column (5) demonstrates that, relative to step-1 variables (that are directly shocked), step-2 variables are understood by fewer people, and step-3-and-above variables by even fewer, confirming what we observe from Figure 3.

Last, column (6) shows the nonlinear estimation and strongly rejects the null hypothesis of $b_2 = 1$. The estimation suggests that $\hat{\lambda} \approx 0.61$. That means, people on average think about only 2.6 steps, even under our parametric assumption that there is a tail distribution of people who are deeper than 3. We take this value as our calibration of λ .¹⁸

We conduct various robustness checks in Appendix B.2. We offer further evidence in Appendix B.3 to suggest that the limited depth of thinking vis-à-vis causal relations is an individual characteristic and reflects limited knowledge about the macroeconomy.

Identification and importance of shallow thinking. We conclude with several remarks. First, we interpret rational expectations as fully thinking through shock propagation. An alternative interpretation is that people may have learned how variables respond to shocks through experience, without understanding the underlying mechanisms. However, our findings suggest that many respondents fail to forecast even the directions, already casting doubt on this view. Learning may certainly occur but their importance cannot be exaggerated considering the past and ongoing efforts by macroeconomists on even identifying shocks (Ramey, 2016). Further, we not only establish belief errors, but a particular theory-predicted declining pattern of correct belief in causal distance.

Second, our findings confirm that among all equilibrium effects, some are better understood than others, a key thesis of our theory. Without a formal theory that distinguishes among equilibrium effects, one may intuitively expect a difference in understanding step-1 variables (that are directly affected) and others. But our empirics say more than that—there is a significant decline in the share of correct directional belief 1_{nvs} from step 2 to step 3 and beyond, substantiating our theory.

¹⁸Alternatively, one could infer λ from the average probability of correctly forecasting step-2 variables, $\mathbb{E}^{population} [1_{nvs} | D_{vs} = 2]$, based on (15). That is approximately 0.5, as shown in Figure 3. The difference relates to a slight discrepancy between Assumption 2—which assumes that everyone correctly understands the directional responses of step-1 variables—and the findings in Figure 3 that they mostly, but not always, do. It may arise from respondents’ occasional misunderstanding of or inattention to, certain survey questions, as any noise only lowers the indicator 1_{nvs} . We proceed with the higher value of λ to be conservative.

Last, the steep declining pattern suggests that our focus on limited depth of thinking is an important one, compared to potential differences in perceived causal models. When apply shallow thinking to macro models, we assume for simplicity that agents know the true causal links and strength. Alternatively, if people could iterate their models infinitely many times but believed in different causal models or strength of causal relations, they have to be wrong in a way that correlates with causal distance in the New Keynesian model to explain our findings, which is far from obvious. Moreover, our estimation of average depth of thinking is less than 3, suggesting that perceived strength of causal relations more distant than that does not matter much. For example, in response to cost-push shocks, they perceive changes only in inflation and interest rates, the latter via the Taylor rule, while ignoring all other causal relations. Later, we present formulas suggesting that the average perceived strength of GE feedback loops declines exponentially with their length under shallow thinking, while other model parameters (such as the Taylor rule coefficient) and any misperceptions thereof only affect it proportionally.

4 Implications for Inflation and Long-Term Rates

We show that shallow thinking leads to belief under- and overreaction and realistic implications for inflation and long-term rates, with transitory news shocks introduced in Section 2. We present analytical results and quantify them based on a standard quarterly calibration of the New Keynesian model, with all parameters listed in Table 3.

We assume that the yield of a 2-period bond $y_0^{(2)}$, i.e., the long-term yield, is determined by the expectations hypothesis with average shallow thinking beliefs as¹⁹

$$y_0^{(2)} = \frac{i_0 + \bar{\mathbb{E}}_0 [i_1]}{2} \quad (18)$$

We note that our results obtained under transitory news shock is general in this 2-period setting: while we compare equilibrium under shallow thinking against rational expectations regarding news about period-1 shocks, the same comparison holds for persistent shocks that last for 2 periods. That is simply because in the log-linearized economy, a 2-period persistent shock is the sum of a period-0 shock and a period-1 shock that be-

¹⁹Our model focuses on beliefs while abstracting away from risk. A microfoundation of this formula is that that an intermediary prices the 2-period bond on behalf of all households by averaging their beliefs. We also assume that the 2-period bond is in zero supply, so that its price does not affect the macro outcomes.

comes known in period 0. As the economy's response to a period-0 shock is independent of agents' belief formation, the comparison across theories of expectations regarding any 2-period persistent shock is solely driven by its news shock component.

Table 3: Quarterly calibration of the New Keynesian economy

Parameter	Description	Value	Estimate/Target
Beliefs			
λ	Continuation rate of depth of thinking	0.61	Our survey evidence
Firms			
θ	Price stickiness	0.75	Average price duration of 1 year
κ	Phillips curve slope	0.086	$\kappa = \theta^{-1}(1 - \theta)(1 - \beta\theta)$
μ	Steady state markup	1.1	
Households			
β	Discount factor	0.99	Steady state annual $\bar{r} = 4\%$
σ^{-1}	Elasticity of intertemporal substitution (EIS)	1	
ν^{-1}	Frisch elasticity of labor supply	0.5	
Central bank			
ϕ	Taylor rule coefficient	1.5	

4.1 Inflation Expectations Overreact to Cost-Push Shocks

Here we study the cost-push shock ϵ_1^π . We characterize the period-1 equilibrium and period-0 expectations in Proposition 4.

Proposition 4. (Period-1 cost-push shock) *The period-1 equilibrium in response to a cost-push shock ϵ_1^π features*

$$\pi_1 = \frac{1}{1 - \underbrace{(1 - \theta)}_{\text{pricing complementarity}} + \underbrace{\phi\theta\kappa}_{\text{monetary policy loop}} \cdot \underbrace{\sigma^{-1}(\nu + \sigma)}_{\text{Keynesian cross}}} \epsilon_1^\pi \quad (19)$$

$$i_1 = \phi\pi_1 \quad (20)$$

whereas the period-0 average expectations upon observing the news about ϵ_1^π are

$$\bar{\mathbb{E}}_0[\pi_1] = \frac{1}{1 - \lambda(1 - \theta) + \lambda^4\phi\theta\kappa K_1(\lambda)} \epsilon_1^\pi \quad (21)$$

$$\bar{\mathbb{E}}_0[i_1] = \lambda\phi\bar{\mathbb{E}}_0[\pi_1] \quad (22)$$

with $K_1(\lambda)$ increasing in λ under our calibration and $K_1(1) = \sigma^{-1}(\nu + \sigma)$.

To understand these results, we start with equilibrium inflation (19) and the average inflation expectation (21). The period-1 equilibrium is independent of agents' belief formation, as (19) is independent of λ . It coincides with rational expectations, i.e., (21) under $\lambda = 1$. Recall that both the equilibrium and the average expectations are the sum of all n -step effects, as established in (9) and (13), with more distant effects dampened more for expectations. We organize all these effects into three groups that involve different loops, color-coded in Figure 4, and inflation π_1 is directly involved in two of them.

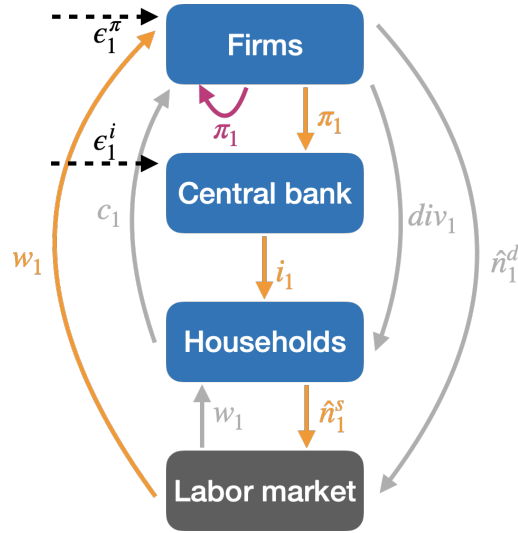


Figure 4: Causal relations and loops of period-1 economy

Notes: This figure illustrates the three loops of causal relations in the period-1 economy in different colors, to accompany the discussion of Propositions 4 and 5. Three loops are the pricing complementarity self-loop (in purple), the monetary policy loop (in orange) and the Keynesian cross (in gray).

The first loop is a self-loop of *pricing complementarity*, shown in purple. A higher inflation π_1 incentivizes all firms to price higher, thus amplifying itself. This effect has a strength of $(1 - \theta)$. As we sum up the infinite geometric series going through this loop, its strength appears in the denominator of π_1 in (19). It is dampened by λ in $\bar{\mathbb{E}}_0[\pi_1]$ in (21) since only a λ share of people understand each loop.

The second one is a 4-step loop involving *monetary policy*, shown in orange. As inflation π_1 rises, the central bank raises the interest rate i_1 , which encourages labor supply $\hat{n}_1^s$, lowering the real wage w_1 . As a lower wage prompts firms to reduce prices, this offsets the inflation response. This loop with a strength of $\phi\theta\kappa$ takes 4 steps to close, meaning that whenever it loops once, only λ^4 share of the people perceive the next loop, resulting in a dampening of λ^4 in $\bar{\mathbb{E}}_0[\pi_1]$ in (21) relative to π_1 in (19).

The final loop concerns the *Keynesian cross*, shown in gray. As the central bank raises the interest rate i_1 , it discourages household consumption c_1 , leading firms to lower dividends div_1 and reduce labor demand $\hat{n}^d$, resulting in a lower wage w_1 . Consequently, households want to consume even less, triggering additional adjustments by firms. This Keynesian cross strengthens any effect that impacts households, thus compounding the monetary policy loop. Once again, summing the infinite geometric series leads the strength of this loop $\sigma^{-1}(\nu + \sigma)$ to appear in the denominator of π_1 in (19), with its dampening for expectations $\bar{\mathbb{E}}_0[\pi_1]$ in (21) captured by a polynomial $K_1(\lambda)$.

Overall, the average inflation expectation $\bar{\mathbb{E}}_0[\pi_1]$ in (21) is modified relative to the true inflation π_1 in (19), with different loops exponentially dampened by lengths. In contrast, other model parameters (such as the Taylor rule coefficient ϕ) and any misperceptions thereof only change the perceived strength proportionally. This distinction highlights the quantitative importance of shallow thinking.

Once we establish the inflation response, the equilibrium interest rate response i_1 in (20) follows directly as the Taylor rule coefficient ϕ times inflation. The average interest expectation $\bar{\mathbb{E}}_0[i_1]$ in (22) is λ times ϕ times the inflation expectation, as it takes one more step for agents to appreciate the response of interest rate to inflation.

In the limit of $\lambda = 0$, all agents take only one step. They do not perceive any feedback on inflation, i.e., $\bar{\mathbb{E}}_0[\pi_1] = \epsilon_1^\pi$, and overlook changes in all other variables, e.g., $\bar{\mathbb{E}}_0[i_1] = 0$.

Figure 5 plots underlying heterogenous inflation expectations $\bar{\mathbb{E}}_0^d[\pi_1]$ in panel (a), and the average inflation expectation $\bar{\mathbb{E}}_0[\pi_1]$ as a function of λ in panel (b), in response to cost-push shocks ϵ_1^π . In panel (b), the blue vertical line indicates our calibrated λ , whereas the green vertical link corresponds to rational expectations, which coincides with the true equilibrium response π_1 .

As shown in Panel 5a, a depth-1 agent anticipates a one-to-one response of inflation to cost-push shock. As an agent thinks more deeply, they acknowledge the amplification of inflation through pricing complementarity and hence forecast higher inflation response. However, once they reason 5 steps, they start to recognize the monetary-policy loop which offsets inflation response. As agents take more steps in reasoning, their inflation expectations fluctuates non-monotonically and converges towards the rational expectation. Under our calibration of $\lambda = 0.61$, people on average understand only about 2.6 steps, leading to an average inflation expectation that exceeds the actual inflation response.

Panel 5b varies λ and shows that the inflation expectation $\bar{\mathbb{E}}_0[\pi_1]$ is non-monotonic. Our calibration implies that the inflation expectation exceeds the size of the direct effect

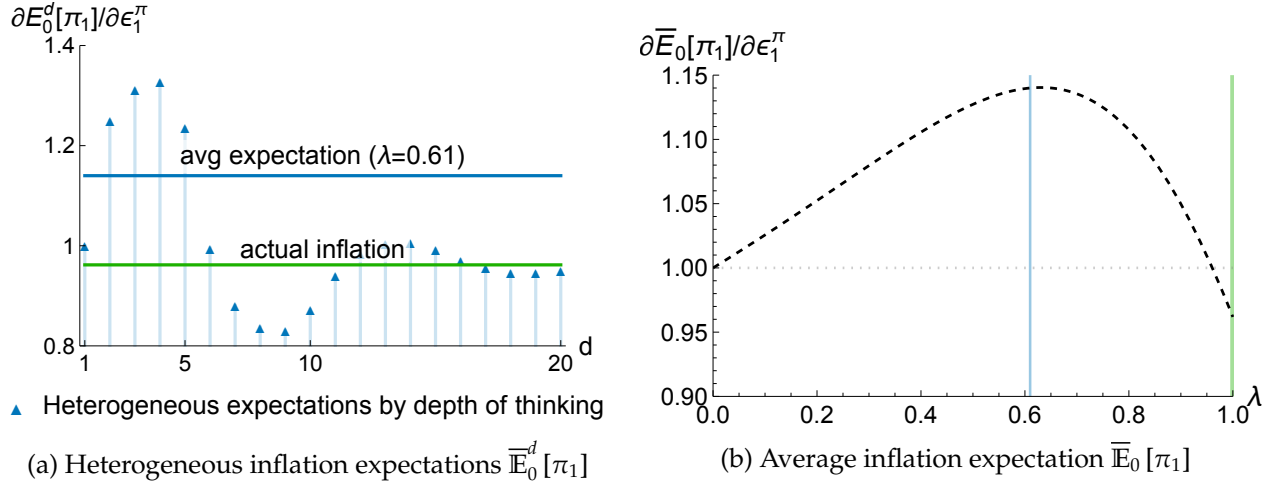


Figure 5: Beliefs in response to cost-push news shocks ϵ_1^π

Notes: Panel (a) plots the heterogeneous inflation expectations $\bar{E}_0^d[\pi_1]$ (relative to the size of the shock) across individuals, in response to news about a cost-push shock ϵ_1^π . Panel (b) plots the average inflation expectation $\bar{E}_0[\pi_1]$ as a function of the shallow thinking parameter λ . The blue line indicates our calibration of λ , while the green line represents rational expectations ($\lambda = 1$).

of one, whereas the true inflation response π_1 is below one. That is, shallow agents think that the cost-push shock will be amplified, though actually it will be offset. The underlying reason is that, in determining inflation (19), there is a shorter amplification loop of pricing complementarity and a much longer offsetting loop via monetary policy response. Shallow agents understand the shorter loop relatively better than the longer loop. Thus, on net, they perceive amplification. That can be true even though the longer offset loop is actually stronger than the shorter amplification loop, meaning that the actual net effect is offsetting.

Strength and length of GE feedback loops jointly determine belief misreaction. We note that the predicted relationship between the inflation expectation $\bar{E}_0[\pi_1]$ and the deviation from rational expectations under shallow thinking differs qualitatively from theories of sticky information (Mankiw and Reis, 2002) and cognitive discounting (Gabaix, 2020). They predict a monotonic increase in the inflation expectation from 0 to the true equilibrium value π_1 , as one moves from full stickiness/myopia to full information rational expectations, and predict underreaction only.

At a high level, these theories predict a monotonic relationship because their agents understand the relative importance of multiple GE feedback loops and only dampen

the net effect in their perception. In contrast, shallow thinking suggests that longer GE feedback loops are harder to understand. Under our calibration, shallow agents on average perceive an amplification of inflation response rather than an offset. Appendix D.3 further examines this mechanism by varying the strength of GE feedback.

4.2 Over- and Underreaction of Long-Term Rate

Here we study the monetary policy shock ϵ_1^i and compare with cost-push shock ϵ_1^π studied before. We characterize the period-1 equilibrium and period-0 expectations in response to monetary policy shocks in Proposition 5.

Proposition 5. (Period-1 monetary policy shock) *The period-1 equilibrium in response to a monetary policy shock ϵ_1^i features*

$$i_1 = \left[1 + \frac{\phi \theta \kappa \sigma^{-1} (\nu + \sigma)}{1 - (1 - \theta)} \right]^{-1} \epsilon_1^i \quad (23)$$

$$\pi_1 = -\frac{\theta \kappa \sigma^{-1} (\nu + \sigma)}{1 - (1 - \theta)} i_1 \quad (24)$$

whereas the period-0 average expectations upon observing the news about ϵ_1^i are

$$\bar{\mathbb{E}}_0 [i_1] = \left[1 + \frac{\lambda^4 \phi \theta \kappa K_1 (\lambda)}{1 - \lambda (1 - \theta)} \right]^{-1} \epsilon_1^i \quad (25)$$

$$\bar{\mathbb{E}}_0 [\pi_1] = -\frac{\lambda^3 \theta \kappa K_1 (\lambda)}{1 - \lambda (1 - \theta)} \bar{\mathbb{E}}_0 [i_1] \quad (26)$$

with $K_1 (\lambda)$ increasing in λ under our calibration and $K_1 (1) = \sigma^{-1} (\nu + \sigma)$.

These results relate to those regarding the cost-push shocks in Proposition 4, but with a subtle and consequential difference. In this case, *all* general equilibrium effects offset the interest response to a monetary policy shock, whereas the inflation response to a cost-push shock involves both amplifying and offsetting feedback loops.

To appreciate that, we analyze the interest rate in (23), which is the sum of all n -step effects, as in (9). These effects belong to three different loops—pricing complementarity, the monetary policy loop, and the Keynesian cross—as previously established and displayed in Figure 4. Among the three loops, the interest rate response is directly involved in *only one*: the monetary policy loop. This loop offsets the interest rate response to a monetary

policy shock in 4 steps: a higher interest rate i_1 encourages labor supply $\hat{n}_1^s$, which then lowers the real wage w_1 , leading to lower inflation π_1 through firms' pricing decisions, ultimately prompting the central bank to lower the interest rate i_1 according to the Taylor rule. This 4-step monetary policy offset loop, with a strength $\phi\theta\kappa$, compounds with the other two loops—pricing complementarity and the Keynesian cross—which correspond to the $\frac{1}{1-(1-\theta)}$ and $\sigma^{-1}(\nu + \sigma)$ terms in (23). That occurs because pricing complementarity strengthens any effect on inflation, while the Keynesian cross reinforces any effect impacting households.

For the interest rate expectation $\bar{\mathbb{E}}_0[i_1]$ in (25), the 4-step monetary policy loop is dampened by λ^4 , the pricing complementarity self-loop is dampened by λ , and the Keynesian cross is also dampened, captured by $K_1(\lambda)$, as in the prior case with cost-push shocks.

The equilibrium inflation response π_1 in (24) depends on the interest rate response i_1 . It is compounded by the pricing complementarity loop $\frac{1}{1-(1-\theta)}$ and the Keynesian cross $\sigma^{-1}(\nu + \sigma)$, as any effect of the interest rate on inflation involves responses of both firms and households. The inflation expectation $\bar{\mathbb{E}}_0[\pi_1]$ derives accordingly from the interest rate expectation $\bar{\mathbb{E}}_0[i_1]$ but is additionally dampened by λ^3 , since it takes 3 steps for the interest rate to affect inflation.

In the limit of $\lambda = 0$, shallow agents perceive no general equilibrium effects, and thus $\bar{\mathbb{E}}_0[i_1] = \epsilon_1^i$, $\bar{\mathbb{E}}_0[\pi_1] = 0$.

Figure 6 plots the average interest rate expectation $\bar{\mathbb{E}}_0[i_1]$ in response to monetary policy and cost-push news shocks respectively, both as functions of λ in dashed black lines. The blue vertical line indicates our calibrated λ , whereas the green vertical link corresponds to rational expectations and the true equilibrium response i_1 .

Panel 6a indicates that the average interest rate expectation $\bar{\mathbb{E}}_0[i_1]$ overreacts to the monetary policy shock compared to the true response, thus suggesting that the period-0 long-term yield $y_0^{(2)}$ overreacts too according to (18). This aligns with a large body of literature on the excess sensitivity of long-term interest rates to monetary policy shocks, including Cochrane and Piazzesi (2002), Gürkaynak, Sack and Swanson (2005), and Hanson and Stein (2015), among others. The underlying mechanism is that, all general equilibrium feedback loops offset the interest rate response to a monetary policy shock, and as they are dampened to varying degrees, shallow agents perceive less offset than the true overall effect. In contrast, panel 6b implies that $\bar{\mathbb{E}}_0[\pi_1]$ underreacts, because agents underappreciate the effect of monetary policy on inflation.

In contrast, panel 6b suggests that the average interest rate expectation $\bar{\mathbb{E}}_0[i_1]$ under-

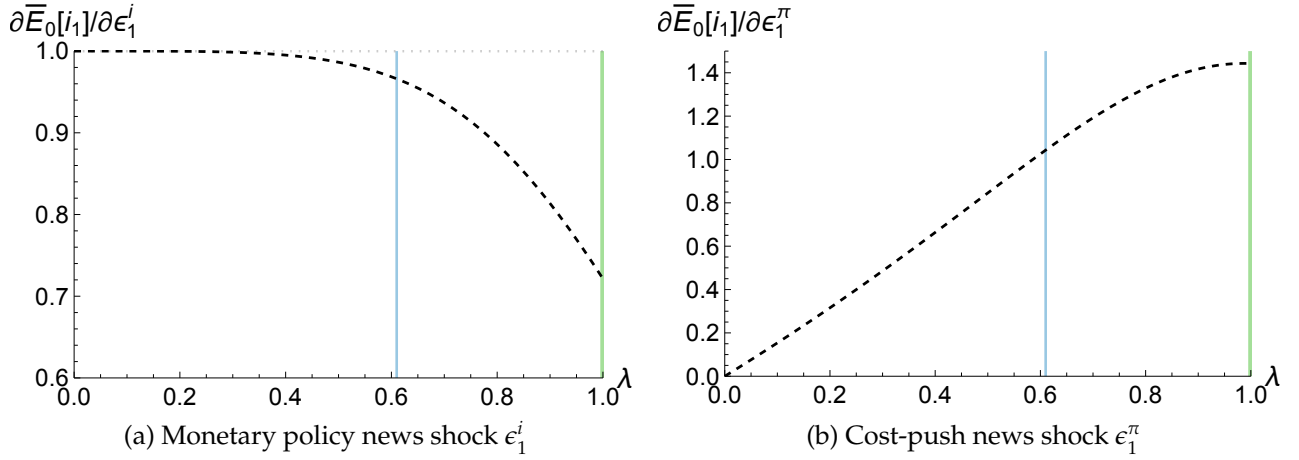


Figure 6: Average interest rate expectations in response to two shocks

Notes: Panel (a) plots the average interest expectation $\bar{E}_0[i_1]$ (relative to the size of the shock) as a function of the shallow thinking parameter λ , in response to news about a monetary policy shock ϵ_1^i , and panel (b) plots its response to news about a cost-push shock ϵ_1^π . The blue line indicates our calibration of λ , while the green line represents rational expectations ($\lambda = 1$).

reacts to the cost-push shock compared to the true response, leading to an underreaction of the period-0 long-term yield $y_0^{(2)}$. This occurs because shallow thinking agents underappreciate the Taylor rule, i.e., $\bar{E}_0[i_1] = \lambda \bar{E}_0[\pi_i]$, despite that the inflation expectations overreact. This aligns with findings that long-term interest rates responded too little to inflation surprises before the March 2022 interest rate hike (Bauer, Pflueger and Sunderam, 2024) and that professional forecasters systematically underappreciate the extent of monetary easing during recessions (Cieslak, 2018).

Taking stock, our theory predicts over- and underreaction of long-term interest rates depending on the shocks. As a result, unconditionally, whether the long-term rate is excessively sensitive compared to the short rate depends on the relative importance of shocks. Indeed, Hanson, Lucca and Wright (2021) show that, after 2000, long-term interest rates are overly sensitive to changes in the short-term rates and predictably revert, suggesting overreaction. However, this pattern does not hold prior to 2000. Through the lens of our theory, one possibility is that there were more supply shocks (such as oil supply shocks) before 2000, contributing to underreaction in the mix and obscuring overreaction to monetary policy shocks.²⁰

²⁰We show the comparative statics of this univariate coefficient of long rate on short rate with respect to the relative importance of monetary policy shocks in Appendix D.4, under persistent shocks.

Mechanism of belief over- and underreaction. So far, we have established a rich pattern of belief misreaction to news shocks, summarized in Table 4. For a variable of interest (such as inflation), we refer to shocks that directly affect it as *own shocks* (e.g., cost-push shocks) and shocks that directly affect other variables as *cross shocks* (e.g., monetary policy shocks). Under our calibration, the average expectations of inflation and interest rate overreact to own shocks and underreact to cross shocks.

Shallow thinking generates differential misreaction to different shocks, contrasting with existing theories that predict underreaction to all shocks, such as sticky information (Mankiw and Reis, 2002) and cognitive discounting (Gabaix, 2020), or those that give rise to overreaction via extrapolation or overconfidence, as reviewed by Barberis (2018).

Table 4: Belief over- and underreaction in the New Keynesian economy

News shock	Inflation expectation $\bar{\mathbb{E}}_0 [\pi_1]$	Interest rate expectation $\bar{\mathbb{E}}_0 [i_1]$
Cost-push ϵ_1^π	Overreaction	Underreaction
Monetary policy ϵ_1^i	Underreaction	Overreaction

Notes: This table summarizes belief misreaction to different news shocks. The average expectations of variable v overreact (compared to the true equilibrium) to shock s if $\left| \frac{\partial \bar{\mathbb{E}}_0[v_1]}{\partial \epsilon_1^s} \right| > \left| \frac{\partial v_1}{\partial \epsilon_1^s} \right|$, and underreact otherwise.

4.3 Predictability of Bond Excess Returns

Taking stock of previous results, we consider an economy impacted by both the cost-push shock ϵ_1^π and the monetary policy shock ϵ_1^i , and study the bond excess returns. We show shallow thinking implies that bond excess returns can be predicted by macroeconomic variables, controlling for current yields, as established empirically by Ludvigson and Ng (2009), Cooper and Priestley (2009), Joslin, Pribsch and Singleton (2014), Cieslak and Povala (2015), and Cieslak (2018), among others.

We consider the excess return of holding a 2-period (i.e., long-term) bond from period 0 to period 1, relative to holding a 1-period (i.e., short-term) bond,²¹

$$xr_{0 \rightarrow 1}^{(2)} \equiv \underbrace{-i_1 + 2y_0^{(2)}}_{\text{return of long-term bond}} - \underbrace{i_0}_{\text{return of short-term bond}} = \bar{\mathbb{E}}_0 [i_1] - i_1 \quad (27)$$

²¹The return of the 2-period bond is the difference between its period-1 log price deviation, $-i_1$, and its period-0 log price derivation, $-2y_0^{(2)}$.

where the equality follows from (18). When the period-0 expectation of period-1 interest rate $\bar{\mathbb{E}}_0[i_1]$ exceeds its actual value i_1 , the long-term bond is undervalued in period 0 and will appreciate in period 1, leading to a positive excess return, and vice versa.

In particular, we study the predictability of bond excess return $xr_{0 \rightarrow 1}^{(2)}$ by the average inflation expectation $\bar{\mathbb{E}}_0[\pi_1]$, as noted by Joslin, Pribsch and Singleton (2014) and Cieslak (2018), controlling for the forward rate $\bar{\mathbb{E}}_0[i_1]$,²²

$$\underbrace{xr_{0 \rightarrow 1}^{(2)}}_{\text{bond excess return}} = \beta_\pi \underbrace{\bar{\mathbb{E}}_0[\pi_1]}_{\text{inflation expectation}} + \beta_i \underbrace{\bar{\mathbb{E}}_0[i_1]}_{\text{forward rate}} + \alpha + \epsilon_{0 \rightarrow 1} \quad (28)$$

Figure 7a illustrates the theory-implied coefficients β_π and β_i as functions of λ , in brown and purple respectively. As long as $\lambda < 1$, our theory predicts a negative β_π , which is what Joslin, Pribsch and Singleton (2014) and Cieslak (2018) find when using inflation expectations or other macroeconomic variables to predict bond excess returns.

To understand the mechanism, we start by examining the limits of $\lambda = 0$ and 1 and build intuition with shock loadings illustrated in Figure 7b. In the limit of $\lambda = 0$, we have $\bar{\mathbb{E}}_0[\pi_1] = \epsilon_1^\pi$, $\bar{\mathbb{E}}_0[i_1] = \epsilon_1^i$ from Propositions 4 and 5. As the interest rate expectation underreacts to the cost-push shock ϵ_1^π and overreacts to the monetary policy shock ϵ_1^i , the excess return $xr_{0 \rightarrow 1}^{(2)}$ loads negatively on the former and positively on the latter, as determined by (27). That is, it lies in the second quadrant (plotted in Figure D2a). Because the inflation expectation and the forward rate are unit vectors along the x- and y-axes in this case, the coefficient β_π is negative and β_i is positive.

In the limit of rational expectations ($\lambda = 1$), $\bar{\mathbb{E}}_0[\pi_1]$ loads positively on the cost-push shock ϵ_1^π and negatively on the monetary policy shock ϵ_1^i , placing it in the fourth quadrant (plotted in Figure D2b). $\bar{\mathbb{E}}_0[i_1]$ loads positively on both shocks, placing it in the first quadrant. However, in this case, and only in this case, the excess return $xr_{0 \rightarrow 1}^{(2)}$ is 0, as there is no expectational error. Thus, both coefficients β_π and β_i are 0.

Generically, the inflation expectation $\bar{\mathbb{E}}_0[\pi_1]$ lies in the fourth quadrant, and the forward rate $\bar{\mathbb{E}}_0[i_1]$ is in the first quadrant, as long as $\lambda > 0$ (i.e., some understanding of shock propagation). The excess return $xr_{0 \rightarrow 1}^{(2)}$ is placed in the second quadrant if $\lambda < 1$ (i.e., not fully understanding shock propagation). Figure 7b illustrates these generic placements. It also shows the residual of $xr_{0 \rightarrow 1}^{(2)}$ projected onto $\bar{\mathbb{E}}_0[i_1]$, denoted $xr_{0 \rightarrow 1}^{(2)} | \bar{\mathbb{E}}_0[i_1]$, and the

²²This regression, with expectations as predictors, is equivalent to a bivariate Coibion and Gorodnichenko (2015) regression. The coefficients reflect the misappreciation of causal relations between variables, which Wu (2023) systematically analyzes across many variable pairs.

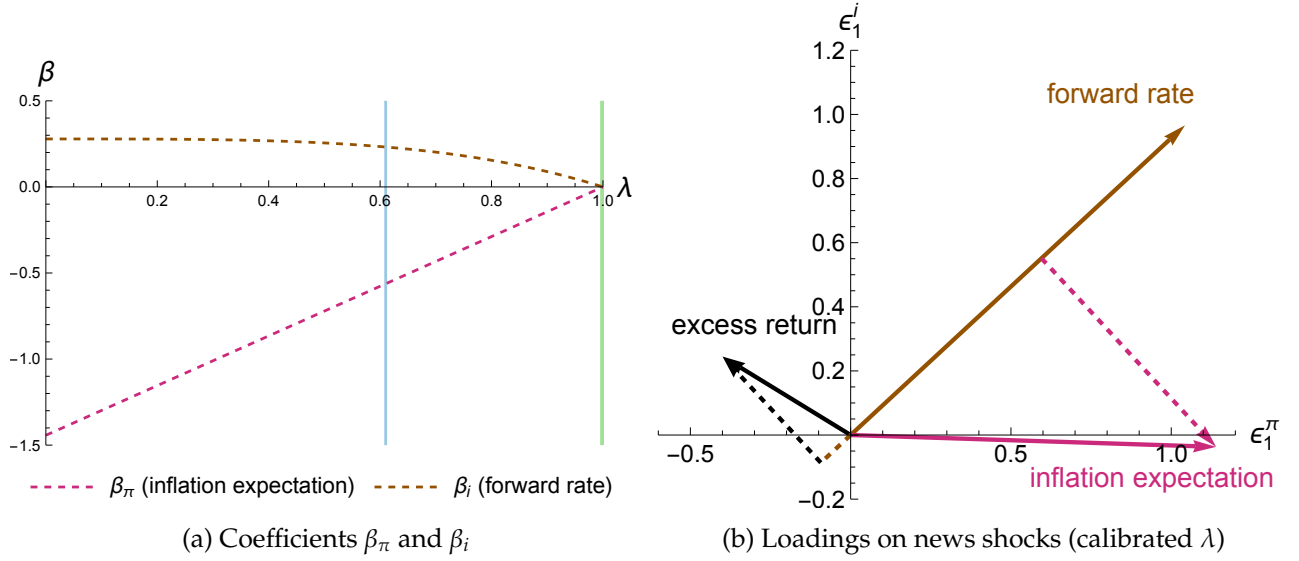


Figure 7: Predictability of bond excess returns, with news shocks

Notes: Panel (a) plots the theory-implied coefficients β_π and β_i from the predictive regression of bond excess returns $xr_{0 \rightarrow 1}^{(2)}$ in (28) on the inflation expectation $\bar{\mathbb{E}}_0[\pi_1]$ and forward rate $\bar{\mathbb{E}}_0[i_1]$, as functions of the shallow thinking parameter λ . The period-0 economy is impacted by news about period-1 cost-push shocks ϵ_1^π and monetary policy shocks ϵ_1^i . The blue vertical line corresponds to our calibration of λ , while the green line represents rational expectations ($\lambda = 1$).

Panel (b) illustrates the loadings of bond excess return $xr_{0 \rightarrow 1}^{(2)}$, inflation expectation $\bar{\mathbb{E}}_0[\pi_1]$ and forward rate $\bar{\mathbb{E}}_0[i_1]$ on the cost-push news shock ϵ_1^π (x-axis) and the monetary policy news shock ϵ_1^i (y-axis), as solid vectors. The vectors are determined under our calibration of λ , but their quadrant placements are generically true when $\lambda \in (0, 1)$. Figures D2a and D2b show the cases with $\lambda = 0, 1$. The two dashed vectors show the residuals of bond excess return $xr_{0 \rightarrow 1}^{(2)}$ and inflation expectation $\bar{\mathbb{E}}_0[\pi_1]$ projected onto the forward rate $\bar{\mathbb{E}}_0[i_1]$. Their opposite directions imply a negative β_π , following the Frisch-Waugh-Lovell theorem.

residual $\bar{\mathbb{E}}_0[\pi_1] | \bar{\mathbb{E}}_0[i_1]$, both shown as dashed vectors. These two residuals point to opposite directions due to the quadrant placements of the three vectors involved. According to the Frisch-Waugh-Lovell theorem, the coefficient β_π from the bivariate regression (28) equals the univariate regression coefficient of the residual $xr_{0 \rightarrow 1}^{(2)} | \bar{\mathbb{E}}_0[i_1]$ on the residual $\bar{\mathbb{E}}_0[\pi_1] | \bar{\mathbb{E}}_0[i_1]$. Opposite directions of these residuals imply a negative coefficient β_π .

In this subsection, we show that that inflation expectations negatively predict bond excess returns, in an economy with two shocks. In reality, the economy is impacted by multiple shocks. Mathematically, in a linear model with N shocks, N independent predictors from the yield curve would span all shocks. However, in practice, the entire

yield curve is well captured by the first three principal components.²³ Our theory suggests that other macro variables may contain additional information about non-monetary-policy macro shocks, and can therefore predict bond excess returns.

4.4 Cost-Push News Shocks Cause Inflation

So far we have focused on belief misreaction and asset prices. Based on these results, here we establish that news about period-1 cost-push shocks is inflationary, rather than deflationary as predicted by rational expectations, and less contractionary.

Period-0 equilibrium. In period 0, the shocks have not materialized, but firms and households are forward-looking. The period-0 equilibrium arises from firms' and households' optimal behavior, given their perfect observation of all period-0 variables and their beliefs about period-1 outcomes, along with the central bank's Taylor rule and the labor market clearing condition.

The period-0 equilibrium consists of seven variables $\{i_0, \pi_0, div_0, n_0^d, c_0, n_0^s, w_0\}$, similar to the period-1 equilibrium. Among these seven variables, three of them (the interest rate i_0 , dividend div_0 and labor demand n_0^d) depend only on the contemporaneous values of the other variables. These contemporaneous causal relations are the same as their period-1 counterparts in Section 2.1. Three variables (inflation π_0 , consumption c_0 and labor supply n_0^s) depend on agents' beliefs about period-1 outcomes, since firms's pricing decisions and households' consumption and labor supply decisions are forward-looking, detailed next. Last, the wage w_0 arises from the labor market clearing condition $n_0^s = n_0^d$.

The period-0 inflation π_0 satisfies

$$\pi_0 = \theta\kappa(w_0 + \beta\theta\bar{\mathbb{E}}_0[w_1]) + (1 - \theta)(\pi_0 + \beta\theta\bar{\mathbb{E}}_0[\pi_1]) \quad (29)$$

which increases in expectations of both future real wage w_1 and inflation π_1 , as firms want to front run a higher future marginal cost.

The period-0 consumption c_0 and labor supply n_0^s follow

$$c_0 = -\sigma^{-1}\beta(i_0 - \bar{\mathbb{E}}_0[\pi_1] + \beta\bar{\mathbb{E}}_0[i_1]) + \frac{(1 - \beta)(\mu - 1)\nu}{\sigma + \mu\nu}(div_0 + \bar{\mathbb{E}}_0[div_1])$$

²³Indeed, in our model, in the presence of multiple shocks with the same persistence, all forward rates are collinear with each other, but not with other macroeconomic variables or expectations thereof.

$$+ \frac{(1-\beta)(1+\nu)}{\sigma + \mu\nu} (w_0 + \bar{\mathbb{E}}_0[w_1]) \quad (30)$$

$$\begin{aligned} n_0^s = & \nu^{-1}\beta(i_0 - \bar{\mathbb{E}}_0[\pi_1] + \beta\bar{\mathbb{E}}_0[i_1]) - \frac{(1-\beta)(\mu-1)\sigma}{\sigma + \mu\nu} (div_0 + \bar{\mathbb{E}}_0[div_1]) \\ & + \nu^{-1}\left[1 - \sigma\frac{(1-\beta)(1+\nu)}{\sigma + \mu\nu}\right]w_0 - \frac{(1-\beta)(1+\nu)\nu^{-1}\sigma}{\sigma + \mu\nu}\bar{\mathbb{E}}_0[w_1] \end{aligned} \quad (31)$$

Households react to expectations of the future interest rate, dividend, and wage, as well as future inflation, as a higher inflation lowers the real interest rate from period 0 to 1.

In determining the period-0 equilibrium, the only decision-relevant beliefs are about inflation π_1 , wage w_1 , interest rate i_1 and dividend div_1 . Figure 8a illustrates the determination of period-0 equilibrium, where these beliefs act like shocks to firms and households.²⁴

In particular, the inflation expectation $\bar{\mathbb{E}}_0[\pi_1]$ serves two roles here: it acts like a cost-push shock for firms as they want to front run future inflation, and it functions as a demand shock for households since a higher inflation lowers the real interest rate. This observation will prove useful when we discuss the effects of cost-push shocks next.

Cost-push news shocks cause inflation. Figure 8b depicts the period-0 inflation π_0 and output y_0 (which is simply the consumption c_0), in response to news about a period-1 cost-push shock ϵ_1^π . Unlike the period-1 equilibrium, which is independent of λ , the period-0 equilibrium does depend on λ . The blue line indicates the equilibrium under our calibration of λ , whereas the green line stands for the rational expectations equilibrium.

A cost-push news shock is *inflationary* in period 0 under the calibrated shallow thinking parameter λ , consistent with empirical evidence (Känzig, 2021), but *deflationary* under rational expectations. The latter occurs because rational agents understand that the central bank will raise the interest rate i_1 in response to higher inflation π_1 , causing a contraction of the period-1 economy. Anticipating that, households cut back on their consumption in period 0, resulting in a period-0 output contraction as well.

In contrast, in the limit of extremely shallow agents ($\lambda = 0$), they only perceive a change in future inflation, $\bar{\mathbb{E}}_0[\pi_1] = \epsilon_1^\pi$, but not in any other variable. As we have established with (29-31), inflation expectations act like a cost-push shock for firms and a demand shock for households. As a result, firms set higher prices and households consume more and work

²⁴In Figure 8a that determines the equilibrium given beliefs, we equalize the actual labor demand and supply n_0^d, n_0^s as the market clearing condition. We only introduce fictitious Walrasian auctioneers for the purpose of characterizing beliefs. In the equilibrium given any beliefs, markets always clear.

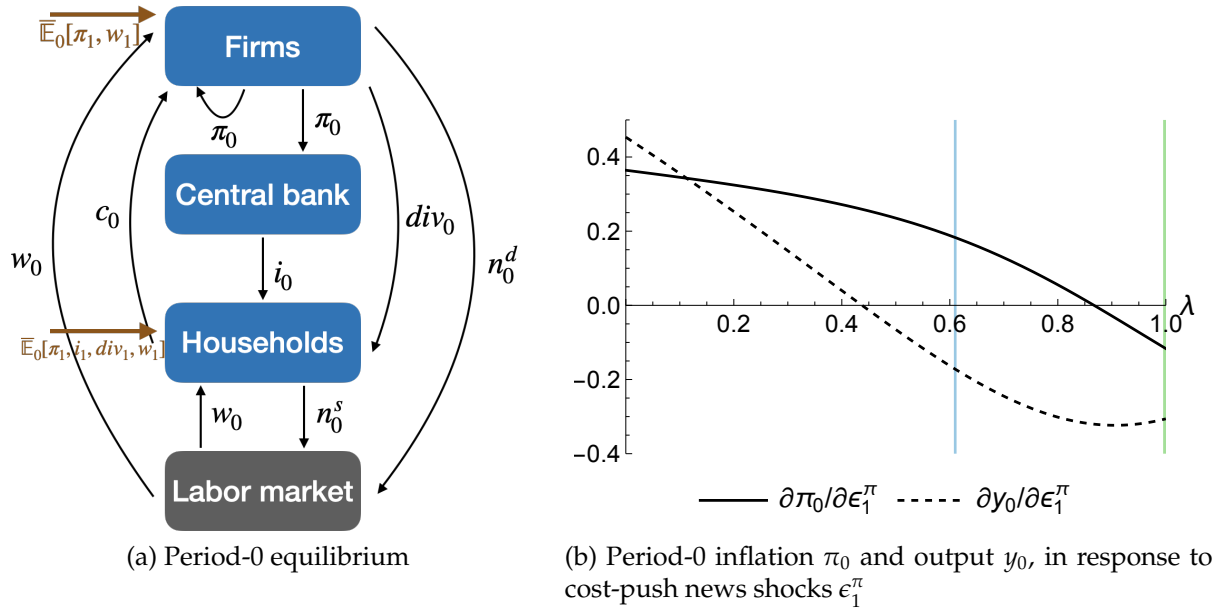


Figure 8: Period-0 equilibrium in response to news shocks

Notes: Panel (a) demonstrates the period-0 equilibrium in response to transitory news shocks, which are period-1 shocks observed in period 0, driven by firms' and households' beliefs about period-1 outcomes. Panel (b) plots the period-0 inflation π_0 and output y_0 (which equals consumption c_0) in response to news about a cost-push shock ϵ_1^π , normalized relative to the size of the shock, as functions of the shallow thinking parameter λ . The solid line stands for inflation π_0 and the dashed line represents output y_0 . The blue vertical line indicates the equilibrium under our calibration of λ , while the green vertical line represents the rational expectations equilibrium with $\lambda = 1$.

less, leading to period-0 inflation and output expansion.

With our calibrated λ , agents partially expect an economic downturn in period 1. In this case, news about period-1 cost-push shocks causes inflation instead of deflation and leads to a smaller output contraction compared to the rational expectations equilibrium.

In terms of the inflation response to cost-push news shocks, shallow thinking reverses the sign relative to the rational expectations equilibrium to align with empirical evidence, as agents better understand certain causal relations than others in the period-1 economy. In contrast, theories of sticky information (Mankiw and Reis, 2002) and cognitive discounting (Gabaix, 2020) preserve the sign of the rational expectations equilibrium but modify only the magnitude of period-0 deflation.

From transitory news shocks to persistent shocks. So far, we have focused on transitory news shocks, but this is much more informative than it may seem. As we have noted at the beginning of this section, comparing theories of beliefs under transitory news shocks

is equivalent to comparing them under 2-period shocks, since the period-0 equilibrium response to contemporaneous shocks is independent of belief formation.

Furthermore, analyzing transitory news shocks helps build intuition for persistent shocks. Under a first-order approximation, a persistent cost-push shock observed in period 0 is a sum of a current shock and a series of future shocks. Shallow thinking implies that a future cost-push shock is inflationary in period 0, whereas rational expectations predict otherwise. This suggests that a persistent cost-push shock is more inflationary under shallow thinking. Additionally, a *more* persistent cost-push shock is a sum of more future shocks. Thus, shallow thinking predicts that a *more* persistent cost-push shock leads to *higher* inflation, contrary to the rational expectations prediction.

Next, we will generalize shallow thinking to persistent shocks to substantiate these two claims. The generalization will address a missing piece from the intuitive analysis above: in responding to a persistent shock, agents must think about how all decision-relevant variables in any future period t_1 depend on shocks in subsequent periods t_2 .

5 Shallow Thinking of Dynamic General Equilibrium

We generalize to persistent shocks in the New Keynesian economy and show that the insights gained from transitory news shocks still hold, while new lessons emerge. In particular, more persistent cost-push shocks lead to higher inflation under shallow thinking, in contrast to lower inflation predicted under rational expectations.

We introduce shallow thinking of dynamic general equilibrium in Section 5.1, by focusing on the cross-variable causal relations and abstracting away from the cross-horizon dimension. This will allow us to formally address belief misreaction to persistent shocks and their effects in Section 5.2. Readers who are more interested in the consequences and less interested in the technical details could skip the first subsection.

We consider persistent shocks $\{\epsilon_t\}_{t \geq 0}$ that are observed at time 0^- , and assume that agents form expectations $\mathbb{E}^d[\cdot]$ once and for all at time 0^- , i.e., $\mathbb{E}_t^d[v_s] \equiv \mathbb{E}^d[v_s]$ for $s > t$. This assumption simplifies the analysis, as it nests the rational expectations benchmark and is a reasonable starting point.²⁵

²⁵We believe this is reasonable beyond convenience for two reasons. First, an important reason for expectation updates over time is that agents gradually learn about shocks. We focus on the rationality of beliefs in the absence of any information frictions, and our survey design mimics this environment. Second, expectations can update as agents learn how aggregate variables respond to shocks *through repeated experiences*, similar to how economists study shock propagation using time series data. We leave this for

5.1 Causal Representation, Shallow Thinking Beliefs, and Equilibrium

We represent the infinite-horizon New Keynesian model as a system of causal relations, which can be similarly represented in a directed graph, generalizing Sections 2.2 and 2.3. This representation of dynamic GE also applies to the RBC model in Appendix C.

The infinite-horizon New Keynesian model is characterized by sequences of seven variables $\{i_t, \pi_t, div_t, n_t^d, c_t, n_t^s, w_t\}_{t \geq 0}$. We will refer to this collection of variables as $\mathcal{V}$. The first six variables are agents' actions, which we collect as $\mathcal{V}^{action}$, whereas the last is a price formed in the competitive labor market. Note that we start with the labor supply and demand n_t^s, n_t^d in order to reinterpret them as supply and demand curves shortly. In that reinterpretation, with only slight abuse of notation, we will use $\mathcal{V}$ and $\mathcal{V}^{action}$ to denote variables of interests with n^s, n^d replaced by $\hat{n}^s, \hat{n}^d$.

Among these, three variables i_t, div_t, n_t^d only depend on contemporaneous values of other variables, laid out in (D1, D2, D6). Three variables π_t, c_t, n_t^s are forward-looking, capturing best responses of forward-looking firms and households. Inflation π_t is a linear function in w_t, π_t and $\{\bar{\mathbb{E}}_t[w_{t+k}], \bar{\mathbb{E}}_t[\pi_{t+k}]\}_{k \geq 1}$, and consumption and labor supply c_t, n_t^s are linear functions in i_t, div_t, w_t and $\{\bar{\mathbb{E}}_t[i_{t+k}], \bar{\mathbb{E}}_t[\pi_{t+k}], \bar{\mathbb{E}}_t[div_{t+k}], \bar{\mathbb{E}}_t[w_{t+k}]\}_{k \geq 1}$, detailed in (D3-D5). The last variable w_t arises from equilibrating labor supply n_t^s and demand n_t^d . These equations completely characterize the equilibrium, given beliefs.

REE as causal relations in sequence space. To determine shallow thinking beliefs, we first characterize the rational expectations equilibrium (REE) in terms of causal relations. For REE, by replacing each expectation $\bar{\mathbb{E}}_t[v_\tau]$ with the true outcome v_τ under rational expectations, the six variables in $\mathcal{V}^{action}$ that agents choose can be represented in the sequence space, following Auclert, Rognlie and Straub (2024) and Auclert et al. (2021), as

$$\mathbf{v}^{REE} = \sum_{\mathbf{u} \in \mathcal{V}} \mathbf{J}_{\mathbf{vu}}^{REE} \mathbf{u}^{REE} + \boldsymbol{\epsilon}^v, \quad \forall \mathbf{v} \in \mathcal{V}^{action} \quad (32)$$

where

$$(\mathbf{J}_{\mathbf{vu}})_{ts} \equiv \begin{cases} \frac{\partial v_t}{\partial u_s} & s \leq t \\ \frac{\partial v_t}{\partial \bar{\mathbb{E}}[u_s]} & s > t \end{cases} \quad (33)$$

future research, but we note that in our survey we consider conventional shocks, such as oil supply shocks and monetary policy shocks. These are age-old shocks, so our estimation already incorporates knowledge that people have accumulated over time. This contrasts with unconventional policy shocks (such as forward guidance), to which people had no prior exposure until recently.

is the Jacobian of the sequence of one variable $\mathbf{v} \equiv (\{v_t\}_{t \geq 0})'$ with respect to the sequence of another variable $\mathbf{u} \equiv (\{u_t\}_{t \geq 0})'$, and $\epsilon^v \equiv (\{\epsilon_t^v\}_{t \geq 0})'$ denotes the sequence of a structural shock. The Jacobians $\mathbf{J}_{\mathbf{v}\mathbf{u}}$ are upper triangular matrices in the New Keynesian model, which is purely forward-looking.²⁶

For labor supply and demand $\mathbf{n}^s, \mathbf{n}^d$, by separating their dependence on the wage $\mathbf{w}$ from the rest, we interpret them as supply and demand curves in the sequence space

$$\mathbf{v}^{REE} = \mathbf{J}_{\mathbf{v}\mathbf{w}} \mathbf{w}^{REE} + \hat{\mathbf{v}}^{REE}, \mathbf{v} \in \{\mathbf{n}^s, \mathbf{n}^d\}$$

with elasticities $\mathbf{J}_{\mathbf{n}^s \mathbf{w}}, \mathbf{J}_{\mathbf{n}^d \mathbf{w}}$ and shifts $\hat{\mathbf{n}}^{s,REE}, \hat{\mathbf{n}}^{d,REE}$ defined as

$$\hat{\mathbf{v}}^{REE} = \sum_{\mathbf{u} \in \mathcal{V} \setminus \{\mathbf{w}\}} \mathbf{J}_{\mathbf{v}\mathbf{u}} \mathbf{u}^{REE} + \epsilon^v, \mathbf{v} \in \{\mathbf{n}^s, \mathbf{n}^d\} \quad (34)$$

In the New Keynesian model, demand elasticity $\mathbf{J}_{\mathbf{n}^d \mathbf{w}}$ is a matrix of zeros in this model since firms only use labor inputs and supply elasticity $\mathbf{J}_{\mathbf{n}^s \mathbf{w}}$ is an upper triangular matrix, as households' labor supply responds to future wages.

By equalizing supply and demand $\mathbf{n}^{s,REE} = \mathbf{n}^{d,REE}$, we can interpret the wage $\mathbf{w}^{REE}$ as resulting from supply and demand shifts $\hat{\mathbf{n}}^{s,REE}, \hat{\mathbf{n}}^{d,REE}$

$$\mathbf{w}^{REE} = (\mathbf{J}_{\mathbf{n}^s \mathbf{w}} - \mathbf{J}_{\mathbf{n}^d \mathbf{w}})^{-1} (\hat{\mathbf{n}}^{d,REE} - \hat{\mathbf{n}}^{s,REE}) \quad (35)$$

This rule of price determination generalizes its counterpart in the period-1 economy (7) and incorporates the response of time- t wage on time- s demand and supply shifts.

Taking stock of the rational expectations equilibrium, formula (32) describes all agents' actions $\mathcal{V}^{action}$, where we use $\hat{\mathbf{n}}^{s,REE}, \hat{\mathbf{n}}^{d,REE}$ as in (34) instead of $\mathbf{n}^{s,REE}, \mathbf{n}^{d,REE}$, and equation (35) characterizes the wage from the competitive labor market. We stack the sequences of outcomes to form a long vector $\mathbf{V} \equiv (\{\mathbf{v}'\}_{\mathbf{v} \in \mathcal{V}})'$, sequences of shocks as $\mathbf{S} \equiv (\{\epsilon^{v'}\}_{\mathbf{v} \in \mathcal{V}})'$, and correspondingly stack the Jacobians as a giant matrix $\mathbf{M} \equiv (\{\mathbf{J}_{\mathbf{v}\mathbf{u}}\}_{\mathbf{v}, \mathbf{u} \in \mathcal{V}})$. $\mathbf{M}$ captures all *causal relations* in the dynamic general equilibrium, which are now defined as the *sequence-space Jacobians*, instead of partial derivatives in the period-1 economy.

Proposition 6. (Dynamic GE as a system of causal relations in the sequence space) *The*

²⁶(32) can be easily generated to accommodate models with state variables that depend on the past, such as the RBC model where firms adjust capital stock. In that case, we interpret v_t as the time-0⁻ planned optimal action at time t as a function of the perceived sequence shocks $\{\bar{\mathbb{E}}[u_s]\}_{s \geq 0}$ and define the Jacobian accordingly.

rational expectations equilibrium in the New Keynesian economy is characterized by the fixed point to the system of causal relations in the sequence space among all agents' actions and competitive prices, $\mathbf{V} \equiv (\mathbf{i}', \boldsymbol{\pi}', \mathbf{div}', \hat{\mathbf{n}}^d, \mathbf{c}', \hat{\mathbf{n}}^s, \mathbf{w}')'$, as

$$\underbrace{\mathbf{V}^{REE}}_{\text{sequence of variables}} = \underbrace{\mathbf{M}}_{\text{causal relations in sequence space}} \mathbf{V}^{REE} + \underbrace{\mathbf{S}}_{\text{sequence of shocks}} \quad (36)$$

The rational expectations equilibrium can be solved to yield

$$\mathbf{V}^{REE} = (\mathbf{I} - \mathbf{M})^{-1} \mathbf{S} = \sum_{n=1}^{\infty} \mathbf{M}^{n-1} \mathbf{S} \quad (37)$$

which is a sum of all equilibrium effects of varying distance, where each $\mathbf{M}^{n-1} \mathbf{S}$ term is an n -step effect of the sequence of a shock on the sequence of a variable via sequences of $n - 1$ intermediate variables.

This proposition generalizes Proposition 1 in the case of period-1 general equilibrium to dynamic general equilibrium with persistent shocks.

Shallow thinking. With Proposition 6, we generalize Assumption 1 as follows to capture shallow thinking about dynamic general equilibrium, by replacing (V, M, S) with the long vector and stacked Jacobian $(\mathbf{V}, \mathbf{M}, \mathbf{S})$. Consequently, the average expectations characterized by Proposition 2 is generalized, also by replacing (V, M, S) with the long vector and stacked Jacobian $(\mathbf{V}, \mathbf{M}, \mathbf{S})$.

As briefly mentioned in the introduction, we focus on the causal relations across variables and abstract away from the horizon-dependent dampening studied by Angeletos and Lian (2018), Farhi and Werning (2019) and Gabaix (2020). We assume that if agents understand how consumption c_t depends on the contemporaneous interest rate i_t , they also understand how c_t depends on future i_s . This is reflected in that the dependence of $\mathbb{E}^d[c_t]$ on $\mathbb{E}^{d-1}[\mathbf{i}]$ is mediated via the causal relations $\mathbf{M}$, which collects all the Jacobians $\mathbf{J}_{vu}$ in (33). One could further generalize our theory to introduce additional dampening across periods, by modifying $\mathbf{M}$.²⁷

²⁷For example, by replacing $(\mathbf{J}_{vu})_{ts}$ with $m^{s-t} (\mathbf{J}_{vu})_{ts}$ for $s > t$ with a factor $m < 1$ and adjusting $\mathbf{M}$ accordingly, we can allow for the possibility that agents underappreciate the dependence of one variable on future variables relative to its dependence on contemporaneous variables.

Equilibrium under shallow thinking. Given the average beliefs $\bar{\mathbb{E}}[\cdot]$, the equilibrium $\{i_t, \pi_t, div_t, n_t^d, c_t, n_t^s, w_t\}_{t \geq 0}$ is characterized period by period by firms' and households' optimal behavior given their perfect observation of time- t variables and expectations about future variables, as in (D1-D5), along with the Taylor rule (D6) and the labor market clearing condition (D7).

5.2 More Persistent Cost-Push Shocks Cause Higher Inflation

We consider the responses of shallow thinking beliefs and equilibrium to a persistent cost-push shock $\epsilon_t^\pi = \rho^t \epsilon^\pi$. In particular, we highlight that a more persistent (higher ρ) cost-push shock leads to higher inflation under shallow thinking, contrary to the prediction of rational expectations.

REE and shallow thinking beliefs. In response to an exponentially decaying shock, both beliefs and equilibrium outcomes decay exponentially at the same rate. Proposition 7 establishes the rational expectations equilibrium (REE) and shallow thinking beliefs, characterizing their time- t values relative to the time- t size of the shock ϵ_t^π .

Proposition 7. (Persistent cost-push shock) *The rational-expectations equilibrium (REE) response to a persistent cost-push shock $\epsilon_t^\pi = \rho^t \epsilon^\pi$ features*

$$\pi_t^{REE} = \left[1 - \frac{(1 - \theta) + (\rho - \phi) \theta \kappa \frac{\sigma^{-1}(v+\sigma)}{1-\rho}}{1 - \beta \theta \rho} \right]^{-1} \epsilon_t^\pi \quad (38)$$

$$i_t^{REE} = \phi \pi_t^{REE} \quad (39)$$

whereas the average expectations under shallow thinking are

$$\bar{\mathbb{E}}[\pi_t] = \left[1 - \frac{\lambda(1 - \theta) + (\lambda^3 \rho - \lambda^4 \phi) \theta \kappa K(\lambda, \rho)}{1 - \beta \theta \rho} \right]^{-1} \epsilon_t^\pi \quad (40)$$

$$\bar{\mathbb{E}}[i_t] = \lambda \phi \bar{\mathbb{E}}[\pi_t] \quad (41)$$

with $K(\lambda, \rho)$ increasing in λ and ρ under our calibration and $K(1, \rho) = \frac{\sigma^{-1}(v+\sigma)}{1-\rho}$.

This proposition shows how exactly the persistence of shock ρ matters, nesting Proposition 4 with $\rho = 0$. A positive ρ gives rise to new terms and modifies existing terms,

which we dissect in order by analyzing REE inflation (38) and shallow expectations (40).

Regarding new terms, four general equilibrium loops across sequences of variables are involved, instead of three, displayed in Figure 9a. Relative to the case with period-1 shocks (Figure 4), persistent shocks give rise to new causal relations—the responses of household consumption and labor supply to inflation π , plotted in green. This results in the fourth loop, the *inflation-on-households loop* (in green), in addition to pricing complementarity (in purple), the monetary policy loop (in orange), and the Keynesian cross (in gray). This inflation-on-household loop takes 3 steps to close and amplifies the inflation response to cost-push shocks, since a higher inflation π simultaneously encourages consumption c and discourages labor supply $\hat{n}^s$, which leads to a higher wage w , feeding into inflation π . As a result, this loop is dampened by λ^3 in expectations (40). Further, the strength of this loop is proportional to ρ , meaning that it only exists when $\rho > 0$ and is stronger when ρ is higher. It is because only future inflation impacts household behavior by changing the real interest rate, and the expected inflation decays with rate ρ .

Concerning the strength of these terms, a positive ρ does two things, in addition to activating the inflation-on-households loop. First, it strengthens the general equilibrium feedbacks by $\frac{1}{1-\beta\theta\rho}$, because firms' pricing decisions are forward-looking and respond to a sum of future changes discounted by $\beta\theta$. Second, it boosts the Keynesian cross via $K(\lambda, \rho)$. This additional boost occurs because households' decisions are also forward-looking. Since the Keynesian cross reflects the feedback between firms and households, the effect of persistence is compounded. As ρ increases, all general equilibrium feedbacks become stronger, and the strengthening of inflation-on-households and monetary policy loops compounds with that of the Keynesian cross.

More persistent cost-push shocks are more inflationary. Figure 9b plots the average inflation expectation $\bar{\mathbb{E}}[\pi_t]$ and the equilibrium inflation π_t , both relative to the time- t size of the shock, as functions of persistence ρ . The dashed lines indicate expectations, and the solid lines represent the equilibrium. Different colors correspond to different values of λ , with $\lambda = 0$ in brown, our calibrated λ in blue, and $\lambda = 1$ (rational expectations) in green. In the case with rational expectations, the beliefs coincide with the equilibrium.

We note from Figure 9b that two insights obtained with transitory news shocks extend here, and a new lesson emerges regarding persistence. First, the average inflation expectation exceeds the equilibrium inflation when shocks are not too persistent, suggesting

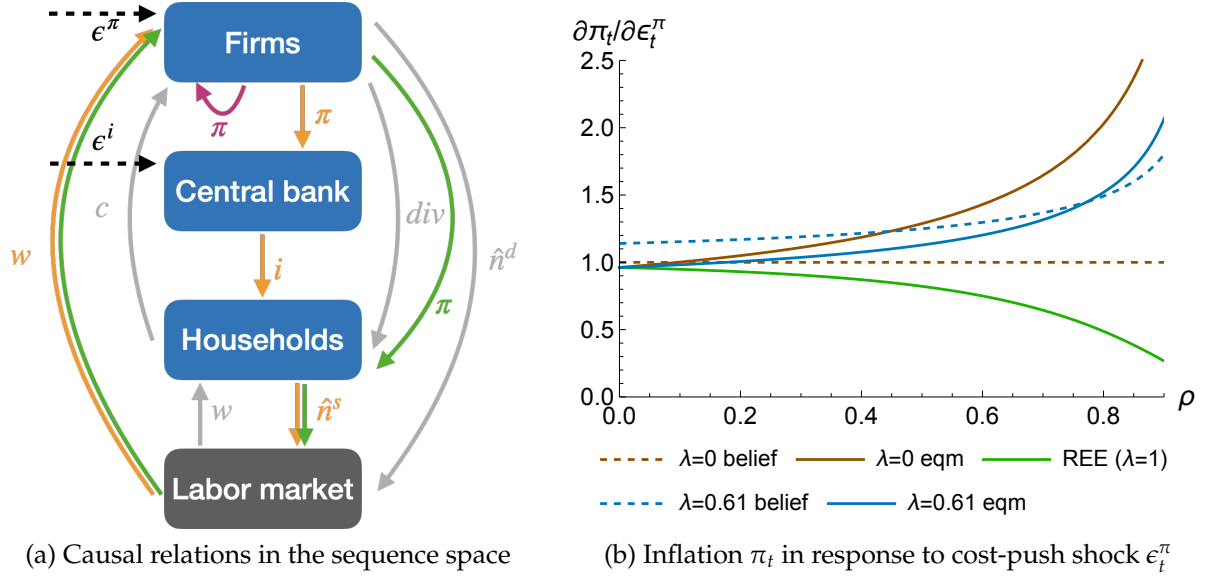


Figure 9: New Keynesian economy with persistent shocks

Notes: Panel (a) illustrates the four loops of causal relations across sequences of variables in the New Keynesian economy with persistent shocks in different colors, to accompany the discussion of Proposition 7. Four loops are the pricing complementarity self-loop (in purple), the inflation-on-households loop (in green), the monetary policy loop (in orange) and the Keynesian cross (in gray).

Panel (b) plots the equilibrium inflation π_t and the average expectation $\mathbb{E}[\pi_t]$ (relative to the size of the shock) in response to a persistent cost-push shock $\epsilon_t^\pi = \rho^t \epsilon^\pi$, as functions of the persistence ρ , under different values of λ . The dashed lines indicate expectations, and the solid lines represent the equilibrium. Different colors stand for different values of λ , with $\lambda = 0$ in brown, our calibration of λ in blue, and $\lambda = 1$ (rational expectations) in green. Under $\lambda = 1$, the beliefs coincide with the rational expectations equilibrium (REE).

overreaction, as the dashed blue line lies above the solid blue line.²⁸ Second, cost-push shocks are more inflationary under shallow thinking than under rational expectations, since the solid blue line lies above the solid green line. Last, a more persistent cost-push shock leads to higher inflation under shallow thinking, but lower inflation under rational expectations, since the solid blue line increases in ρ while the solid green line decreases.

To understand this new lesson regarding persistence, we start with the limiting cases of $\lambda = 0$ and $\lambda = 1$ (rational expectations). In the limit of $\lambda = 0$, inflation expectations always have the same size as the shock, with the dashed brown line being flat, as agents perceive no general equilibrium effects. As inflation expectations affect agents' behavior by acting like a cost-push shock for firms and a demand shock for households, a more persistent shock leads to a larger cumulative effects of future disturbance, resulting in

²⁸We also find that the average interest rate expectation underreacts to cost-push shocks but overreacts to monetary policy shocks (Figure D5), extending the case with transitory news shocks.

higher inflation through households' and firms' responses.

In contrast, in the limit of $\lambda = 1$ (rational expectations), as analyzed based on Proposition 7, a higher ρ strengthens all general equilibrium effects and, in particular, boosts the inflation-on-households and monetary policy loops relative to the pricing complementarity self-loop. The monetary policy loop offsets the inflation response, whereas the other two amplify it. As the monetary policy loop is the strongest among them and rational agents appreciate that, a higher ρ strengthens the offset and leads to a lower inflation.

Under our calibrated λ , the average inflation expectation increases in ρ . The mechanism is that, while the monetary policy offset loop is objectively the strongest, it is also the longest and gets dampened the most in expectations. Thus, as shallow agents better understand the shorter amplification loops than the monetary policy offset loop, they believe that a more persistent shock leads to more amplification. Through their responses to expectations, the equilibrium inflation is higher when ρ is higher. That increasing relationship is less drastic than in the limit case of $\lambda = 0$, as shallow agents partially understand the monetary policy reaction and its effects on the economy.

6 Conclusion

This paper develops a theory of *shallow thinking* as the structure of belief formation, validates its empirical content using a customized qualitative survey, and illustrates its implications for macro and finance. We develop a systematic representation of general equilibrium as a system of causal relations. We introduce shallow thinking by assuming that people only take finite steps on the causal chain starting from shocks. The key implication of our theory is that more distant causal relations and longer feedback loops have less influence on beliefs. We find that that, on average, people think about only 2.6 steps of shock propagation—far below infinity assumed by rational expectations.

Our theory predicts that, in a New Keynesian model, long-term nominal rates underreact to cost-push shocks but overreact to monetary policy shocks. It occurs as agents underappreciate shock propagation and offsetting feedback loops and reconciles multiple bond market puzzles. Moreover, news about future cost-push shocks leads to inflation—consistent with empirical findings—whereas rational expectations predict deflation. Additionally, more persistent cost-push shocks result in higher inflation, again contrary to rational expectations. These differences arise because shallow agents overweigh short feedback loops that amplify cost-push shocks, relative to a long offset loop via monetary

policy response. In an RBC model, shallow thinking amplifies and prolongs fluctuations in response to productivity shocks and leads to a stock market boom and crash.

At a high level, we acknowledge the immense complexity of the economy. If it has taken decades for our best economists to understand how it works—or if they are still figuring it out—we must carefully consider how much actual economic agents understand. We hope this paper is a useful step towards modeling people’s understanding of the economy. While we focus on the qualitative and positive consequences of shallow thinking, our theory can be applied to answer quantitative and normative questions, such as business cycle accounting and optimal stabilization policy. By modeling expectation formation in an iterative manner rather than as a fixed point, it brings better computational tractability in addition to its empirical relevance (Moll, 2025). It may also be fruitful to explore how people form their mental models through learning and how they understand the economy at more disaggregated levels, such as production networks and spatial economics.

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Online Appendix for

“Thinking about the Economy, Deep or Shallow?”

Pierfrancesco Mei Lingxuan Wu

November 25, 2025

A Survey Details

A.1 Survey Design and Sample

Detailed survey design. We design our survey to elicit the general public’s directional beliefs about changes in a host of macroeconomic variables (such as prices, labor hours, and interest rates) in response to a set of hypothetical macroeconomic shocks (such as oil supply shocks and monetary policy shocks).

Our survey builds on [Andre et al. \(2022\)](#), which ask respondents to forecast changes (in levels) of inflation and the unemployment rate in response to hypothetical macroeconomic shocks.²⁹ Grounded in our theory, two innovative features of our design are to inquire about a number of major macroeconomic variables and to only elicit beliefs about the directional responses. Inquiring about a host of macroeconomic variables traces out the path of shock propagation and tests if perception fails sequentially as suggested by our theory. For example, one of our questions concerns labor hours and asks respondents whether the average worker will work for more, fewer, or the same number of hours during a typical week in response to shocks. Comparing people’s beliefs about labor hours to those about aggregate demand sheds light on their understanding of firms’ input choice. Eliciting directional assessments instead of level forecasts lowers the cognitive strain and lets us focus on the qualitative aspect of people’s mental models. For instance, their responses about directional price change reveal whether participants understand the Phillips curve, rather than their potentially different perceptions about its slope.³⁰

²⁹[Haaland, Roth and Wohlfart \(2023\)](#) discuss the potential and limitations of hypothetical vignettes.

³⁰[Andre et al. \(2022\)](#) further ask participants to select the relevant ones from a list of potential channels and show that the selected channels predict their forecasts. For example, one channel in the oil supply shock vignette is “due to lower incomes or job loss, households cut back on their spending.” We simply ask for directional assessments for a set of variables, without showing them any directional statement that is objectively true or false, which may influence their responses.

Respondents are randomized into three groups with probability one half, one quarter and one quarter, each receiving two shock scenarios in random order. Groups 1, 2, and 3 receive the following pairs of shocks respectively: oil supply shock and monetary policy shock, government spending shock and personal income tax shock, corporate income tax shock and transfer payment shock. In each scenario, we introduce a shock that realizes now and persists for 12 months (except for the transfer payment shock), and then elicit respondents' beliefs about changes in the economy over the next 12 months. We stress the exogeneity of the shocks and state clearly that the shocks are publicly announced or broadcasted and are common knowledge to everyone in the US. Appendix A.2 shows the phrasing of these hypothetical shocks.

Our questions cover a large set of macroeconomic variables, divided into four blocks presented in random order, each on a separate page. The four blocks correspond to choices and decision-relevant variables of firms, households, the central bank, and the federal government's fiscal policy. We ask people's opinions about the average US business and household, to avoid any potential peculiarity of their own situations. Each block contains variables that the block either responds to or decides on, summing up to 12 to 16 distinct variables for each shock scenario, listed in Table B1. Several key variables, such as price and total labor hours, are included in more than one block (as one type of agents' choices and as other agents' decision-relevant variables), resulting in a total of about 22 questions for each shock scenario. For each question, respondents select among "up," "down," "unchanged," or "I don't know" to indicate their perceived directional changes of the specific variable in response to the shocks. The directional responses of most variables we elicit to these shocks are well-established in the empirical literature, as surveyed in Table B2.

In addition, to contrast our depth of thinking against level- k thinking in the game theoretical literature, we play the popular game of "guess 2/3 of the average" à la Nagel (1995) to measure respondents' game theoretical sophistication. Each respondent selects a number between 0 and 100, and they are informed that the number closest to the 2/3 of the average wins the game.³¹ Based on each respondent's answer g_i , we compute their

³¹Our baseline design runs this game without incentives for two reasons. First, offering a prize requires collecting respondents' email addresses, which some may be averse to, leading to selection bias. Second, since we cannot monitor whether respondents use search engines or other sources (though we ask them not to), we choose not to link compensation to performance and inform them of this. Nonetheless, we conducted an incentivized version with a subsample of 300 respondents. In this subsample, we confirm our additional finding that depth of thinking correlates across shock scenarios but does not correlate with level- k thinking.

level- k as $k_i \equiv \log\left(\frac{g_i}{50}\right) / \log\left(\frac{2}{3}\right)$, assuming that a level-0 player randomly selects a number.

We also assess respondents' financial literacy using questions from [Lusardi and Mitchell \(2011\)](#) and collect other information, including gender, age, race, ZIP code, household composition, education, main occupation and additional employment, political affiliation, household income, assets, and debts. Figure A1 illustrates the flow of our survey.

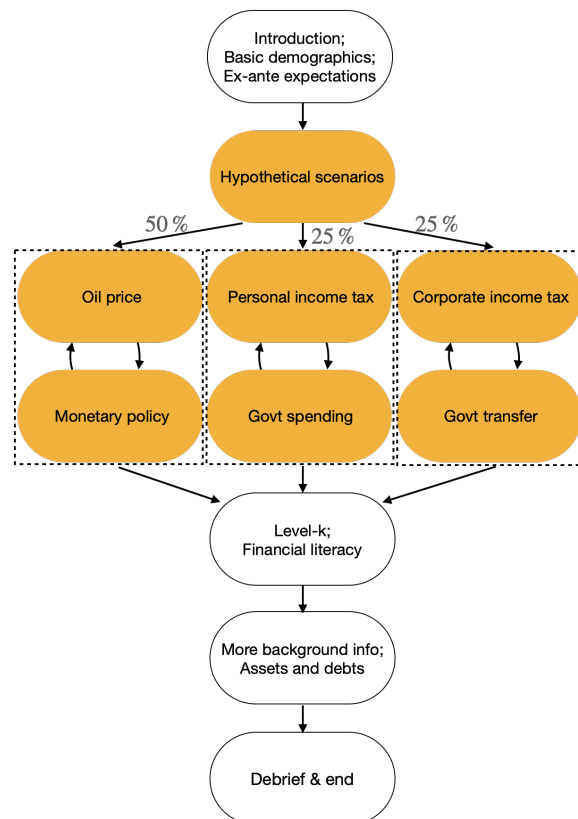


Figure A1: Survey structure

Sample. We conducted an online survey of 1,000 US households in June and October 2024. The survey was distributed through LUCID Marketplace, a platform that is widely used for research and is made up of hundreds of suppliers with a diverse set of recruitment methodologies, ensuring that the sample does not overweigh any particular segment of the population. The vast majority are double opt-in suppliers. Suppliers incentivize their respondents with loyalty reward points or gift cards or cash payments. The median completion time for our survey is 16 minutes.

We focus on US residents who are in the labor force at the time of the interview, and are aged between 25 and 65. Conditional on these characteristics, the survey is constrained

through quotas to be broadly representative of the US population along the dimensions of gender, age, total gross household income, and race, based on the Annual Social and Economic Supplement data of the Current Population Survey in 2022. Table A1 notes that our sample is largely representative of the US population.³²

We drop people who fail the attention or sanity checks placed in our survey or spend too much or too little time completing the survey to form our main sample. Our results are robust to various sample selection criteria.

Table A1: Sample statistics

	US population	Survey sample
<i>Gender</i>		
Male	.53	.48
<i>Age</i>		
25-29 years old	.13	.13
30-39 years old	.28	.29
40-49 years old	.25	.29
50-59 years old	.24	.21
60-65 years old	.10	.07
<i>Household income</i>		
\$0-\$19,999	.04	.11
\$20,000-\$39,999	.11	.17
\$40,000-\$69,999	.20	.24
\$70,000-\$124,999	.29	.34
\$125,000+	.36	.15
<i>Race</i>		
White	.61	.72
Black/African-American	.12	.13
Hispanic/Latino	.18	.08
Asian/Asian-American	.07	.03
Other	.02	.03
<i>Employment status</i>		
Full time employed	.78	.72
Part time employed	.09	.12
Self-employed	.10	.09
Unemployed	.03	.07

Notes: Shares may not exactly add up to 1 due to rounding errors.

A.2 Hypothetical Scenarios and Other Questions

Group 1.

Oil supply shock (oil). “Since the beginning of 2024, the average price for one barrel of WTI crude oil, which is a major benchmark for oil prices in the US, has been around \$80.

³²As known in the literature (Stantcheva, 2023a), online samples are hard to reach the tails of income distribution and tend to skew towards white and non-Hispanic respondents. Further, we show in Appendix B.3 household income does not correlate with depth of thinking.

Now, imagine that the price of crude oil unexpectedly increases due to production problems in the Middle East. For the next 12 months, the price for one barrel of crude oil will be, on average, \$20 higher than its current level.

This price increase is publicly broadcasted by major news outlets and is common knowledge to everyone in the US.

We will now ask you a few short questions to understand how you think the US economy would be affected by such an increase in oil price."

Monetary policy shock (MP). *"The Federal Reserve, often referred to simply as "the Fed," is the central bank of the United States that conducts the nation's monetary policy to help regulate the economy. It sets a key interest rate known as the Federal Funds Rate. This is the rate at which banks lend to each other, and it affects the economy in many ways. The Federal Funds Rate influences the interest rates for savings accounts, credit card balances, mortgages, loans, and others. As of now, the Federal Funds Rate set by the Federal Reserve is 4.75%.*

Now, imagine that the Federal Reserve unexpectedly raises the Federal Funds Rate by 0.5 percentage points, changing it to 5.25%, and announces that it will maintain this rate for the next 12 months.

This interest rate raise is publicly announced and is common knowledge to everyone in the US. The Federal Reserve clarifies that this decision is made with no changes in their assessment of economic conditions.

We will now ask you a few short questions to understand how you think the US economy would be affected by this raise in the interest rate."

Group 2.

Government spending shock (G). *"Since 2000, US federal government spending has averaged about 25% of the US Gross Domestic Product (GDP), which is the total value of all goods and services produced in the country.*

Now, imagine that the US federal government unexpectedly announces a new defense program, leading to an increase in federal government spending over the next 12 months. And the additional spending will be directed domestically. Specifically, federal government spending relative to US GDP will increase by about 2% over the next 12 months.

This increase is publicly announced and is common knowledge to everyone in the US. The government clarifies that this change is temporary and occurs without any alterations in its

assessment of national security or economic conditions.

We will now ask you a few short questions to understand how you think the US economy would be affected by this increase in federal government spending."

Personal income tax shock (PIT). *"In 2023, a typical household earning the median income is subject to a 12% federal personal income tax rate. Collectively, all households paid about 2.2 trillion dollars in federal personal income taxes in 2023, which is about 8% of the US Gross Domestic Product (GDP), the total value of all goods and services produced in the country.*

Now, imagine that the US federal government unexpectedly announces a 2% increase in the federal personal income tax rate over the next 12 months.

This increase is publicly announced and is common knowledge to everyone in the US. The government clarifies that the change is temporary and occurs without any changes in its assessment of economic conditions.

We will now ask you a few short questions to understand how you think the US economy would be affected by this increase in the federal personal income tax rate."

Group 3.

Transfer payment shock (TP). *"Imagine that today the US federal government unexpectedly announces that each taxpayer will receive a one-time transfer payment worth, on average, \$1,200. This one-time payment, which will not be taxed, will be available in bank accounts or as a check in mailboxes within three months.*

Taking into account that around 200 million US taxpayers will receive the payment, the total payments disbursed will be approximately 240 billion dollars, which is about 1% of the US Gross Domestic Product (GDP), the total value of all goods and services produced in the country.

The payment is publicly announced and is common knowledge to everyone in the US. The government clarifies that the payment is a one-time event and occurs without any changes in its assessment of economic conditions.

We will now ask you a few short questions to understand how you think the US economy would be affected by this transfer payment by the federal government"

Corporate income tax shock (CIT). *"Since 2017, there has been a 21% federal corporate income tax rate in place. The taxable income is a business's revenue minus expenses. In 2023, all*

corporations together paid about 420 billion dollars, which is about 1.5% of the US Gross Domestic Product (GDP), the total value of all goods and services produced in the country.

Now, imagine that the US federal government unexpectedly announces a 2% increase in the federal corporate income tax rate over the next 12 months.

The increase is publicly announced and is common knowledge to everyone in the US. The government clarifies that the change is temporary and occurs without any changes in its assessment of economic conditions.

We will now ask you a few short questions to understand how you think the US economy would be affected by the increase in federal corporate income tax rate."

Financial literacy. Three questions are asked for each respondent.

1. "Imagine that the interest rate on your savings account was 1% per year and inflation was 2% per year. After 1 year, how much would you be able to buy with the money in this account?: more than today; less than today; exactly the same; don't know."
2. "Do you think that the following statement is 'true' or 'false'? Buying a company stock usually provides a safer return than a stock mutual fund.: true; false; don't know."
3. "Suppose you had \$100 in a savings account and the interest rate was 2% per year. After 5 years, how much do you think you would have in the account if you left the money to grow?: more than \$102; less than \$102; exactly the same; don't know."

Level-k thinking ("guess 2/3 of the average" game). "Imagine you are playing a game with about 300 other people chosen randomly from across the United States.

Please choose a number between 0 and 100, inclusive.

We will take your number, as well as the numbers chosen by other participants, to calculate the average number. The winning number will be the number that is closest to two-thirds ($\frac{2}{3}$) of the average number. Specifically, we sum the chosen numbers by everyone and divide by the number of participants. Multiply the result by $\frac{2}{3}$. The winning number is the one closest to the last result.

The winner will get an electronic gift card at any popular merchant worth \$30, which will be split when there are multiple winners. The winner will be contacted in a few days at the conclusion of this study, using the email provided below."

B Appendix to Survey Findings

B.1 Correct Directional Responses and Causal Distance

Table B1: Variables elicited in forecast part of our survey

Variable	Abbrev.	Group 1		Group 2		Group 3	
		Oil ↑	MP ↑	G ↑	PIT ↑	CIT ↑	TP ↑
Firms-related	bus						
Nominal marginal cost	mc	✓	✓	✓	✓	✓	✓
Demand	Y	✓	✓	✓	✓	✓	✓
Interest rate	i	✓	✓	✓	✓	✓	✓
Corporate income tax rate	CIT	✓	✓	✓	✓	✓	✓
Prices	p	✓	✓	✓	✓	✓	✓
Intermediate inputs	x	✓	✓	✓	✓	✓	✓
Investment	I	✓	✓	✓	✓	✓	✓
Total hours	N	✓	✓	✓	✓	✓	✓
Unemployment rate	u	✓	✓	✓	✓	✓	✓
Dividends/post-tax profits	div	✓	✓	✓	✓	✓	✓
Households-related	hh						
Interest rate	i	✓	✓	✓	✓	✓	✓
Prices	p	✓	✓	✓	✓	✓	✓
Hours	N	✓	✓	✓	✓	✓	✓
Personal income tax rate	PIT	✓	✓	✓	✓	✓	✓
Pre-tax nominal wage	W	✓	✓	✓	✓	✓	✓
Durable consumption	D	✓	✓	✓	✓	✓	✓
Non-durable consumption	ND	✓	✓	✓	✓	✓	✓
Central-bank-related	fed						
Unemployment rate	u	✓	✓	✓	✓	✓	✓
Inflation	p	✓	✓	✓	✓	✓	✓
Interest rate	i	✓	×	✓	✓	✓	✓
Government-related	gov						
Borrowing/repayment	B	✓	✓	✓	✓	✓	✓
Tax revenue	TR	✓	✓	✓	✓	✓	✓

Table B2: Literature review of directional impulse responses

	Oil price ↑	Monetary policy (MP) ↑	Transfer payment (TP)† ↑
Output	Down (Känzig)	Down (Ramey)	Up (Pennings)
Interest rate	Up (Känzig)	Up (Ramey, as shock itself)	
Price	Up (Känzig)	Down (MAR), insignificant or up (Ramey, classic price puzzle)	
Unemployment	Up (Känzig)	Up (Ramey)	
Labor hours	Down (BG)	Down (MAR)	Down (MMNPF)
Nonresidential investment	Down (Känzig)	Down (BKM)	
Durable consumption	Down (Känzig)	Down (BKM)	Up (EHMNW)
Nondurables & services	Down (Känzig)	Down (BKM)	Up (EHMNW)
Nominal wage	Up (BG)	Down (OT, as real wage and price both down), insignificantly down (MAR)	Up (EHMNW)
Dividend/post-tax profits			
References	Känzig (2021, figs 8-10/A.7) Blanchard and Gali (2010, fig 7.6.A)	Ramey (2016, figs 2-3) Miranda-Agrippino and Ricco (2021, fig 7) Boivin, Kiley and Mishkin (2010, fig 4, post-84) Olivei and Tenreiro (2007, figs 10-14)	Pennings (2021, tab 1) Mendes et al. (2024, tab 5) Egger et al. (2022, tabs 1, 3)
	Government spending (G) ↑	Personal income tax (PIT) ↑	Corporate income tax (CIT) ↑
Output	Up (Ramey16)	Down (MR)	Down (MR)
Interest rate	Insignificant (Ramey11)	Insignificant (MR)	Insignificant (MR)
Price	Up (Ramey16 unreported result)	Down (CMMS23), insignificantly down (MR)	Insignificantly up (CMMS23, MR)
Unemployment	Down (Ramey13*)	Up (MR)	Up (CKS), insignificantly up (MR)
Labor hours	Up (Ramey11*)	Down (CMMS24, MR)	Insignificantly down (CMMS24), insignificant (MR)
Nonresidential investment	Down (Ramey11)	Down (MR)	Down (MR)
Durable consumption	Down (Ramey11**)	Down (MR)	Down (CMMS24†), insignificantly up (MR)
Nondurables & services	Down (Ramey11**)	Down (CMMS24†), insignificantly down (MR)	Down (CMMS24†), insignificantly up (MR)
Nominal wage	Up (Ramey11)	Down (CMMS24, as real wage and price both down)	Down (CKS)
Dividend/post-tax profits		Up (MR, as shock itself)	Insignificantly down (MR)
Personal income tax rate	Up (Ramey11)	Insignificantly down (MR)	Up (MR, as shock itself)
Corporate income tax rate	Up (Ramey16**)	Up (CMMS24†, MR)	Up (CMMS24†), insignificant (MR)
Tax revenue	Up (Ramey16, as shock itself)	Insignificant (MR)	Insignificant (MR)
Government spending	Up (CMM)	Down (CMMS24, MR)	Insignificant (CMMS24, MR)
Government debt			
References	Ramey (2016, fig 5) Ramey (2011, fig X) Ramey (2013, figs 1.11-17) Corsetti, Meier and Müller (2012, fig 1)	Mertens and Ravn (2013, figs 2-4/9/10) Cloyne et al. (2023, fig 1) Cloyne et al. (2024, figs 1-2/B.1/H.8) Cloyne, Kurt and Surico (2023, figs 2/3B)	

Notes: This table lists directional impulse responses at about 1-year horizon across variables to shocks. Shocks considered are oil supply shocks, contractionary monetary policy (MP) shocks, transfer payment (TP) shocks, government spending (G) shocks, and positive shocks in personal income tax (PIT) rate and corporate income tax (CIT) rate. As we are only interested in the directions rather than the magnitudes of responses, we only impose a weak assumption that each variable responds to positive and negative shocks with opposite signs. This is weaker than assuming that the multipliers are the same for positive and negative shocks. Nonetheless, it is worth noting that while some earlier papers advocate for asymmetric multipliers, more recent papers argue that the evidence is weak (e.g., Kilian and Vigfusson (2011) for oil supply shocks and Ben Zeev, Ramey and Zubairy (2023) for government spending shocks). The abbreviation in parentheses indicates the main reference, usually the most recent or most cited paper. All references are listed at the bottom of each column.

† For the transfer payment shock, Pennings (2021) and Mendes et al. (2024) provide cross-sectional estimates instead of aggregate ones, in the US and Brazil respectively. Egger et al. (2022) study transfers in Kenya, but these transfers are funded from outside the economy. We drop this shock in a robustness version in Table B3.

* Ramey (2013) shows that the unemployment rate falls in response to government spending shocks. It is mainly driven by government employment, with the response of private employment either insignificant or negative in different specifications. We drop this variable in a robustness version in Table B3, since we elicit participants' opinions about private businesses.

** Ramey (2011, 2016) discusses extensively potential issues with previous works (e.g., Blanchard and Perotti, 2002; Galí, López-Salido and Vallés, 2007) that find a positive consumption response to government spending shocks. We drop this variable in a robustness version in Table B3.

*** Ramey (2016, fig 5) suggests that the average tax rate goes up in response to government spending shocks, calculated as federal current receipts divided by nominal GDP. † Cloyne et al. (2024, fig 2) show positive consumption responses to decreases in PIT and CIT, but do not split into durables vs. nondurables.

†† Cloyne et al. (2024, fig H.8) show negative primary surplus response (tax revenue minus government spending) to decreases in PIT and CIT.

Table B3: Correct directional responses with robustness versions

Version	Oil ↑			MP ↑			G ↑			PIT ↑			CIT ↑			TP ↑		
	Va	Vb	Vc	Va	Vb	Vc	Va	Vb	Vc	Va	Vb	Vc	Va	Vb	Vc	Va	Vb	Vc
Y	↓	↓	↓	↓	↓	↓	↑	↑	↑	↓	↓	↓	↓	↓	↓	↑	↑	↑
i	↑	↑	↑	↑	↑	↑	↑	↑	↑	↓	↓	↓	↓	↓	↓	↑	↑	↑
P	↑	↑	↑	↓	↓	↓	↑	↑	↑	↓	↓	↓	↓	↓	↓	↑	↑	↑
u	↑	↑	↑	↑	↑	↑	↓	↓	↓	↑	↑	↑	↑	↑	↑	↓	↓	↓
N	↓	↓	↓	↓	↓	↓	↑	↑	↑	↓	↓	↓	↓	↓	↓	↑	↑	↑
I	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↑	↑	↑
D	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↑	↑	↑
ND	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↑	↑	↑
W	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↑	↑	↑
div	↑	↑	↑	↓	↓	↓	↑	↑	↑	↓	↓	↓	↓	↓	↓	↑	↑	↑
mc																		
PIT							↑	↑	↑	↑	↑	↑						
CIT							↑	↑	↑	↑	↑	↑	↑	↑	↑			
TR							↑	↑	↑	↑	↑	↑	↑	↑	↑	↑	↑	↑
B							↑	↑	↑	↓	↓	↓	↓	↓	↓			

Notes: Three versions are constructed as follows. Version a is the baseline specification in Table 1.

1. Version a is most closely based on most up-to-date empirical estimates with clear directions.
2. Version b drops from Version a variables for which estimates are noisy or controversy exists. A few noteworthy choices are price response to monetary policy shocks (classic price puzzle), unemployment and consumption responses to government spending shocks (as discussed in the notes of Table B2).
3. Version c makes additional predictions based on theoretical predictions, relative to Version a.

Table B4: Model-implied causal distance with robustness versions

Model	Oil			MP			G			PIT			CIT			TP		
	M1	M2	M3	M1	M2	M3	M1	M2	M3	M1	M2	M3	M1	M2	M3	M1	M2	M3
Y	3	3	3	2	2	2	1	1	1	2	2	2	3	3	3	2	2	2
i	2	2	2	1	1	1	5	3	3	5	3	4	6	4	3	5	4	3
p	1	1	1	4	3	4	4	2	2	4	3	3	5	4	2	4	3	2
u	3	3	3	2	2	2	2	2	2	2	2	2	3	3	2	2	3	2
N	3	3	3	2	2	2	2	2	2	2	2	2	3	3	2	2	3	2
I			3			2			2			3			2			2
D	3	3	3	2	2	2	2	2	2	2	2	2	3	3	3	2	2	2
ND	3	3	3	2	2	2	2	2	2	2	2	2	3	3	3	2	2	2
W			2			3			3						3			3
div	4	4	3	3	3	3	2	2	2	3	3	3	2	2	2	2	2	2
mc	1	1	1	3	3	3	3	2	2	3	3	3	4	4	2	3	3	2
PIT							1	1	1	1	1	1				1	1	1
CIT							1	1	1				1	1	1	1	1	1
B							1	1	1	1	1	1	1	1	1	1	1	1

Notes: Three sets of causal distance D_{vs} are constructed as follows, with the oil supply shock interpreted as a cost-push shock, as is standard in the literature. Model 2 is the baseline specification in Table 1.

1. Model 1 is the textbook New Keynesian model in the main text of this paper.
2. Model 2 extends Model 1 to feature decreasing-returns production (so that the marginal cost is increasing in quantity) and a Taylor rule of monetary policy that depends on both inflation and unemployment.
3. Model 3 extends Model 1 with capital investment by firms, price and wage rigidity (via a labor union instead of a competitive labor market).

Our directed graph representation of mental models (Figure 1) only includes labor demand and supply shifts $\hat{n}_1^d, \hat{n}_1^s$ and assumes that the real wage w_1 requires one additional step beyond demand and supply shifts. For labor hours, we assume that people perceive the change as soon as they perceive a change in either $\hat{n}_1^d$ or $\hat{n}_1^s$. Alternatively, one could assume that labor hours require an additional step beyond these shifts to be recognized as the equilibrium outcome. This difference is absorbed by variable fixed effects in our regression of correct directional belief on causal distance, and is irrelevant when applying shallow thinking in macroeconomic models, since only beliefs about the real wage are decision-relevant, while those about labor hours are not.

B.2 Additional Results of Predicting Correct Directional Belief

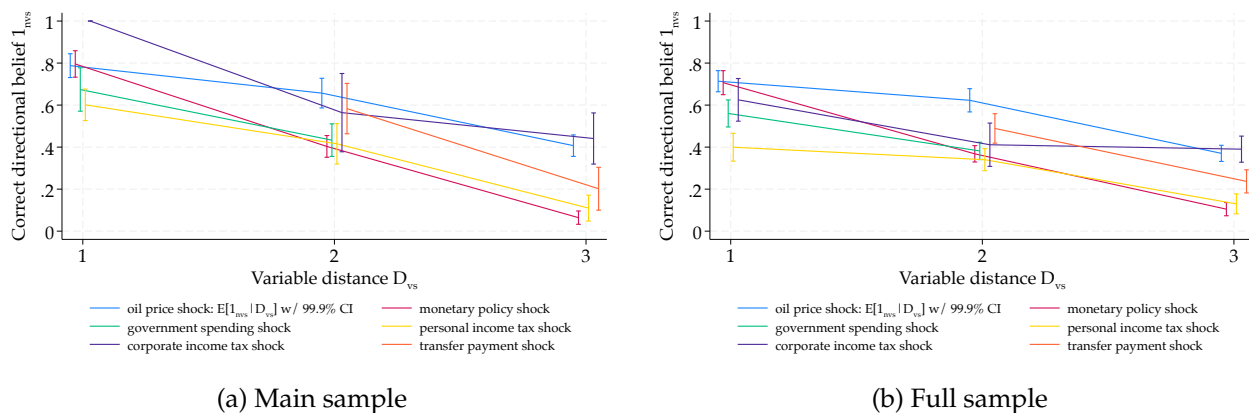


Figure B1: Expectation of correct directional belief 1_{nvs} conditional on causal distance D_{vs}

Notes: This figure shows the expectation of correct directional belief 1_{nvs} , conditional on causal distance D_{vs} , together with the 99.9% confidence interval, by each shock. Panel (a) shows the main sample of our analysis that filters attention and completion time and panel (b) shows the full sample.

Table B5: Predicting correct directional belief 1_{nvs} on subsets of variables and shocks

	Main		Oil & MP		Decision-relevant		Decision-irrelevant	
Correct directional belief 1_{nvs}	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Variable distance D_{vs}	-0.24*** (0.01)	-0.27*** (0.01)	-0.23*** (0.01)	-0.25*** (0.01)	-0.37*** (0.01)	-0.33*** (0.01)	-0.12*** (0.01)	-0.32*** (0.07)
Constant	0.99*** (0.02)	1.04*** (0.02)	0.99*** (0.02)	1.03*** (0.02)	1.24*** (0.02)	1.18*** (0.02)	0.71*** (0.03)	1.17*** (0.16)
Observations	10920	10763	7775	7775	1560	1560	4137	3902
R^2	0.24	0.63	0.26	0.66	0.56	0.85	0.24	0.63
Individual FE	Yes	Absorbed	Yes	Absorbed	Yes	Absorbed	Yes	Absorbed
Individual-variable FE		Yes		Yes		Yes		Yes
Individual-shock FE		Yes		Yes		Yes		Yes

Standard errors in parentheses

Standard errors clustered at individual level

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Notes: Columns (1, 2) present the specification for the main sample that filters attention and completion time. Columns (3, 4) focus on respondents who receive the oil and monetary policy (MP) shocks. Columns (5, 6) include only responses to the household block in the survey, which covers four decision-relevant variables: price, wage, interest rate, and personal income tax. Columns (7, 8) include only responses outside the household block and exclude the four aforementioned decision-relevant variables.

Table B6: Predicting correct directional belief 1_{nvs} with robustness versions

Correct directional belief 1_{nvs}	Main (1)	Full (2)	D_{vs}^{M1} (3)	D_{vs}^{M3} (4)	1_{nvs}^{Vb} (5)	1_{nvs}^{Vc} (6)
Variable distance D_{vs}	-0.27*** (0.01)	-0.23*** (0.01)	-0.19*** (0.01)	-0.19*** (0.01)	-0.20*** (0.01)	-0.26*** (0.01)
Constant	1.04*** (0.02)	0.91*** (0.02)	0.89*** (0.01)	0.87*** (0.01)	0.93*** (0.02)	1.06*** (0.02)
Observations	10763	22023	10763	12479	8335	14357
R^2	0.63	0.58	0.61	0.62	0.25	0.60
Individual FE	Absorbed	Absorbed	Absorbed	Absorbed	Yes	Absorbed
Individual-variable FE	Yes	Yes	Yes	Yes		Yes
Individual-shock FE	Yes	Yes	Yes	Yes		Yes

Standard errors in parentheses
Standard errors clustered at individual level
* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Notes: Column (1) presents the main specification with individual-variable and individual-shock fixed effects on the main sample that filters attention and completion time. Column (2) examines the full sample, i.e., all respondents who completed the survey. Columns (3, 4) use the distance implied by Model 1 and Model 3 in Table B4, instead of Model 2. Columns (5, 6) use the shock-variable combinations from Version b and Version c in Table B3, instead of Version a. In the case of Version b, which includes fewer shock-variable combinations, there is not enough variation to apply individual-variable and individual-shock fixed effects.

B.3 Depth of Thinking is Individual Characteristic and Domain-Specific

We further show that the ability to understand shock propagation is indeed an individual characteristic and does not correlate with a classic measure of level- k thinking.

The previous finding that individual fixed effects matter for correct directional beliefs 1_{nvs} already suggests that some people get more variables correct than others. To further investigate this, we measure individual n 's overall understanding of shock s by a *total depth score (TDS)* as

$$TDS_{ns} \equiv \sum_v D_{vs} \cdot 1_{nvs} \quad (B1)$$

To receive a higher TDS, a respondent needs to correctly forecast directional changes in more variables and especially more distant variables.³³

To the extent that depth of thinking is an individual characteristic as we postulate, we expect each respondent's TDSs to correlate strongly across shocks. To test this, we rank the TDSs from the lowest to the highest for each shock, and correlate the two TDS rank measures across individuals in Table B7. Column (1) confirms this prediction.

³³This TDS is a more robust measure to noise than the maximum distance of variables whose changes are understood correctly, as we have several variables for each step and respondents may coincidentally get some correct.

Table B7: Total depth score is individual characteristic and domain-specific

	Main				Incentivized	Full	
TDS of 2nd shock (rank)	(1)	(2)	(3)	(4)	(5)	(6)	(7)
TDS of 1st shock (rank)	0.31*** (0.05)				0.24*** (0.05)	0.30** (0.11)	0.20*** (0.04)
Level k (rank)		-0.08 (0.05)			-0.04 (0.05)	0.05 (0.11)	-0.02 (0.03)
Financial literacy (rank)			0.28*** (0.05)		0.23*** (0.05)	0.11 (0.11)	0.20*** (0.03)
Education (rank)			0.02 (0.05)		-0.02 (0.06)	-0.04 (0.12)	-0.04 (0.04)
Male			-0.03 (0.03)		-0.03 (0.03)	-0.10 (0.06)	-0.01 (0.02)
Net asset (rank)				0.05 (0.06)	0.05 (0.06)	0.04 (0.11)	0.08* (0.04)
Income (rank)				0.06 (0.06)	0.06 (0.07)	0.11 (0.13)	0.03 (0.05)
Constant	0.38*** (0.03)	0.59*** (0.03)	0.37*** (0.05)	0.49*** (0.04)	0.25*** (0.06)	0.29* (0.14)	0.26*** (0.04)
Observations	383	383	383	383	383	118	828
R^2	0.09	0.01	0.09	0.02	0.15	0.17	0.12
Shock group FE					Yes	Yes	Yes
Age group FE					Yes	Yes	Yes

Standard errors in parentheses

Standard errors are heteroscedasticity-consistent

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Notes: Columns (1-5) use the main sample that filters attention and completion time. Column (6) focuses on a subsample of the main sample, where the “guess 2/3 of the average” game is incentivized. Column (7) studies the full sample, i.e., all respondents who completed the survey.

In contrast, column (2) suggests that TDS does not correlate with a classic measure of strategic sophistication (level k), via a “guess 2/3 of the average” game we play with survey respondents. This connects to findings in the macroeconomic literature that the measured level k does not predict differential consumption response to inflation news by Dutch households (Coibion et al., 2023) or first- and higher-order inflation expectations of New Zealand firm managers (Coibion et al., 2021).³⁴ We remark that shallow thinking

³⁴In the experimental literature, Dal Bó, Dal Bó and Eyster (2018) show that voters prefer policy changes that bring in direct benefits but induce larger indirect costs, but their voting behavior is not correlated with level k . Georganas, Healy and Weber (2015) study stability of level k using two families of games: beauty contest games à la Nagel (1995) and undercutting games similar to Arad and Rubinstein (2012). They find that the participants’ levels are consistent within the beauty contest family, but do not correlate within the

likely reflects people’s limited knowledge about the macroeconomy, a distinct aspect of bounded rationality from limited strategic sophistication. After all, a chess master who could anticipate opponents well may not know macroeconomics, and vice versa.

Column (3) suggests that a measure of financial literacy, based on [Lusardi and Mitchell \(2011\)](#), but not general education, correlates with TDS. This supports the idea that shallow thinking reflects individuals’ economic knowledge, for which general education may be too noisy a proxy.

Column (4) shows that households’ net asset positions or income do not significantly correlate with TDS. The two significant predictors, TDS and financial literacy, remain significant when all variables are pooled together, as indicated in column (5).

Columns (6) and (7) demonstrate that our findings hold true over a subsample where the “guess 2/3 of the average” game is incentivized, as well as in the full sample, i.e., all respondents who completed the survey.

C Shallow Thinking in an RBC Economy

Here, we consider an RBC economy, and show that shallow thinking amplifies its responses to persistent productivity shocks and produces a stock market boom and crash.³⁵

We outline the model and shallow thinking thereof in Section [C.1](#), with details in Appendix [E](#), and present the effects of productivity shocks on the macroeconomy and asset prices in Section [C.2](#).

C.1 Shallow Thinking in an RBC Economy

The RBC economy consists of two types of agents (firms and households) and two competitive markets (the goods market and the labor market). We study a first-order approximation around the steady state. Differing from the convention in the New Keynesian model, here we use capital letters for variables in levels, so that different components of GDP are in the same units.

As in the prior analysis of the New Keynesian economy, we assume that agents observe the sequence of shocks at time 0^- and form their expectations $\mathbb{E}^d[\cdot]$ once and for all at time

undercutting game family or across two families.

³⁵As we analytically applied shallow thinking to the New Keynesian economy, studying an RBC economy is also useful for illustrating its applicability to a broader class of models that can be solved using the sequence-space Jacobian developed by [Auclert et al. \(2021\)](#). We present the procedure in Appendix [F](#).

0^- , i.e., $\mathbb{E}_t^d[v_s] \equiv \mathbb{E}^d[v_s]$ for $s > t$.

Firms. There is a continuum of firms that produce using capital K_t and labor N_t^d with a production function $Y_t = Z_t \left(\frac{K_t}{\alpha}\right)^\alpha \left(\frac{N_t^d}{1-\gamma}\right)^{1-\gamma}$. Firms own capital and make investment I_t to increase the capital stock in the next period, $K_{t+1} = (1 - \delta) K_t + I_t$. In addition, firms are subject to capital adjustment cost $\Psi(I_t, K_t) = \frac{\psi}{2} \left(\frac{I_t}{K_t} - \delta\right)^2 K_t$, which gives rise to an investment- q relation as in Hayashi (1982), and pay dividends $DIV_t = Y_t - W_t N_t^d - I_t - \Psi(I_t, K_t)$. Firms maximize their values, i.e., the sum of dividends discounted by the gross interest rate. We use $1 + r_t$ to denote the gross interest rate from period t to $t + 1$ and assume that it is known in period t , as the return of a 1-period bond in zero supply.

In each period, firms choose investment I_t , output Y_t , and labor demand N_t^d , responding to current productivity Z_t , the prevailing wage W_t , and interest rate r_t , as well as future productivities $\{Z_{t+k}\}_{k \geq 1}$ and their expectations $\{\bar{\mathbb{E}}[W_{t+k}], \bar{\mathbb{E}}[r_{t+k}]\}_{k \geq 1}$.

Households. There is a continuum of households who live infinitely and maximize their lifetime utility, discounted by β , which is separable in consumption and labor supply. In each period, households choose consumption C_t and labor supply N_t^s , taking as given the prevailing wage W_t , interest rate r_t and dividend DIV_t , as well as their expectations of the future values of these variables $\{\bar{\mathbb{E}}[W_{t+k}], \bar{\mathbb{E}}[r_{t+k}], \bar{\mathbb{E}}[DIV_{t+k}]\}_{k \geq 1}$. The household side of the RBC economy is the same as that of the New Keynesian economy, except that they invest in a real bond as opposed to a nominal bond.

Goods and labor markets. The interest rate $\{r_t\}_{t \geq 0}$ and wage $\{w_t\}_{t \geq 0}$ arise to clear the goods and labor markets, $Y_t = I_t + \Psi(I_t, K_t) + C_t, N_t^d = N_t^s$.

Shallow thinking. In terms of belief formation, the system of causal relations consists of firms and households' best responses, as well as the determination of the interest rate and wage by two fictitious auctioneers for the goods and labor markets, respectively. The interest rate depends on the shifts in firms' output and investment and in households' consumption ($\hat{Y}, \hat{I}$ and $\hat{C}$), each of which is a function of decision-relevant variables for firms and households other than the interest rate. Similarly, the wage is determined by the shifts in firms' labor demand and households' labor supply ($\hat{N}^d, \hat{N}^s$), both of which depend on decision-relevant variables other than the wage. Figure E1 illustrates these causal relations, which pin down the average expectations $\bar{\mathbb{E}}[\cdot]$.

Equilibrium. Given the average expectations, the equilibrium is determined by agents' best responses and market clearing conditions, period by period.

We adopt a quarterly calibration of the RBC economy, with all parameters listed in Table E1. We assume slight decreasing returns to scale in production ($\alpha < \gamma$) and the existence of a small fringe of rational agents, which we discuss in greater detail in Appendix E.

C.2 Effects of Productivity Shocks on Macroeconomy and Asset Prices

We study the effects of a persistent productivity shock, with $\rho = 0.979$ following King and Rebelo (1999), on the macroeconomy and asset prices. In particular, we show that shallow thinking amplifies the economy's responses and leads to a stock market boom and crash.

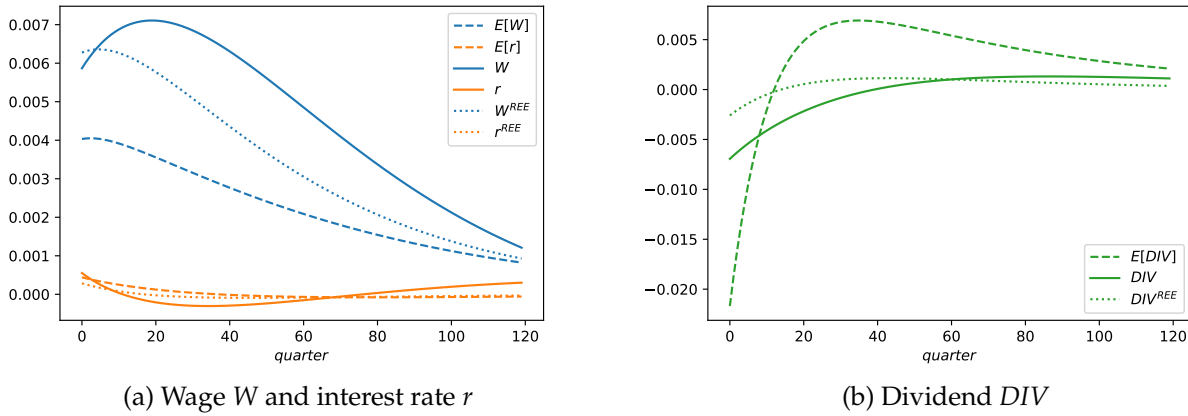


Figure C1: Decision-relevant beliefs in response to a productivity shock Z_t

Notes: Two panels plot the shallow thinking beliefs (in dashed lines), the shallow thinking equilibrium outcomes (in solid lines) and the rational expectations equilibrium outcomes (in dotted lines) of wage W , interest rate r , and dividend DIV , in response to a persistent productivity shock Z .

Figure C1 illustrates in dashed lines beliefs about the interest rate r , wage W , and dividend DIV , which are relevant for firms' and households' decisions. In comparison, the solid lines indicate their equilibrium values, and the dotted lines indicate the rational expectations equilibrium (REE). Panel C1a suggests that shallow agents underappreciate the wage response. That occurs because agents believe that the firms will produce more by hiring and investing more in response to a productivity shock, but fail to recognize that firms' behavior will push up wages in the economy. Panel C1b shows that shallow agents perceive a much more volatile stream of dividends compared to the equilibrium

values or the REE. This is because they believe firms will invest more early on by cutting dividends and pay more dividends later. However, in equilibrium, facing elevated wages and changes in interest rates, firms will not invest as much as shallow agents believe.

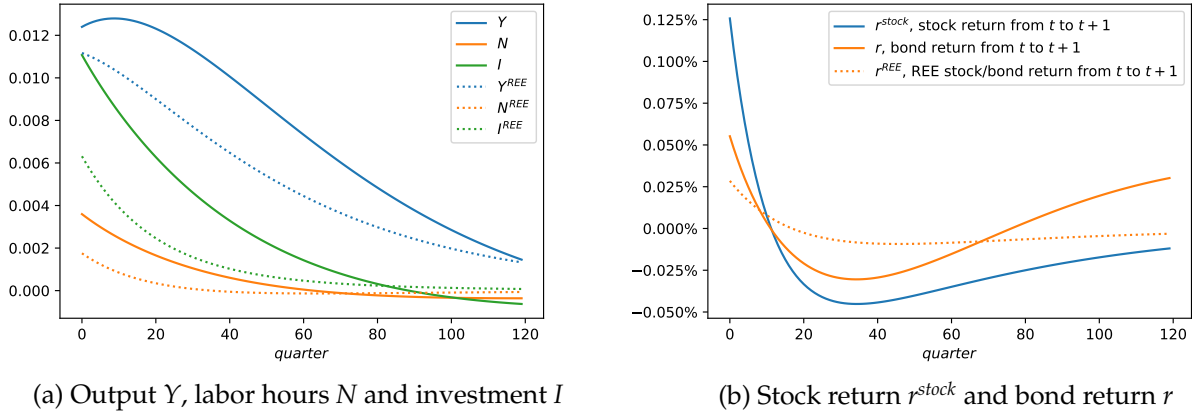


Figure C2: Equilibrium outcomes in response to productivity shock Z_t

Notes: Two panels plot the shallow thinking equilibrium outcomes (in solid lines) and the rational expectations equilibrium outcomes (in dotted lines) of output Y , labor hours N , investment I , stock return r^{stock} and bond return r (which is the interest rate), in response to a persistent productivity shock. In the rational expectations equilibrium, the stock and bond returns coincide.

Figure C2 shows the responses of macroeconomic variables and asset returns in solid lines, compared to the rational expectations equilibrium in dotted lines. Panel C2a demonstrates that the output response (in blue) under shallow thinking is hump-shaped and more persistent. The responses of investment and labor hours (in green and orange) are almost twice as large.

Upon observing the productivity shock, the stock market experiences a positive revaluation at time 0^- . Panel C2b suggests that, after the revaluation, the stock excess return relative to bond from t to $t+1$, $xr_t^{stock} = r_t^{stock} - r_t$, is initially positive but turns negative after a few quarters. This occurs because agents underestimate dividends in the short term but overestimate them in the long term, as shown in Figure C1b. This pattern of positive stock excess returns turning negative is consistent with the classic [Kindleberger \(1978\)](#) narrative of crises and many real-world episodes, in which technological or financial innovations lead to booms and crashes in asset markets.

Table C1 summarizes the key moments of the impulse responses to a persistent productivity shock, with the rational expectations equilibrium in column (1) and shallow thinking equilibrium in (2). Quantitatively, shallow thinking leads to amplified and more

Table C1: Moments of RBC impulse responses

	(1) REE	(2) ST	(3) ST firms	(4) ST households
<i>Shock (same for all columns)</i>				
Half life of productivity shock Z_t	33	33	33	33
Shock size $dZ_0/\bar{Z}$	1.0%	1.0%	1.0%	1.0%
<i>Macroeconomy</i>				
Half life of Y_t	49	69	71	66
Quarter of peak output Y_t (hump shape)	0	9	3	11
Peak output $dY_t/\bar{Y}$	1.12%	1.28%	1.22%	1.98%
Peak hours $dN_t/\bar{N}$	0.18%	0.36%	0.32%	1.15%
Peak investment $dI_t/\bar{I}$	3.54%	6.2%	5.66%	17.68%
Peak consumption $dC_t/\bar{C}$	0.8%	0.83%	0.79%	1.33%
<i>Asset prices</i>				
Peak stock price $dP_t/\bar{P}$	0.85%	2.55%	2.59%	2.59%
Quarter of excess return xr_t^{stock} turning negative	$xr_t^{stock} = 0$ always	12	9	$xr_t^{stock} < 0$ always

Notes: Four columns correspond to the rational expectations equilibrium (REE), the shallow thinking equilibrium (ST), a model in which firms are shallow but households are rational, and a model in which households are shallow but firms are rational. In the latter two cases which feature belief disagreement, to illustrate the stock market dynamics, we simply assume that the stock is priced by the shallow agents.

persistent responses in labor hours, investment and the stock price relative to the rational expectations equilibrium. Qualitatively, it produces hump-shaped responses in consumption and output, as well as a stock market boom and crash, which are absent under rational expectations. Columns (3) and (4) further examine cases where only firms or households exhibit shallow thinking, suggesting that both parties' beliefs are important.

D Appendix to the New Keynesian Model

D.1 The Infinite-Horizon New Keynesian Model

Firms. There is a continuum of firms indexed by $j \in [0, 1]$ in this economy subject to Calvo price rigidity. Each firm chooses labor demand N_{jt}^s , pays dividend DIV_{jt} , and sets price P_{jt}^* when it can, taking as given the aggregate inflation rate π_t , the real wage W_t , and the aggregate demand C_t . They agree on the steady state of the economy but may have heterogeneous beliefs about the economy's response to shocks.

Each firm produces a differentiated good, using the same constant-returns production technology using labor hours $Y_{jt} = N_{jt}^d$, which together forms a bundle with constant elasticity ε of substitution (CES) that the households consume. At the steady state, they

each charge a markup $\mu \equiv \frac{\varepsilon}{\varepsilon-1}$. The log-linearized real dividend and aggregate labor demand are

$$div_t = c_t - \frac{1}{\mu-1} w_t \quad (D1)$$

$$n_t^d = c_t \quad (D2)$$

since the price dispersion only introduces second-order changes as in Galí (2015).

Each firm resets its price with independent probability $1 - \theta$ in any period and fulfills its demand period by period. When considering its reset price P_{jt}^* , each firm maximizes its discounted sum of profits

$$\max_{P_{jt}^*} \sum_{k=0}^{\infty} \theta^k \mathbb{E}_{jt} \left[\beta^k \frac{C_{t+k}^{-\sigma}}{C_t^{-\sigma}} \frac{P_{jt}^* - W_{t+k} P_{t+k}}{P_{t+k}} Y_{j,t+k|t} \right]$$

where $\beta^k \frac{C_{t+k}^{-\sigma}}{C_t^{-\sigma}}$ is the discount factor and $W_{t+k} P_{t+k}$ is the nominal marginal cost, subject to the sequence of demand constraints

$$Y_{j,t+k|t} = \left(\frac{P_{jt}^*}{P_{t+k}} \right)^{-\varepsilon} Y_{t+k}$$

The first-order condition is

$$0 = \sum_{k=0}^{\infty} \theta^k \mathbb{E}_{jt} \left[\beta^k \frac{C_{t+k}^{-\sigma}}{C_t^{-\sigma}} \frac{(1-\varepsilon) P_{jt}^* + \varepsilon W_{t+k} P_{t+k}}{P_{t+k}} Y_{j,t+k|t} \right]$$

and thus

$$p_{jt}^* = p_{t-1} + (1-\beta\theta) \sum_{k=0}^{\infty} (\beta\theta)^k \mathbb{E}_{jt} \left[\sum_{l=0}^k \pi_{t+l} + w_{t+k} \right]$$

Aggregate inflation emerges from the pricing behavior of the $(1-\theta)$ share of resetting firms as $\pi_t = (1-\theta)(p_{jt}^* - p_{t-1})$. Following the tradition, we consider a cost-push shock ϵ_t^π for inflation

$$\pi_t = \theta\kappa w_t + (1-\theta)\pi_t + \theta\kappa \sum_{k=1}^{\infty} (\beta\theta)^k \bar{\mathbb{E}}_t[w_{t+k}] + (1-\theta) \sum_{k=1}^{\infty} (\beta\theta)^k \bar{\mathbb{E}}_t[\pi_{t+k}] + \epsilon_t^\pi \quad (D3)$$

with $\kappa \equiv \frac{(1-\theta)(1-\beta\theta)}{\theta}$ capturing the slope of the Phillips curve and $\bar{\mathbb{E}}_t[\cdot]$ being the average

expectations. Importantly, we do not move π_t on the right-hand side to the left. We intentionally preserve the dependence of π_t on itself, which encapsulates the within-period complementarity in individual price setting as each firm takes aggregate inflation as given.

Households. There is a continuum of households who live infinitely indexed by $h \in [0, 1]$. Each household chooses consumption C_{ht} and labor supply N_{ht}^s , taking as given the gross nominal interest rates R_{t-1} , the inflation rate $\pi_t \equiv P_t/P_{t-1} - 1$, the real wage W_t , and the real dividend DIV_t . They agree on the steady state of the economy but may have heterogeneous beliefs about the economy's response to shocks.

They each maximize

$$\max_{\{C_{ht}, N_{ht}^s\}_{t \geq 0}} \mathbb{E}_{h,t=0} \sum_{t=0}^{\infty} \beta^t \left(\frac{C_{ht}^{1-\sigma} - 1}{1-\sigma} - \frac{(N_{ht}^s)^{1+\nu}}{1+\nu} \right)$$

subject to the budget constraint

$$C_{ht} + A_{ht} = \frac{R_{t-1}}{1 + \pi_t} A_{h,t-1} + W_t N_{ht}^s + DIV_t$$

where C_{ht} is a CES bundle of goods in the economy, A_{ht} is the period- t saving.

We derive the log-linearized aggregate consumption and labor supply functions as follows. The log-linearized life-time budget constraint is

$$\mathbb{E}_{ht} \sum_{k=0}^{\infty} \beta^k \left(\underbrace{c_{h,t+k} - \frac{\overline{WN}}{\overline{C}}}_{\mu^{-1}} (w_{t+k} + n_{h,t+k}^s) - \underbrace{\frac{\overline{DIV}}{\overline{C}}}_{1-\mu^{-1}} div_{t+k} \right) - \beta^{-1} a_{h,t-1} = 0$$

where the lower-case variables denote the log deviation from the corresponding steady-state values, and $\mu = \frac{\overline{C}}{\overline{WN}}$ denotes the ratio of consumption to labor income at the steady state (which equals firms' steady state markup μ).

Log-linearizing the consumption-labor FOC $W_t C_{ht}^{-\sigma} = (N_{ht}^s)^{\nu}$ yields

$$n_{ht}^s = \frac{1}{\nu} (w_t - \sigma c_{ht})$$

Plugging this into the budget constraint gives

$$\begin{aligned}\mathbb{E}_{ht} \sum_{k=0}^{\infty} \beta^k \left((1 + \mu^{-1} \sigma \nu^{-1}) c_{h,t+k} - \mu^{-1} (1 + \nu^{-1}) w_{t+k} - (1 - \mu^{-1}) div_{t+k} \right) - \beta^{-1} a_{h,t-1} &= 0 \\ \mathbb{E}_{ht} \sum_{k=0}^{\infty} \beta^k \left(c_{h,t+k} - \frac{\mu^{-1} (1 + \nu)}{\mu^{-1} \sigma + \nu} w_{t+k} - \frac{(1 - \mu^{-1}) \nu}{\mu^{-1} \sigma + \nu} div_{t+k} \right) - \frac{\beta^{-1} \nu}{\mu^{-1} \sigma + \nu} a_{h,t-1} &= 0\end{aligned}$$

Log-linearizing the Euler condition $C_{ht}^{-\sigma} = \mathbb{E}_{ht} \left[\frac{\beta R_t}{1 + \pi_{t+1}} C_{h,t+1}^{-\sigma} \right]$ gives

$$c_{ht} = \mathbb{E}_{ht} \left(c_{h,t+1} - \sigma^{-1} (i_t - \pi_{t+1}) \right)$$

Combining this with the budget constraint to substitute $c_{h,t+k}$ gives rise to

$$\begin{aligned}\mathbb{E}_{ht} \sum_{k=0}^{\infty} \beta^k \left(c_{ht} + \sum_{l=0}^{k-1} \sigma^{-1} (i_{t+l} - \pi_{t+l+1}) - \frac{\mu^{-1} (1 + \nu)}{\mu^{-1} \sigma + \nu} w_{t+k} - \frac{(1 - \mu^{-1}) \nu}{\mu^{-1} \sigma + \nu} div_{t+k} \right) - \frac{\beta^{-1} \nu}{\mu^{-1} \sigma + \nu} a_{h,t-1} &= 0 \\ \frac{1}{1 - \beta} c_{ht} - \sum_{k=0}^{\infty} \beta^k \mathbb{E}_{ht} \left(\frac{(1 + \nu)}{\sigma + \mu \nu} w_{t+k} + \frac{(\mu - 1) \nu}{\sigma + \mu \nu} div_{t+k} \right) + \frac{\sigma^{-1} \beta}{1 - \beta} \sum_{k=0}^{\infty} \beta^k \mathbb{E}_{ht} (i_{t+k} - \pi_{t+k+1}) - \frac{\beta^{-1} \mu \nu}{\sigma + \mu \nu} a_{h,t-1} &= 0\end{aligned}$$

which leads to the aggregate consumption function, once aggregated across all households

$$\begin{aligned}c_t &= -\sigma^{-1} \beta i_t - \sigma^{-1} \beta \sum_{k=1}^{\infty} \beta^k \bar{\mathbb{E}}_t [i_{t+k}] + \sigma^{-1} \sum_{k=1}^{\infty} \beta^k \bar{\mathbb{E}}_t [\pi_{t+k}] + (1 - \beta) \left[\frac{(\mu - 1) \nu}{\sigma + \mu \nu} div_t + \frac{(1 + \nu)}{\sigma + \mu \nu} w_t \right] \\ &+ (1 - \beta) \sum_{k=1}^{\infty} \beta^k \bar{\mathbb{E}}_t \left[\frac{(\mu - 1) \nu}{\sigma + \mu \nu} div_{t+k} + \frac{(1 + \nu)}{\sigma + \mu \nu} w_{t+k} \right]\end{aligned}\tag{D4}$$

Using the consumption-labor FOC $n_{ht}^s = \frac{1}{\nu} (w_t - \sigma c_{ht})$ again, we get the aggregate labor supply function

$$\begin{aligned}n_t^s &= \nu^{-1} \beta i_t + \nu^{-1} \beta \sum_{k=1}^{\infty} \beta^k \bar{\mathbb{E}}_t [i_{t+k}] - \nu^{-1} \sum_{k=1}^{\infty} \beta^k \bar{\mathbb{E}}_t [\pi_{t+k}] - (1 - \beta) \frac{(\mu - 1) \sigma}{\sigma + \mu \nu} div_t + \nu^{-1} \left(1 - \sigma \frac{(1 - \beta) (1 + \nu)}{\sigma + \mu \nu} \right) w_t \\ &- (1 - \beta) \sum_{k=1}^{\infty} \beta^k \bar{\mathbb{E}}_t \left[\frac{(\mu - 1) \sigma}{\sigma + \mu \nu} div_{t+k} + \frac{(1 + \nu) \sigma \nu^{-1}}{\sigma + \mu \nu} w_{t+k} \right]\end{aligned}\tag{D5}$$

where $\bar{\mathbb{E}}_t [\cdot]$ denotes the average expectations.

Central bank. The central bank follows a Taylor rule with a monetary policy shock ϵ_t^i ,

$$i_t = \phi \pi_t + \epsilon_t^i \quad (\text{D6})$$

Labor market. Last, to close the model, the wage arises by equilibrating labor supply and demand

$$n_t^s = n_t^d \quad (\text{D7})$$

Equilibrium. We study a *temporary equilibrium* in which agents maximize their utilities, taking as given the average expectations $\bar{\mathbb{E}}_t[\cdot]$, and markets clear. In this New Keynesian model, equations (D1-D7) characterizes the equilibrium given the average expectations.

D.2 3-Variable Representation

A popular way of representing the textbook New Keynesian model is to use three variables—the interest rate i_1 , inflation π_1 , and consumption c_1 (i.e., output). To derive this representation from (1-7), we first solve for the other variables $V_1^{other} = (div_1, \hat{n}_1^d, \hat{n}_1^s, w_1)$ in (8) in terms of (i_1, π_1, c_1) . Substituting V_1^{other} into the consumption function (4) yields $c_1 = -\sigma i_1$. Then, substituting V_1^{other} into the Phillips curve (3) and using $c_1 = -\sigma i_1$ to eliminate i_1 , we obtain $\pi_1 = \kappa(\sigma + \nu)c_1 + \theta^{-1}\epsilon_1^\pi$, which expresses inflation solely as a function of output c_1 . The Taylor rule (6) is kept unchanged.

$$\begin{pmatrix} i_1 \\ \pi_1 \\ c_1 \end{pmatrix} = \begin{pmatrix} 0 & \phi & 0 \\ 0 & 0 & \kappa(\sigma + \nu) \\ -\sigma & 0 & 0 \end{pmatrix} \begin{pmatrix} i_1 \\ \pi_1 \\ c_1 \end{pmatrix} + \begin{pmatrix} \epsilon_1^i \\ \theta^{-1}\epsilon_1^\pi \\ 0 \end{pmatrix} \quad (\text{D8})$$

This representation expresses inflation π_1 as a function of consumption c_1 , rather than in the best response form (3), by canceling the pricing complementarity term π_1 on the right-hand side with the left-hand side in (3) and implicitly substituting the equilibrium value of the wage, thereby failing to reflect firms' best responses. While (D8) shares the same fixed point in terms of (i_1, π_1, c_1) as our representation (8), it is not a system of best responses and involves a different iterative process to arrive at the fixed point. Furthermore, this 3-variable representation is still incomplete in that it does not take a stance on how other variables, e.g., wage w_1 and dividend div_1 , depend on these three variables.

D.3 Belief Misreaction and Strength of GE Feedback

We offer a synthesis of belief misreaction under shallow thinking. To start with, consider a simple model with a single general equilibrium (GE) loop. Beliefs about a variable underreact (or overreact) to its own shocks if the single GE loop amplifies (or offsets) the direct effect of these shocks, as shallow agents underappreciate the GE loop, in line with existing theories reviewed by [Angeletos and Lian \(2023a\)](#). Additionally, our theory suggests that beliefs about a variable underreact to cross shocks (e.g., inflation in response to cost-push shocks), because shallow agents underperceive shock propagation. This latter prediction extends to more complex models with multiple GE loops.

In models with multiple GE loops, such as our workhorse macroeconomic models, whether beliefs over- or underreact to own shocks depends jointly on the strength and length of these GE loops. For example, consider the inflation response to a cost-push shock ϵ_1^π . Under our calibration, shallow agents perceive the cost-push shock ϵ_1^π as amplified, even though it is actually offset, i.e., $\frac{\partial \mathbb{E}_0[\pi_1]}{\partial \epsilon_1^\pi} > 1 > \frac{\partial \pi_1}{\partial \epsilon_1^\pi}$. Shallow thinking flips the sign of the perceived net GE effect, because shallow agents underappreciate the long, strong offset loop involving the monetary policy reaction. The order of operations is key because GE feedbacks of varying length are dampened differently, due to the Jensen's inequality. The implication could differ if one ignored length by first collapsing multiple GE loops in the model into a single net effect (which, in this case, is offset) and then applying dampening.

To illustrate, we study inflation expectations, varying the strength of the offset loop by adjusting the Taylor rule coefficient ϕ , as depicted in Figure [D1](#). If, instead of $\phi = 1.5$, ϕ takes on an intermediate value, e.g., $\phi = 1$, the long offset loop is too weak to turn the true net effect into offset. In that case, the true net effect is amplification, but shallow agents perceive even *more* amplification, i.e., $\frac{\partial \mathbb{E}_0[\pi_1]}{\partial \epsilon_1^\pi} > \frac{\partial \pi_1}{\partial \epsilon_1^\pi} > 1$, as they overweigh the short amplification loop relative to the long offset loop. Again, if one naively collapsed multiple GE loops into a net amplification effect and dampened it, the perceived amplification would be less rather than more.

With an even lower Taylor rule coefficient ϕ , e.g., $\phi = 0$ in a case where the policy rate is constrained at the zero lower bound (ZLB) in response to a deflationary shock,³⁶ the long offset loop is weak or non-existent, leaving the short amplification loop as the

³⁶The Taylor principle requires $\phi > 1$ for determinacy of the infinite-horizon New Keynesian model under rational expectations. In our 2-period setting, determinacy is not a concern. With persistent shocks, shallow thinking beliefs are uniquely defined by a formula similar to [\(10\)](#), which helps select an equilibrium. For further discussion on determinacy, which is not our focus, see [Farhi and Werning \(2019\)](#), [Gabaix \(2020\)](#) and [Angeletos and Lian \(2023b\)](#).

dominant force. That leads us towards a simple case with only one GE loop that amplifies the direct effect, which we began this discussion with. As a result, shallow agents perceive less amplification on net than the true effect, i.e., $\frac{\partial \pi_1}{\partial \epsilon_1^\pi} > \frac{\partial \bar{\mathbb{E}}_0[\pi_1]}{\partial \epsilon_1^\pi} > 1$.

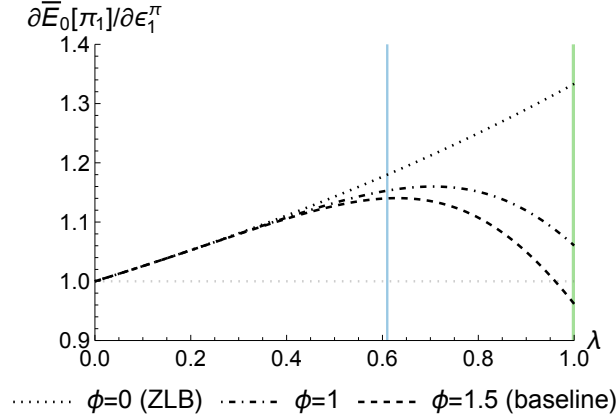


Figure D1: Inflation expectation $\bar{\mathbb{E}}_0[\pi_1]$ in response to cost-push news shocks ϵ_1^π , under varying Taylor rule coefficient ϕ

Notes: This figure plots the average interest expectation $\bar{\mathbb{E}}_0[i_1]$ (relative to the size of the shock) as a function of the shallow thinking parameter λ , in response to news about a cost-push shock ϵ_1^π , under different values of the Taylor rule coefficient ϕ . The blue line indicates our calibration of λ , while the green line represents rational expectations ($\lambda = 1$).

D.4 Additional Results of Bond Returns

Bond excess returns with transitory news shocks. Figure D2 shows the loadings of bond excess returns and predictors on transitory news shocks, under $\lambda = 0, 1$, complementing Figure 7b.

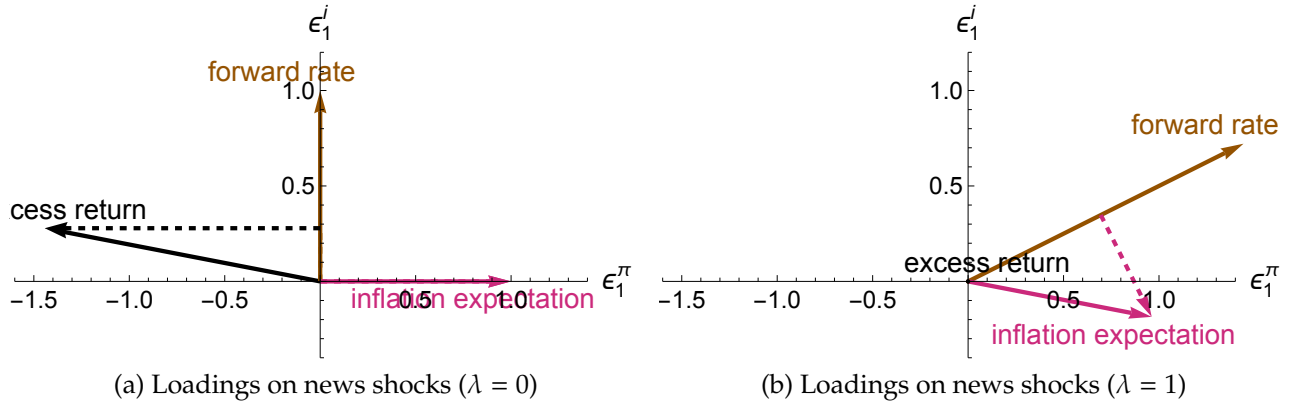


Figure D2: Predictability of bond excess returns, with transitory news shocks (cont'd)

Notes: These panels plot the loadings of bond excess returns and predictors on transitory news shocks, under $\lambda = 0, 1$, complementing Figure 7b. In the case of $\lambda = 0$, the forward rate and inflation expectation point along the y- and x-axes, respectively. In the case of $\lambda = 1$, the excess return is 0.

Sensitivity of long-term rates to short-term rates with persistent shocks. We consider an economy impacted by persistent monetary policy shocks $\epsilon_t^i = \rho^t \epsilon^i$ and cost-push shocks $\epsilon_t^\pi = \rho^t \epsilon^\pi$. Suppose that in each period, the two shocks have variances σ_i^2 and σ_π^2 , and are independently distributed.

In this environment, the yield of a T -period bond, $y_t^{(T)}$, is determined by the expectations hypothesis as

$$y_t^{(T)} = \frac{i_t + \sum_{k=1}^{T-1} \bar{\mathbb{E}}[i_{t+k}]}{T} \quad (\text{D9})$$

We consider a univariate regression of the long-term yield $y_t^{(T)}$ on the short-term interest rate i_t à la [Hanson, Lucca and Wright \(2021\)](#)

$$y_t^{(T)} = \beta^{OLS} i_t + \alpha + \epsilon_0 \quad (\text{D10})$$

Since the economy is affected by two different shocks, the univariate regression coefficient β^{OLS} depends on the mix of these shocks. To shed light on this, we express the responses of interest rate expectation $\bar{\mathbb{E}}[i_{t+k}]$ and equilibrium interest rate i_t to shocks as

$$\begin{aligned} \bar{\mathbb{E}}[i_{t+k}] &= \rho^k \gamma_i^{exp} \epsilon_t^i + \rho^k \gamma_\pi^{exp} \epsilon_t^\pi, \quad \forall k \geq 1 \\ i_t &= \gamma_i^{ST} \epsilon_t^i + \gamma_\pi^{ST} \epsilon_t^\pi \end{aligned}$$

where the γ coefficients are implicit functions of model parameters including λ , and ρ^k stems from the fact that beliefs are mean-reverting at the same rate ρ . The γ^{exp} coefficients agree with γ^{ST} coefficients under $\lambda = 1$ (rational expectations). We denote the relative importance of monetary policy shocks in driving the short-term interest rate as $\zeta_i \equiv \frac{(\gamma_i^{ST})^2 \sigma_i^2}{(\gamma_i^{ST})^2 \sigma_i^2 + (\gamma_\pi^{ST})^2 \sigma_\pi^2} \in [0, 1]$, and the relative importance of cost-push shocks as $1 - \zeta_i$.

According to (D9), the long-term yield follows

$$y_t^{(T)} = T^{-1} \left(\gamma_i^{ST} + \rho \frac{1 - \rho^{T-1}}{1 - \rho} \gamma_i^{exp} \right) \epsilon_t^i + T^{-1} \left(\gamma_\pi^{ST} + \rho \frac{1 - \rho^{T-1}}{1 - \rho} \gamma_\pi^{exp} \right) \epsilon_\pi^i$$

Therefore, the univariate regression coefficient β^{OLS} from (D10) is

$$\beta^{OLS} = \frac{\text{Cov}(y_t^{(T)}, i_t)}{\text{Var}(i_t)} = T^{-1} \underbrace{\left(1 + \rho \frac{1 - \rho^{T-1}}{1 - \rho} \frac{\gamma_i^{exp}}{\gamma_i^{ST}} \right)}_{\equiv \beta_i^{IV}} \zeta_i + T^{-1} \underbrace{\left(1 + \rho \frac{1 - \rho^{T-1}}{1 - \rho} \frac{\gamma_\pi^{exp}}{\gamma_\pi^{ST}} \right)}_{\equiv \beta_\pi^{IV}} (1 - \zeta_i) \quad (\text{D11})$$

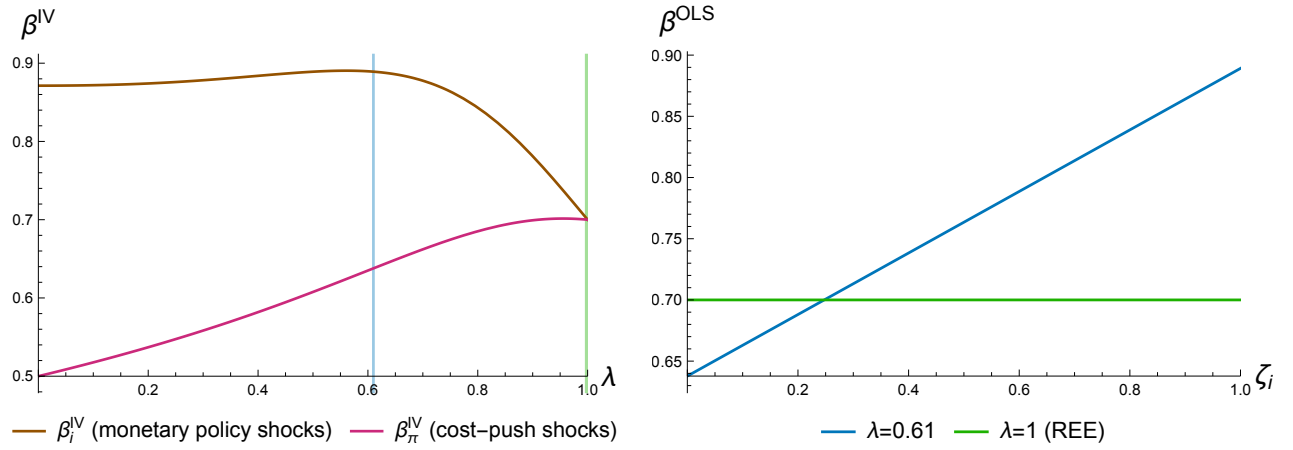
which linearly interpolates between β_i^{IV} , the conditional sensitivity under monetary shocks, and β_π^{IV} , the conditional sensitivity under cost-push shocks, based on the relative importance ζ_i .

As $\gamma_i^{exp} = \gamma_i^{ST}$ and $\gamma_\pi^{exp} = \gamma_\pi^{ST}$ under rational expectations, we have

$$\beta^{OLS,REE} \equiv T^{-1} \frac{1 - \rho^T}{1 - \rho} \quad (\text{D12})$$

which is independent of the shock mix ζ_i .

Figure D3 plots these univariate regression coefficients β_i^{IV} , β_π^{IV} and β^{OLS} , for a 2-period bond ($T = 2$) under a mild shock persistence $\rho = 0.4$. Panel D3a suggests that the conditional sensitivity to monetary policy shocks β_i^{IV} (in brown) is higher than the conditional sensitivity to cost-push shocks β_π^{IV} (in purple), as long as $\lambda < 1$. Consequently, under our calibrated λ , the unconditional sensitivity β^{OLS} increases in the relative importance of the monetary policy shocks ζ_i , as shown in panel D3b.



(a) Conditional sensitivities β_i^{IV} and β_π^{IV} as functions of shallow thinking parameter λ (b) Unconditional sensitivity β^{OLS} as a function of relative importance of monetary policy shocks ζ_i

Figure D3: Univariate regression coefficients of long-term rates on short-term rates

Notes: Panel (a) plots the conditional sensitivities $\beta_i^{IV}, \beta_\pi^{IV}$ of the long-term rate with $T = 2$ on the short-term rate, conditional on monetary policy shocks (in brown) and cost-push shocks (in purple) respectively, under $\rho = 0.4$, as functions of the shallow thinking parameter λ .

Panel (b) plots the unconditional sensitivity β^{OLS} of the long-term rate with $T = 2$, under $\rho = 0.4$, as a function of the relative importance of monetary policy shocks ζ_i . The blue line indicates our calibrated λ , while the green line represents the rational expectations equilibrium (REE) with $\lambda = 1$.

D.5 Additional Results of Persistent Shocks

The effect of more persistent cost-push shocks on equilibrium inflation. Figure 38 suggests that under our calibrated λ , a more persistent cost-push shock leads to lower inflation, contrary to the rational expectations equilibrium. Figure D4 shows the equilibrium inflation response to a persistent cost-push shock, varying the shallow thinking parameter λ and the persistence of the shock ρ . Panel D4a suggests that the inflation response increases in ρ when λ is low and decreases in ρ when λ is high. Panel D4b shows, in the gray meshed area, the combination of (λ, ρ) values under which a more persistent cost-push shock leads to lower inflation than a purely transitory shock, i.e., $\frac{\partial \pi_t}{\partial \epsilon_t^\pi} < \frac{\partial \pi_t}{\partial \epsilon_t^\pi} |_{\rho=0}$. In the case of purely transitory shock ($\rho = 0$), the equilibrium is independent of λ , as agents observe all variables in the same period. The vertical blue line (our calibrated λ) is completely outside the meshed region, whereas the vertical green line (rational expectations equilibrium) is entirely within it.

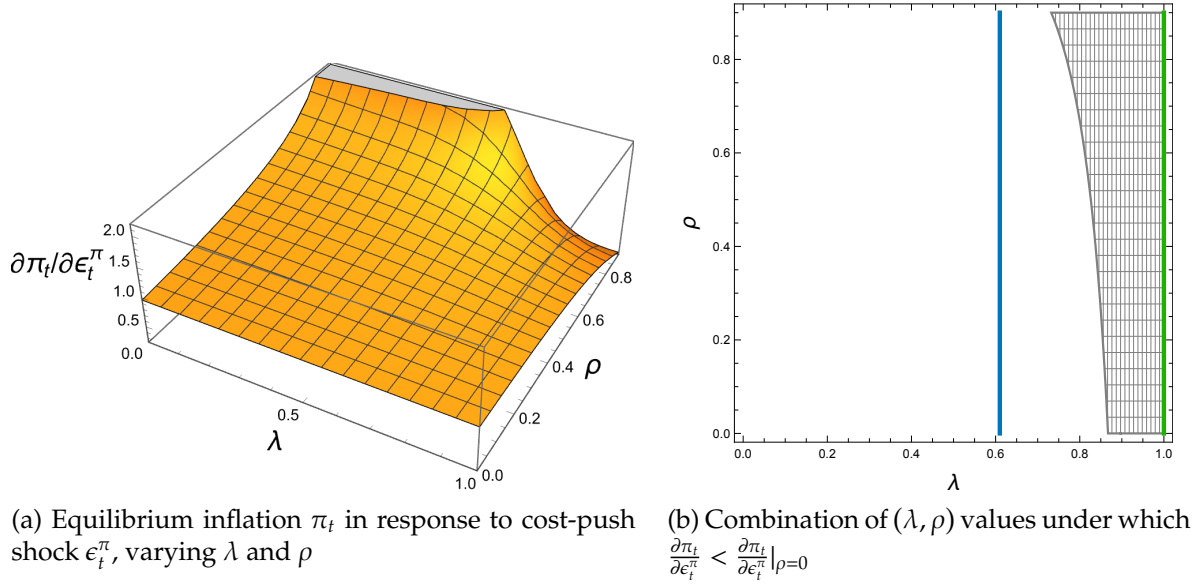


Figure D4: Inflation response to persistent cost-push shock ϵ_t^π

Notes: Panel (a) plots the equilibrium inflation π_t (relative to the size of the shock) in response to a persistent cost-push shock $\epsilon_t^\pi = \rho^t \epsilon^\pi$, as a function of the shallow thinking parameter λ and the shock persistence ρ . Panel (b) shows, in the gray meshed area, the (λ, ρ) combinations under which a persistent cost-push shock leads to lower inflation than a purely transitory shock, i.e., $\frac{\partial \pi_t}{\partial \epsilon_t^\pi} < \frac{\partial \pi_t}{\partial \epsilon_t^\pi} \big|_{\rho=0}$. In the case of purely transitory shock ($\rho = 0$), the equilibrium is independent of λ , as agents observe all variables in the same period. The vertical line in blue indicates our calibration of λ , which is completely outside the gray area. In contrast, the vertical green line representing the rational expectations equilibrium lies entirely within the area.

Persistent monetary policy shocks.

Proposition D1. (Persistent monetary policy shock) *The rational-expectations equilibrium (REE) response to a persistent monetary policy shock $\epsilon_t^i = \rho^t \epsilon^i$ features*

$$i_t^{REE} = \left[1 + \frac{\frac{\phi \theta \kappa}{1-\beta \theta \rho} \frac{\sigma^{-1}(v+\sigma)}{1-\rho}}{1 - \frac{(1-\theta)+\rho \theta \kappa \frac{\sigma^{-1}(v+\sigma)}{1-\rho}}{1-\beta \theta \rho}} \right]^{-1} \epsilon_t^i \quad (\text{D13})$$

$$\pi_t^{REE} = - \frac{\frac{\theta \kappa}{1-\beta \theta \rho} \frac{\sigma^{-1}(v+\sigma)}{1-\rho}}{1 - \frac{(1-\theta)+\rho \theta \kappa \frac{\sigma^{-1}(v+\sigma)}{1-\rho}}{1-\beta \theta \rho}} i_t^{REE} \quad (\text{D14})$$

whereas the average expectations under shallow thinking are

$$\bar{\mathbb{E}}[i_t] = \left[1 + \frac{\frac{\lambda^4 \phi \theta \kappa K(\lambda, \rho)}{1 - \beta \theta \rho}}{1 - \frac{\lambda(1 - \theta) + \lambda^3 \rho \theta \kappa K(\lambda, \rho)}{1 - \beta \theta \rho}} \right]^{-1} \epsilon_t^i \quad (\text{D15})$$

$$\bar{\mathbb{E}}[\pi_t] = - \frac{\frac{\lambda^3 \theta \kappa K(\lambda, \rho)}{1 - \beta \theta \rho}}{1 - \frac{\lambda(1 - \theta) + \lambda^3 \rho \theta \kappa K(\lambda, \rho)}{1 - \beta \theta \rho}} \bar{\mathbb{E}}[i_t] \quad (\text{D16})$$

with $K(\lambda, \rho)$ increasing in λ and ρ under our calibration and $K(1, \rho) = \frac{\sigma^{-1}(\nu + \sigma)}{1 - \rho}$.

This proposition nests Proposition 5 with $\rho = 0$. Figure D5 shows the responses of interest rate expectations and equilibrium interest rate to persistent cost-push shocks and monetary policy shocks.

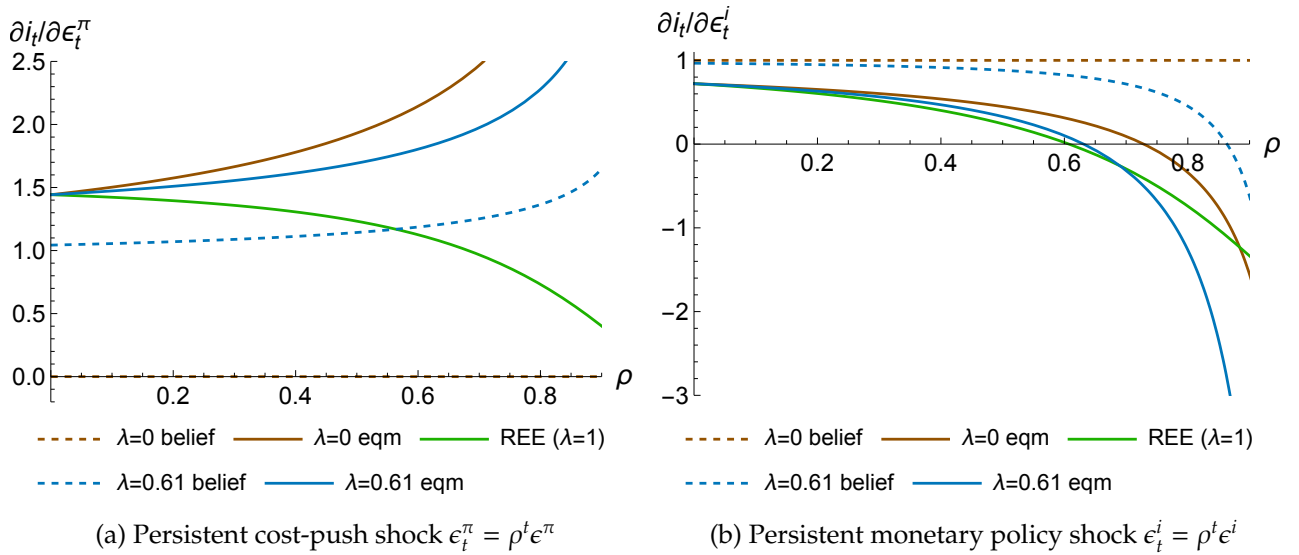


Figure D5: Interest rate i_t in response to persistent shocks

Notes: These panels plot the responses of the interest rate to persistent cost-push shocks ϵ_t^π and monetary policy shocks ϵ_t^i , parallel to Figure 9b. Under our calibration, the average interest rate expectation is higher than its equilibrium value in response to monetary policy shocks, but is lower in response to cost-push shocks, regardless of ρ .

D.6 Proofs

Proof of Proposition 1. It follows directly from Section 2.1. We explicitly write down the system as

$$\underbrace{\begin{pmatrix} i_1 \\ \pi_1 \\ div_1 \\ \hat{n}_1^d \\ c_1 \\ \hat{n}_1^s \\ w_1 \end{pmatrix}}_{V_1} = \underbrace{\begin{pmatrix} 0 & \phi & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 - \theta & 0 & 0 & 0 & 0 & \theta\kappa \\ 0 & 0 & 0 & 0 & 1 & 0 & -\frac{1}{\mu-1} \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ -\sigma^{-1}\beta & 0 & \frac{(1-\beta)(\mu-1)\nu}{\sigma+\mu\nu} & 0 & 0 & 0 & \frac{(1-\beta)(1+\nu)}{\sigma+\mu\nu} \\ \nu^{-1}\beta & 0 & -\frac{(1-\beta)(\mu-1)\sigma}{\sigma+\mu\nu} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & e^{s-1} & 0 & -e^{s-1} & 0 \end{pmatrix}}_M \underbrace{\begin{pmatrix} i_1 \\ \pi_1 \\ div_1 \\ \hat{n}_1^d \\ c_1 \\ \hat{n}_1^s \\ w_1 \end{pmatrix}}_{V_1} + \underbrace{\begin{pmatrix} \epsilon_1^i \\ \epsilon_1^\pi \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}}_{S_1} \quad (D17)$$

with $e^s = \nu^{-1} \left[1 - \sigma \frac{(1-\beta)(1+\nu)}{\sigma+\mu\nu} \right]$. $\square$

Proof of Proposition 2. Assumptions 1 and 2 imply that the average beliefs are

$$\begin{aligned} \bar{\mathbb{E}}_0[V_1] &\equiv \sum_{d=1}^{\infty} \mathbb{P}(d=n) \mathbb{E}_0^d[V_1] \\ &= \sum_{d=1}^{\infty} \mathbb{P}(d=n) \sum_{n=1}^d M^{n-1} S_1 = \sum_{n=1}^{\infty} \mathbb{P}(d \geq n) M^{n-1} S_1 = \sum_{n=1}^{\infty} \lambda^{n-1} M^{n-1} S_1 \end{aligned}$$

which can be recast as (14). $\square$

Proof of Proposition 3. (15) follows from Assumptions 2 and 3 and Definition 2. Since the causal distance D_{vs} is bounded above, an ordinary least squares estimation identifies a negative slope γ . Further, the conditional expectation (15) minimizes the mean squared error of $\mathbb{E}^{population}(1_{nvs} - h(D_{vs}))$ among all possible predictor $h(D_{vs})$. Since it is a case of the exponential family $b_1 \cdot b_2^{D_{vs}-1} + b_0$, we can exactly identify λ with b_2 . $\square$

Proofs of Propositions 4 and 5. Directly solving (1-7) in response to the cost-push shock ϵ_1^π and the monetary policy shock ϵ_1^i yields the equilibrium. Solving the fixed point of (1-7) with cross-variable relations dampened by λ as in (14) gives the average expectations.

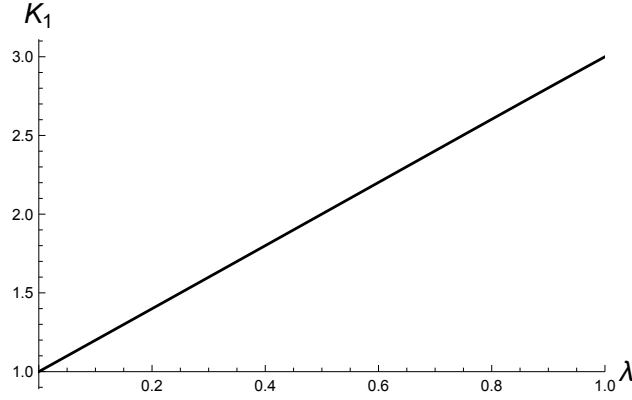


Figure D6: $K_1(\lambda)$ under our calibration

The dampening of Keynesian cross is

$$K_1(\lambda) = \frac{\frac{\beta(\sigma + \lambda\nu)}{\sigma \left(1 - \frac{(1-\beta)(1+\nu)\sigma}{(\mu\nu + \sigma)}\right)}}{1 - \lambda^2 \frac{(1-\beta)(\mu-1)\nu}{(\mu\nu + \sigma)} - \lambda^3 \frac{(1-\beta)\nu \left(1 + \nu - \sigma - \lambda\nu + \lambda(\mu-1) \frac{(1-\beta)(1+\nu)\sigma}{(\mu\nu + \sigma)}\right)}{(\mu\nu + \sigma) - (1-\beta)(1+\nu)\sigma}} \quad (\text{D18})$$

which depends on β, μ, σ, ν but no other parameters when $\lambda \in [0, 1)$ and obtains $\sigma^{-1}(\nu + \sigma)$ at $\lambda = 1$. Under our calibration, $K_1(\lambda)$ increases in λ as illustrated in Figure D6. $\square$

Proof of Proposition 6. It follows direction from Section 5.1 and Appendix D.1. $\square$

Proof of Proposition ??. The proof is the same as that of Proposition 2. $\square$

Proof of Propositions 7 and D1. Though the proposition only concerns a cost-push shock ϵ_t^π , we also consider a monetary policy shock ϵ_t^i in the proof, which does not add much complexity. This proof will nest the Proofs of Proposition 4 and 5 by setting $\rho = 0$.

We solve the average expectations using (??), which will nest the rational expectations equilibrium with $\lambda = 1$. We guess and verify that beliefs mean-revert at the same rate ρ .

Agents' actions (D1-D6) characterize $\{\bar{\mathbb{E}}[div_t], \bar{\mathbb{E}}[\hat{n}_t^d], \bar{\mathbb{E}}[\pi_t], \bar{\mathbb{E}}[c_t], \bar{\mathbb{E}}[\hat{n}_t^s], \bar{\mathbb{E}}[i_t]\}$ as

$$\bar{\mathbb{E}}[div_t] = \lambda \bar{\mathbb{E}}[c_t] - \lambda \frac{1}{\mu - 1} \bar{\mathbb{E}}[w_t] \quad (\text{D19})$$

$$\bar{\mathbb{E}}[\hat{n}_t^d] = \lambda \bar{\mathbb{E}}[c_t] \quad (\text{D20})$$

$$\bar{\mathbb{E}}[\pi_t] = \lambda \theta \kappa \frac{\bar{\mathbb{E}}[w_t]}{1 - \beta \theta \rho} + \lambda (1 - \theta) \frac{\bar{\mathbb{E}}[\pi_t]}{1 - \beta \theta \rho} + \epsilon_t^\pi \quad (\text{D21})$$

$$\bar{\mathbb{E}}[c_t] = -\lambda \sigma^{-1} \beta \left(\frac{\bar{\mathbb{E}}[i_t]}{1 - \beta \rho} - \frac{\rho \bar{\mathbb{E}}[\pi_t]}{1 - \beta \rho} \right) + \lambda (1 - \beta) \frac{(\mu - 1) \nu \bar{\mathbb{E}}[div_t]}{\sigma + \mu \nu} \frac{1}{1 - \beta \rho} + \lambda (1 - \beta) \frac{(1 + \nu) \bar{\mathbb{E}}[w_t]}{\sigma + \mu \nu} \frac{1}{1 - \beta \rho} \quad (\text{D22})$$

$$\bar{\mathbb{E}}[\hat{n}_t^s] = \lambda \nu^{-1} \beta \left(\frac{\bar{\mathbb{E}}[i_t]}{1 - \beta \rho} - \frac{\rho \bar{\mathbb{E}}[\pi_t]}{1 - \beta \rho} \right) - \lambda (1 - \beta) \frac{(\mu - 1) \sigma \bar{\mathbb{E}}[div_t]}{\sigma + \mu \nu} \frac{1}{1 - \beta \rho} \quad (\text{D23})$$

$$\bar{\mathbb{E}}[i_t] = \lambda \phi \bar{\mathbb{E}}[\pi_t] + \epsilon_t^i \quad (\text{D24})$$

Note that in constructing $\bar{\mathbb{E}}[\hat{n}_t^d]$ and $\bar{\mathbb{E}}[\hat{n}_t^s]$, we take away their dependence on the wage w from n^d and n^s in (D2, D5).

Last, the labor market clearing condition (D7) combined with the dependence of n^d and n^s on w from (D2, D5) determines $\bar{\mathbb{E}}[w_t]$

$$\nu^{-1} \left(1 - \sigma \frac{(1 - \beta)(1 + \nu)}{(1 - \beta \rho)(\sigma + \mu \nu)} \right) \bar{\mathbb{E}}[w_t] = \lambda \bar{\mathbb{E}}[\hat{n}_t^d] - \lambda \bar{\mathbb{E}}[\hat{n}_t^s] \quad (\text{D25})$$

Solving (D19-D25) gives the average expectations (40) and (41), which nest the rational expectations equilibrium (38) and (39) with $\lambda = 1$.

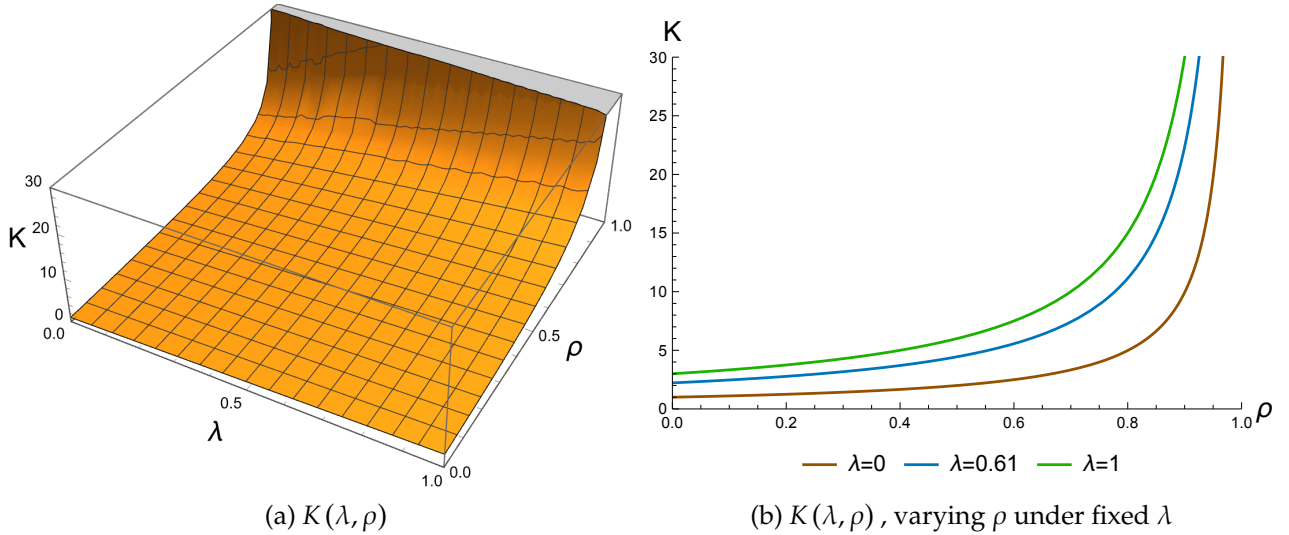


Figure D7: $K(\lambda, \rho)$ under our calibration

The dampening of Keynesian cross under persistent shocks is

$$K(\lambda, \rho) = \frac{\frac{\beta(\sigma + \lambda\nu)}{\sigma(1-\beta\rho)\left(1 - \frac{(1-\beta)(1+\nu)\sigma}{(1-\beta\rho)(\mu\nu + \sigma)}\right)}}{1 - \lambda^2 \frac{(1-\beta)(\mu-1)\nu}{(1-\beta\rho)(\mu\nu + \sigma)} - \lambda^3 \frac{(1-\beta)\nu(1+\nu-\sigma-\lambda\nu + \lambda(\mu-1)\frac{(1-\beta)(1+\nu)\sigma}{(1-\beta\rho)(\mu\nu + \sigma)})}{(1-\beta\rho)(\mu\nu + \sigma) - (1-\beta)(1+\nu)\sigma}} \quad (\text{D26})$$

which depends on $\rho, \beta, \mu, \sigma, \nu$ but no other parameters when $\lambda \in [0, 1)$ and obtains $\frac{\sigma^{-1}(\nu + \sigma)}{1-\rho}$ at $\lambda = 1$. Under our calibration, $K(\lambda, \rho)$ increases both in λ and ρ as shown in Figure D7. $\square$

E Appendix to the RBC Model

E.1 The RBC Model

Under our assumption that agents observe all variables in each period and have expectations formed once and for all after observing the shocks, i.e., $\mathbb{E}_t^d[v_s] \equiv \mathbb{E}^d[v_s]$ for $s > t$, we omit the time subscript on expectations for simplicity.

Firms. There is a continuum of firms with identical production technology and capital stock at the steady state but differing beliefs about impulse responses to shocks. To simplify notation, we consider a generic firm without firm-specific indexing, noting that once we aggregate across firms, the average expectations determine aggregate behavior under a first-order approximation around the steady state.

Firms maximize

$$\max_{\{N_t^d, I_t, Y_t\}} \mathbb{E} \sum_{t=0}^{\infty} \prod_{k=0}^{t-1} (1 + r_k)^{-1} DIV_t$$

subject to

$$DIV_t = Y_t - W_t N_t^d - I_t - \frac{\psi}{2} \left(\frac{I_t}{K_t} - \delta \right)^2 K_t \quad (\text{E1})$$

$$Y_t = Z_t \left(\frac{K_t}{\alpha} \right)^\alpha \left(\frac{N_t^d}{1-\gamma} \right)^{1-\gamma} \quad (\text{E2})$$

$$K_{t+1} = (1 - \delta) K_t + I_t \quad (\text{E3})$$

In each period, firms choose labor demand N_t^d to satisfy

$$Z_t \left(\frac{K_t}{\alpha} \right)^\alpha \left(\frac{N_t^d}{1-\gamma} \right)^{-\gamma} = W_t$$

$$N_t^d = (1-\gamma) \left(\frac{Z_t}{W_t} \right)^{\frac{1}{\gamma}} \left(\frac{K_t}{\alpha} \right)^{\frac{\alpha}{\gamma}} \quad (\text{E4})$$

We form the Lagrangian after plugging in the optimal labor demand as

$$\mathbb{E} \sum_{t=0}^{\infty} \prod_{k=0}^{t-1} R_k^{-1} \left[\gamma Z_t^{\frac{1}{\gamma}} W_t^{1-\frac{1}{\gamma}} \left(\frac{K_t}{\alpha} \right)^{\alpha/\gamma} - I_t - \frac{\psi}{2} \left(\frac{I_t}{K_t} - \delta \right)^2 K_t \right] + \prod_{k=0}^{t-1} R_k^{-1} q_t [(1-\delta) K_t + I_t - K_{t+1}]$$

Its FOC w.r.t. I_t is

$$- \left[1 + \psi \left(\frac{I_t}{K_t} - \delta \right) \right] + q_t = 0$$

which pins down the investment as

$$I_t = \left(\frac{q_t - 1}{\psi} + \delta \right) K_t \quad (\text{E5})$$

Its FOC w.r.t. K_{t+1} is

$$\mathbb{E} \left[Z_{t+1}^{\frac{1}{\gamma}} W_{t+1}^{1-\frac{1}{\gamma}} \left(\frac{K_{t+1}}{\alpha} \right)^{\frac{\alpha}{\gamma}-1} - \frac{\psi}{2} \left(\frac{I_{t+1}}{K_{t+1}} - \delta \right)^2 + \psi \left(\frac{I_{t+1}}{K_{t+1}} - \delta \right) \frac{I_{t+1}}{K_{t+1}} + q_{t+1} (1-\delta) \right] - (1+r_t) q_t = 0$$

and thus

$$q_t = (1+r_t)^{-1} \mathbb{E} \left[Z_{t+1}^{\frac{1}{\gamma}} W_{t+1}^{1-\frac{1}{\gamma}} \left(\frac{K_{t+1}}{\alpha} \right)^{\frac{\alpha}{\gamma}-1} - \frac{\psi}{2} \left(\frac{I_{t+1}}{K_{t+1}} - \delta \right)^2 + \psi \left(\frac{I_{t+1}}{K_{t+1}} - \delta \right) \frac{I_{t+1}}{K_{t+1}} + (1-\delta) q_{t+1} \right] \quad (\text{E6})$$

(E1-E6) implicitly characterize the dependence of output Y_t , investment (including adjustment cost) $I_t + \Psi_t$ and labor demand N_t^d on the past and current interest rates $\{r_s\}_{0 \leq s \leq t}$ and wages $\{W_s\}_{0 \leq s \leq t}$, as well as expectations of future interest rates $\{\mathbb{E}[r_s]\}_{s>t}$ and wages $\{\mathbb{E}[W_s]\}_{s>t}$. As we adopt a first-order approximation around the steady state, such implicit dependence is linear. As a result, once aggregated across all firms, the aggregate behavior depends on the average expectations of future interest rates $\{\bar{\mathbb{E}}[r_s]\}_{s>t}$ and wages

$$\{\bar{\mathbb{E}}[W_s]\}_{s>t}.$$

Households. The household side of the RBC economy is the same as that of the New Keynesian economy, except that they invest in a real bond as opposed to a nominal bond. Thus the consumption and labor supply functions follow from (D4, D5) by replacing $i_{t+k} - \pi_{t+k+1}$ with $\frac{r_{t+k} - \bar{r}}{1 + \bar{r}}$, where the normalization translates a level change of the real interest rate into a log change. Consumption and labor supply C_t, N_t^s depend on the interest rate, wage and dividend $\{r_t, W_t, DIV_t\}_{t \geq 0}$ as follows

$$\begin{aligned} \frac{C_t - \bar{C}}{\bar{C}} &= -\sigma^{-1} \beta \frac{r_t - \bar{r}}{1 + \bar{r}} - \sigma^{-1} \beta \sum_{k=1}^{\infty} \beta^k \bar{\mathbb{E}} \left[\frac{r_{t+k} - \bar{r}}{1 + \bar{r}} \right] + (1 - \beta) \left[\frac{(\mu - 1) \nu}{\sigma + \mu \nu} \frac{DIV_t}{\bar{DIV}} + \frac{(1 + \nu)}{\sigma + \mu \nu} \frac{W_t}{\bar{W}} \right] \\ &\quad + (1 - \beta) \sum_{k=1}^{\infty} \beta^k \bar{\mathbb{E}} \left[\frac{(\mu - 1) \nu}{\sigma + \mu \nu} \frac{DIV_{t+k}}{\bar{DIV}} + \frac{(1 + \nu)}{\sigma + \mu \nu} \frac{W_{t+k}}{\bar{W}} \right] \end{aligned} \quad (E7)$$

$$\begin{aligned} \frac{N_t^s - \bar{N}}{\bar{N}} &= \nu^{-1} \beta \frac{r_t - \bar{r}}{1 + \bar{r}} + \nu^{-1} \beta \sum_{k=1}^{\infty} \beta^k \bar{\mathbb{E}} \left[\frac{r_{t+k} - \bar{r}}{1 + \bar{r}} \right] - (1 - \beta) \frac{(\mu - 1) \sigma}{\sigma + \mu \nu} \frac{DIV_t}{\bar{DIV}} + \nu^{-1} \left(1 - \sigma \frac{(1 - \beta)(1 + \nu)}{\sigma + \mu \nu} \right) \frac{W_t}{\bar{W}} \\ &\quad - (1 - \beta) \sum_{k=1}^{\infty} \beta^k \bar{\mathbb{E}} \left[\frac{(\mu - 1) \sigma}{\sigma + \mu \nu} \frac{DIV_{t+k}}{\bar{DIV}} + \frac{(1 + \nu) \sigma \nu^{-1}}{\sigma + \mu \nu} \frac{W_{t+k}}{\bar{W}} \right] \end{aligned} \quad (E8)$$

Goods and labor markets. The interest rates and wages arise from clearing the goods and labor markets,

$$N_t^s = N_t^d \quad (E9)$$

and

$$Y_t = I_t + \Psi_t + C_t \quad (E10)$$

Equilibrium. We study a *temporary equilibrium* in which agents maximize their utilities, taking as given the average expectations $\bar{\mathbb{E}}_t[\cdot]$, and markets clear. In this RBC model, equations (E1-E10) characterizes the equilibrium given the average expectations.

We adopt a quarterly calibration of the RBC economy, with all parameters listed in Table E1. We assume slight decreasing returns to scale in production ($\alpha < \gamma$) and the existence of a small fringe of rational agents, discussed in detail after introducing shallow thinking.

Table E1: Quarterly calibration of the RBC economy

Parameter	Description	Value	Estimate/Target
Beliefs			
λ	Continuation rate of depth of thinking	0.61	Our survey evidence
ϑ	Share of rational agents	0.1	
Firms			
Z	Productivity	0.33	Steady state $\bar{Y} = 1$
ρ	Persistence of productivity shocks	0.979	King and Rebelo (1999)
α	Capital share	0.25	
$1 - \gamma$	Labor share	0.67	
δ	Capital depreciation rate	0.025	
ψ	Capital adjustment cost	1	
Households			
β	Discount factor	0.99	Steady state annual $\bar{r} = 4\%$
σ^{-1}	Elasticity of intertemporal substitution (EIS)	1	
ν^{-1}	Frisch elasticity	0.5	
φ	Labor disutility scale	0.81	Steady state $\bar{N} = 1$

E.2 Shallow Thinking and Modeling Choices

Similar to Section 5.1, we represent the RBC economy by the sequences of eight variables $\{r_t, Y_t, I_t, DIV_t, N_t^d, C_t, N_t^s, W_t\}_{t \geq 0}$, which we refer to as $\mathcal{V}$.³⁷ Six of them are agents' actions, which we collect as $\mathcal{V}^{action}$, whereas the other two, $\{r_t, W_t\}_{t \geq 0}$, are prices formed in the competitive goods and labor market.

We characterize the rational expectations equilibrium (REE) and introduce shallow thinking beliefs accordingly. For REE, each action $\mathbf{v} = (\{v_t\}_{t \geq 0})'$ in $\mathcal{V}^{action}$ can be represented in the sequence space as

$$\mathbf{v}^{REE} = \sum_{\mathbf{u} \in \mathcal{V}} \mathbf{J}_{\mathbf{v}\mathbf{u}} \mathbf{u}^{REE} + \boldsymbol{\epsilon}^v, \forall \mathbf{v} \in \mathcal{V}^{action} \quad (\text{E11})$$

where $\mathbf{J}_{\mathbf{v}\mathbf{u}}$ encodes the dependence among sequences of variables. Households' consumption $\{C_t\}_{t \geq 0}$ and labor supply $\{N_t^s\}_{t \geq 0}$ depend on the interest rate $\{r_t\}_{t \geq 0}$, dividend $\{DIV_t\}_{t \geq 0}$, and wage $\{W_t\}_{t \geq 0}$. Firms' output $\{Y_t\}_{t \geq 0}$, investment $\{I_t\}_{t \geq 0}$, dividend $\{DIV_t\}_{t \geq 0}$, and labor demand $\{N_t^d\}_{t \geq 0}$ are functions of the interest rate $\{r_t\}_{t \geq 0}$ and wage $\{W_t\}_{t \geq 0}$, subject to the productivity shock $\{Z_t\}_{t \geq 0}$. Differing from the textbook New Keynesian model with no state variable, in the RBC economy firms have a stock of capital. Thus firms' actions depend on not only current and future interest rates and wages, but also past ones. Yet

³⁷For brevity, we ignore the adjustment cost Ψ_t , which is 0 to the first order around the steady state since the adjustment cost is quadratic.

(E11) is general enough to allow for it.

Instead of using (E11) together with market clearing conditions to solve for the REE, as done in Auclert et al. (2021), we use it to represent the causal relations and reinterpret the market clearing conditions as such. We pedagogically write down these expressions by associating each price with a competitive market, i.e., wages with the labor market and interest rates with the goods market.

For labor supply and demand N^s, N^d , by separating their dependence on the wage W from the rest, we interpret them as supply and demand curves in the sequence space

$$v^{REE} = J_{vW} W^{REE} + \hat{v}^{REE}, \quad v \in \{N^s, N^d\}$$

with elasticities $J_{N^s W}, J_{N^d W}$ and shifts $\hat{N}^{s, REE}, \hat{N}^{d, REE}$ defined as

$$\hat{v}^{REE} = \sum_{u \in \mathcal{V} \setminus \{W\}} J_{vu} u^{REE} + \epsilon^v, \quad v \in \{N^s, N^d\} \quad (E12)$$

In the RBC economy, both supply and demand elasticities $J_{N^s W}, J_{N^d W}$ are non-zero. By equalizing $N^{s, REE} = N^{d, REE}$, we can interpret the wage W^{REE} as resulting from shifts $\hat{N}^{d, REE}, \hat{N}^{s, REE}$

$$W^{REE} = (J_{N^s W} - J_{N^d W})^{-1} (\hat{N}^{d, REE} - \hat{N}^{s, REE}) \quad (E13)$$

For output, investment and consumption Y, I, C , by separating their dependence on the interest rate r from the rest, we interpret them as supply and demand curves as well

$$v^{REE} = J_{vr} r^{REE} + \hat{v}^{REE}, \quad \forall v \in \{Y, I, C\}$$

with elasticities J_{Yr}, J_{Ir}, J_{Cr} and shifts $\hat{Y}, \hat{I}, \hat{C}$ defined as

$$\hat{v}^{REE} = \sum_{u \in \mathcal{V} \setminus \{r\}} J_{vu} u^{REE} + \epsilon^v, \quad v \in \{Y, I, C\} \quad (E14)$$

By equalizing $Y = I + C$, we can interpret the interest rate as dependent on shifts $\hat{I}^{REE}, \hat{C}^{REE}, \hat{Y}^{REE}$

$$r^{REE} = (J_{Yr} - J_{Ir} - J_{Cr})^{-1} (\hat{I}^{REE} + \hat{C}^{REE} - \hat{Y}^{REE}) \quad (E15)$$

To determine beliefs, we reason with $\hat{v}$ as in (E12, E14) instead of v in (E11). Taking stock, (E11) and (E13, E15) characterize the REE. We can similarly express it as a linear system and apply Propositions 6 and ??, as in Section 5.1. Figure E1 illustrates the system

of causal relations that represent the RBC economy.

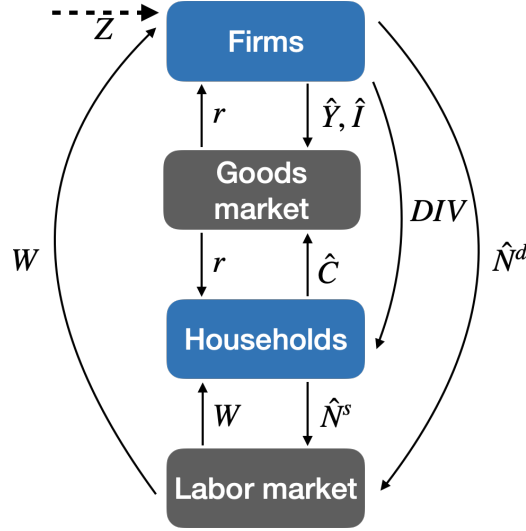


Figure E1: Causal relations of RBC economy in the sequence space

Modeling choices. We make two modeling choices regarding the returns to scale in production and the existence of a fringe of rational agents, to ensure a stationary equilibrium. We leave it for future research to explore other approaches.

Regarding firms' production, we set decreasing returns to scale with $\alpha < \gamma$. The reason is that if firms operate under constant returns, a finite-depth agent who knows the productivity shock $\{Z_t\}_{t \geq 0}$ but does not anticipate changes in wages $\{W_t\}_{t \geq 0}$ or interest rates $\{r_t\}_{t \geq 0}$ would believe that firms expand their capital stock to a different steady level.³⁸ That introduces nonstationary beliefs, leading to nonstationary equilibrium responses. In such a case, agents' beliefs will be perpetually wrong, unlike in a stationary case where agents correctly understand the steady state but misjudge the temporary impulse responses. For this reason, we consider decreasing returns to production as the generic case, with constant returns approximated by a small degree of decreasing returns.

Regarding the existence of rational agents, we first note that households' forward-looking behavior is necessary for the existence of a stationary equilibrium in the RBC

³⁸If there is no adjustment cost ($\psi = 0$), then the agent expects firms to infinitely expand their capital stock and production. With a positive adjustment cost ($\psi > 0$), the agent expects firms to expand their capital stock to a steady level different from before.

economy. Suppose we study a production economy with labor as a factor in fixed supply and firms like those in our RBC economy. No equilibrium with stationary $\{r_t, W_t\}_{t \geq 0}$ exists to meet the market clearing conditions $N_t^d = \bar{N}$ and $Y_t = I_t$, even if we assume that firms are rational. Hence, it is the forward-looking behavior of households that supports the RBC economy under rational expectations. In modeling shallow thinking of dynamic general equilibrium (Proposition 6, Assumptions ?? and 2), we nest rational expectations with $\lambda = 1$, but otherwise do not allow beliefs to be directly related to the actual equilibrium for simplicity. That is because the causal relations used in Assumption ?? are those of the rational expectations equilibrium rather than the actual equilibrium. In the RBC economy, if households have beliefs that are entirely disconnected from the actual equilibrium, no stationary equilibrium exists. Thus, we assume that a small ϑ share of agents (both firms and households) are rational, giving rise to average expectations (for a generic variable v) $\bar{\mathbb{E}}^{mix}[v_t] = \vartheta v_t + (1 - \vartheta) \bar{\mathbb{E}}[v_t]$ that determine the equilibrium, where v_t is the actual equilibrium outcome and $\bar{\mathbb{E}}[v_t]$ represents the average shallow thinking expectations from Proposition ?. Future work may improve on this ad hoc assumption by modeling shallow thinking as a fixed point, using the causal relations of the actual dynamic general equilibrium.

F Implementing Shallow Thinking with the Sequence-Space Jacobian

We pedagogically write down a 3-step procedure to implement shallow thinking in the RBC economy using the sequence-space Jacobians (SSJ) toolkit developed by [Auclert et al. \(2021\)](#). This procedure can also be applied to other models that can be solved in the sequence space, including models with incomplete markets and idiosyncratic shocks.

Step 1: characterizing REE. In the first step, we determine the causal relations as the Jacobians of the rational expectations equilibrium (REE) in terms of agents' best responses, i.e., (E11). We strictly adhere to expressions of agents' choices as functions of decision-relevant variables. For example, household consumption $\{C_t\}_{t \geq 0}$ and labor supply $\{N_t^s\}_{t \geq 0}$ respond to wages $\{W_t\}_{t \geq 0}$ and dividends $\{DIV_t\}_{t \geq 0}$, instead of aggregate output $\{Y_t\}_{t \geq 0}$. The REE is characterized by these Jacobians together with market clearing conditions.

Step 2: determining shallow thinking beliefs. In the second step, we characterize the average shallow thinking expectations $\bar{\mathbb{E}}[\cdot]$ of the sequences of eight variables $\mathcal{V} = \{\mathbf{r}, \mathbf{Y}, \mathbf{I}, \mathbf{DIV}, \mathbf{N}^d, \mathbf{C}, \mathbf{N}^s, \mathbf{W}\}$ with the REE Jacobians. In order to use the SSJ toolkit for Proposition ??, we construct a *modified model* as a directed *acyclic* graph (DAG) based on the REE Jacobians. We divide the system of equations in Proposition ?? into two groups, one concerning agents' actions and one about prices from competitive markets.

Based on (E11), the first group of equations concerning agents is

$$\bar{\mathbb{E}}[\mathbf{v}] = \lambda \sum_{\mathbf{u} \in \mathcal{V}} \mathbf{J}_{\mathbf{vu}} \bar{\mathbb{E}}[\mathbf{u}] + \epsilon^v, \quad \forall \mathbf{v} \in \mathcal{V}^{action} \equiv \mathcal{V} \setminus \{\mathbf{r}, \mathbf{W}\} \quad (\text{F1})$$

where $\{\mathbf{J}_{\mathbf{vu}}\}$ are the Jacobians of the REE determined in the first step. To embed this in the SSJ toolkit, we construct the *modified households and firms blocks* as follows. We replicate the households and firms blocks with REE Jacobians and modify all the Jacobians by λ .³⁹ Then, we create Jacobians of shifts $\bar{\mathbb{E}}[\hat{\mathbf{N}}^d], \bar{\mathbb{E}}[\hat{\mathbf{N}}^s]$ w.r.t. decision-relevant variables by replicating the Jacobians of $\bar{\mathbb{E}}[\mathbf{N}^d, \mathbf{N}^s]$ and setting their dependence on $\bar{\mathbb{E}}[\mathbf{W}]$ to zero. Similarly, we create Jacobians of shifts $\bar{\mathbb{E}}[\hat{\mathbf{Y}}, \hat{\mathbf{I}}, \hat{\mathbf{C}}]$ w.r.t. decision-relevant variables by replicating those of $\bar{\mathbb{E}}[\mathbf{Y}, \mathbf{I}, \mathbf{C}]$ and setting their dependence on $\bar{\mathbb{E}}[\mathbf{r}]$ to zero.

According to (E13, E15), the second group about markets consists of

$$\begin{aligned} \bar{\mathbb{E}}[\mathbf{W}] &= \lambda (\mathbf{J}_{\mathbf{N}^s \mathbf{W}} - \mathbf{J}_{\mathbf{N}^d \mathbf{W}})^{-1} (\bar{\mathbb{E}}[\hat{\mathbf{N}}^d] - \bar{\mathbb{E}}[\hat{\mathbf{N}}^s]) \\ \bar{\mathbb{E}}[\mathbf{r}] &= \lambda (\mathbf{J}_{\mathbf{Yr}} - \mathbf{J}_{\mathbf{Ir}} - \mathbf{J}_{\mathbf{Cr}})^{-1} (\bar{\mathbb{E}}[\hat{\mathbf{I}}] + \bar{\mathbb{E}}[\hat{\mathbf{C}}] - \bar{\mathbb{E}}[\hat{\mathbf{Y}}]) \end{aligned}$$

which can be rewritten as

$$\lambda (\bar{\mathbb{E}}[\hat{\mathbf{N}}^d] - \bar{\mathbb{E}}[\hat{\mathbf{N}}^s]) - (\mathbf{J}_{\mathbf{N}^s \mathbf{W}} - \mathbf{J}_{\mathbf{N}^d \mathbf{W}}) \bar{\mathbb{E}}[\mathbf{W}] = \mathbf{0} \quad (\text{F2})$$

$$\lambda (\bar{\mathbb{E}}[\hat{\mathbf{I}}] + \bar{\mathbb{E}}[\hat{\mathbf{C}}] - \bar{\mathbb{E}}[\hat{\mathbf{Y}}]) - (\mathbf{J}_{\mathbf{Yr}} - \mathbf{J}_{\mathbf{Ir}} - \mathbf{J}_{\mathbf{Cr}}) \bar{\mathbb{E}}[\mathbf{r}] = \mathbf{0} \quad (\text{F3})$$

We set up *fictitious Walrasian auctioneer blocks* as simple blocks that take $\bar{\mathbb{E}}[\hat{\mathbf{N}}^d, \hat{\mathbf{N}}^s, \hat{\mathbf{Y}}, \hat{\mathbf{I}}, \hat{\mathbf{C}}]$ and $\bar{\mathbb{E}}[\mathbf{W}, \mathbf{r}]$ as inputs and produce the residuals of (F2, F3) as outputs.

Putting together the modified households and firms blocks and the fictitious Walrasian auctioneer blocks forms a DAG, with $\bar{\mathbb{E}}[\mathbf{W}, \mathbf{r}]$ as unknowns and (F2, F3) as targets. Solving this DAG yields the average shallow thinking expectations $\bar{\mathbb{E}}[\cdot]$ of the eight variables

³⁹In the RBC model, the households and firms blocks have no cyclic dependence. If there were, one could turn a cycle into a target to use the SSJ toolkit.

$$\mathcal{V} = \{\mathbf{r}, \mathbf{Y}, \mathbf{I}, \mathbf{DIV}, \mathbf{N}^d, \mathbf{C}, \mathbf{N}^s, \mathbf{W}\}.$$

Step 3: calculating equilibrium given beliefs. In the last step, we determine the equilibrium given the average shallow thinking expectations. Here we allow for a slight generalization that ϑ share of agents are rational, hence the average expectations that matter for the equilibrium is

$$\bar{\mathbb{E}}^{mix}[v_t] = \vartheta v_t + (1 - \vartheta) \bar{\mathbb{E}}[v_t] \quad (\text{F4})$$

where v_t is the actual equilibrium outcome and $\bar{\mathbb{E}}[v_t]$ represents the average shallow thinking expectations determined above. The REE is nested by either $\vartheta = 1$ or $\lambda = 1$ (hence $\bar{\mathbb{E}}[v_t] = v_t$). Outside the limits, forward-looking agents with beliefs (F4) are surprised in each period t and change their behavior when a decision-relevant variable turns out different from their beliefs ($v_t \neq \bar{\mathbb{E}}^{mix}[v_t]$). They behave as

$$\mathbf{v} = \sum_{\mathbf{u} \in \mathcal{V}} \left(\mathbf{J}_{\mathbf{vu}} \bar{\mathbb{E}}^{mix}[\mathbf{u}] + \check{\mathbf{J}}_{\mathbf{vu}} \left(\mathbf{u} - \bar{\mathbb{E}}^{mix}[\mathbf{u}] \right) \right) + \epsilon^v, \quad \forall \mathbf{v} \in \mathcal{V}^{action} \quad (\text{F5})$$

where

$$(\check{\mathbf{J}}_{\mathbf{vu}})_{ts} \equiv \begin{cases} (\mathbf{J}_{\mathbf{vu}})_{t-s,0} & s \leq t \\ 0 & s > t \end{cases}$$

is the myopic Jacobian that captures responses to variables observed in the past.⁴⁰ Rearranging terms and plugging in (F4), we get for any $\mathbf{v} \in \mathcal{V}^{action}$

$$\begin{aligned} \mathbf{v} &= \sum_{\mathbf{u} \in \mathcal{V}} \check{\mathbf{J}}_{\mathbf{vu}} \mathbf{u} + \sum_{\mathbf{u} \in \mathcal{V}} (\mathbf{J}_{\mathbf{vu}} - \check{\mathbf{J}}_{\mathbf{vu}}) \bar{\mathbb{E}}^{mix}[\mathbf{u}] + \epsilon^v \\ &= \sum_{\mathbf{u} \in \mathcal{V}} \check{\mathbf{J}}_{\mathbf{vu}} \mathbf{u} + \sum_{\mathbf{u} \in \mathcal{V}} (\mathbf{J}_{\mathbf{vu}} - \check{\mathbf{J}}_{\mathbf{vu}}) (\vartheta \mathbf{u}_t + (1 - \vartheta) \bar{\mathbb{E}}[\mathbf{u}_t]) + \epsilon^v \\ &= \sum_{\mathbf{u} \in \mathcal{V}} \underbrace{(\check{\mathbf{J}}_{\mathbf{vu}} + \vartheta (\mathbf{J}_{\mathbf{vu}} - \check{\mathbf{J}}_{\mathbf{vu}}))}_{\equiv \tilde{\mathbf{J}}_{\mathbf{vu}}} \mathbf{u} + \sum_{\mathbf{u} \in \mathcal{V}} \underbrace{(1 - \vartheta) (\mathbf{J}_{\mathbf{vu}} - \check{\mathbf{J}}_{\mathbf{vu}}) \bar{\mathbb{E}}[\mathbf{u}_t]}_{\equiv \tilde{\epsilon}^v} + \epsilon^v \end{aligned} \quad (\text{F6})$$

where the first part $\tilde{\mathbf{J}}_{\mathbf{vu}} \mathbf{u}$ encodes the dependence among equilibrium outcomes and the second part $\tilde{\epsilon}^v$ is determined independent of the equilibrium. Thus $\tilde{\epsilon}^v$ is equivalent to a

⁴⁰See Auclert et al. (2021) and Auclert, Rognlie and Straub (2020) for the validity of this expression to the first order.

shock to forward-looking agents, which we call *pseudo shocks*.

To determine the equilibrium with the SSJ toolkit, we construct a model by replacing the REE Jacobians for forward-looking agents by $\tilde{J}_{vu}$ and adding pseudo shocks $\tilde{\epsilon}^v$ in addition to the true shocks ϵ^v . We keep the market clearing conditions as they are. Solving for this model yields the equilibrium given shallow thinking beliefs (mixed with a ϑ share of rational beliefs).